

Prevention and Management of Acute Kidney Injury

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KEY POINTS

- Acute kidney injury (AKI) is a pleomorphic syndrome defined by rapid loss of kidney function that has multiple etiologies and clinical manifestations.
- Clinically, acute kidney injury is categorized in broad pathophysiologic categories as prerenal or hemodynamically mediated AKI; postrenal or obstructive AKI; and intrinsic AKI, which itself may be due to a variety of pathophysiologic processes including acute tubular injury and acute interstitial nephritis. While these categories are useful conceptually, individual patients may have overlapping presentations.
- Specific therapies are not available for the majority of etiologies of intrinsic AKI; management is therefore predominantly supportive.
- In severe AKI, kidney replacement therapy is used to treat volume overload and metabolic and electrolyte complications. Early initiation of and intensive delivery of kidney replacement therapy, however, are not associated with improved clinical outcomes.

DEFINITION OF ACUTE KIDNEY INJURY

Acute kidney injury (AKI) is a heterogeneous syndrome defined by rapid (occurring over hours to days) decline in the glomerular filtration rate (GFR), resulting in the retention of metabolic waste products, including urea and creatinine, and dysregulation of fluid, electrolyte, and acid-base homeostasis.¹ Although often considered a

discrete syndrome, AKI represents a broad constellation of pathophysiologic processes of varied severity and etiology. These include decreases in GFR as the result of hemodynamic perturbations that disrupt normal renal perfusion without causing parenchymal injury, partial or complete obstruction to urinary flow, and a range of processes with characteristic patterns of glomerular, interstitial, tubular, or vascular parenchymal injury. AKI occurs in heterogeneous patient populations (genetics, age, kidney functional status,

accompanying comorbidities), and the etiology is often multifactorial.

The term AKI has largely supplanted older terminology of acute renal failure (ARF). This change reflects recognition of serious shortcomings with the older terminology. *Acute renal failure* suggested a dichotomous relationship between normal kidney function and overt organ failure; in contrast, *acute kidney injury* acknowledges the association of small to moderate acute and transient decrements in kidney function with risks of subsequent progressive kidney disease, cardiovascular and other nonrenal morbidity, and an increased mortality risk. While the AKI terminology does emphasize the graded aspect of acute kidney disease, it should be recognized that this terminology is also imperfect. The term “injury” can be construed to imply the presence of parenchymal organ damage, which may be absent in a variety of settings associated with an acute decline in kidney function, such as early obstructive disease and prerenal azotemia associated with volume depletion. Although the term “acute kidney dysfunction” might better characterize the entire spectrum of the syndrome, acute kidney injury is the term that has been adopted by consensus and is now increasingly used in the medical literature.^{2,3} In this chapter, the term AKI will be used to describe the entire spectrum of the syndrome. Although in clinical practice the terms “acute tubular necrosis” (ATN) and “acute tubular injury” (ATI) are often used synonymously with AKI, these terms should not be used interchangeably. Though ATN is the most common form of intrinsic AKI, particularly in critically ill patients, it represents only one of multiple forms of AKI. In addition, there may be a lack of concordance between the clinical syndrome and the classic histopathologic findings of ATN, which has led some to prefer the term ATI.⁴⁻⁶

Decreased urine output is a cardinal (though not universal) manifestation of AKI, and patients are often classified on the basis of urine flow rates as nonoliguric (urine output >400 mL/day), oliguric (urine output <400 mL/day), or anuric (urine output <100 mL/day).⁷ Transient oliguria may occur in the absence of significant decrements in kidney function, as increased tubular salt and water reabsorption is a normal physiologic response to volume depletion. In contradistinction, persistent oliguria despite the presence of adequate intravascular volume is virtually always a manifestation of AKI, with lower urine volume typically associated with more severe initial renal injury. The categorization of AKI based on urine volume has clinical implications for the development of volume overload, severity of electrolyte disturbances, and overall prognosis. Although oliguric AKI is associated with higher mortality risk than nonoliguric AKI, therapeutic interventions to augment urine output (*vide infra*) have not been shown to improve patient outcomes.⁸

AKI can develop *de novo* in the setting of normal kidney function or can be superimposed on underlying chronic kidney disease (CKD) (acute-on-chronic kidney injury). The presence of underlying impaired kidney function has been shown to be one of the most important risk factors for the development of AKI.^{9,10} Multiple mechanisms may contribute to this increased susceptibility, including diminished renal functional reserve, impaired salt and water conservation predisposing to intravascular volume contraction, decreased activity of detoxification mechanisms

increasing susceptibility to cytotoxic injury, impaired clearance of potential nephrotoxins increasing the risk for and/or duration of exposure, and associated macrovascular and microvascular disease increasing the risk of ischemic injury.

The etiologies of AKI are usually divided into three broad pathophysiologic categories:

1. Prerenal AKI—diseases characterized by effective hypoperfusion of the kidneys in which there is no parenchymal damage to the kidney (Table 28.1)
2. Intrinsic AKI—diseases involving the renal parenchyma (Table 28.2)
3. Postrenal (obstructive) AKI—diseases associated with acute obstruction of the urinary tract (Table 28.3)

Although these categories are useful for didactic purposes and help inform the initial clinical assessment of patients presenting with AKI, there is often overlap between them. For example, renal hypoperfusion may cause a spectrum of renal injury ranging from prerenal azotemia to overt ATI depending on its severity and duration. As a result, precise categorization of the etiology of AKI in these three groups

Table 28.1 Causes of Prerenal Acute Kidney Injury

Intravascular volume depletion

Hemorrhage—trauma, surgery, postpartum, gastrointestinal
 Gastrointestinal losses—diarrhea, vomiting, nasogastric loss
 Renal losses—diuretics, osmotic diuresis, diabetes insipidus
 Skin and mucous membrane losses—burns, hyperthermia
 Nephrotic syndrome
 Cirrhosis
 Capillary leak

Reduced cardiac output

Cardiogenic shock
 Pericardial diseases—restrictive/constrictive/tamponade
 Congestive heart failure
 Valvular diseases
 Pulmonary diseases—pulmonary hypertension; pulmonary embolism
 Sepsis

Systemic vasodilation

Sepsis
 Cirrhosis
 Anaphylaxis
 Medications

Renal vasoconstriction

Early sepsis
 Hepatorenal syndrome
 Acute hypercalcemia
 Drugs—norepinephrine, vasopressin, nonsteroidals, angiotension-converting enzyme, calcineurin inhibitors, iodinated contrast agents

Increased intraabdominal pressure

Abdominal compartment syndrome

Table 28.2 Major Causes of Intrinsic Acute Kidney Injury

Tubular injury	
Ischemia due to hypoperfusion	Hypovolemia, sepsis, hemorrhage, cirrhosis, CHF.
Endogenous toxins	See Table 28.1
Exogenous toxins	Myoglobin, hemoglobin, paraproteinemia, uric acid. See Table 28.5
	Antibiotics, chemotherapy agents, radiocontrast, phosphate preps. See Table 28.5
Tubulointerstitial injury	
Acute allergic interstitial nephritis	Nonsteroidal antiinflammatories, antibiotics
Infections	Viral, bacterial, and fungal
Infiltration	infections
Allograft rejection	Lymphoma, leukemia, sarcoid
Glomerular injury	
Inflammation	Anti-GBM disease, ANCA-associated GN, postinfectious, cryoglobulinemia, membranoproliferative glomerulonephritis, IgA nephropathy, SLE, Henoch-Schonlein purpura, polyarteritis nodosa
Hematologic	Hemolytic uremic syndrome, thrombotic thrombocytopenic purpura, medications
Renal microvasculature	
	Malignant hypertension, toxemia of pregnancy, hypercalcemia, radiocontrast, scleroderma, medications
Large vessels	
Arterial	Thrombosis, vasculitis, dissection, thromboembolism, atheroembolism, trauma
Venous	Thrombosis, compression, trauma

ANCA, Anti-neutrophil cytoplasmic antibody; CHF, congestive heart failure; GBM, glomerular basement membrane; GN,

Table 28.3 Causes of Postrenal Acute Kidney Injury

Upper urinary tract extrinsic causes	
Retroperitoneal space—lymph nodes, tumors	
Pelvic or intraabdominal tumors—cervix, uterus, ovary, prostate	
Fibrosis—radiation, drugs, inflammatory	
Ureteral ligation or surgical trauma	
Granulomatous diseases	
Hematoma	
Lower urinary tract causes	
Prostate—benign prostatic hypertrophy, carcinoma, infection	
Bladder—neck obstruction, calculi, carcinoma, infection (schistosomiasis)	
Functional—neurogenic bladder secondary to spinal cord injury, diabetes, multiple sclerosis, stroke, pharmacologic side effects of drugs (anticholinergics, antidepressants)	
Urethral—posterior urethral valves, strictures, trauma, infections, tuberculosis, tumors	
Upper urinary tract intrinsic causes	
Nephrolithiasis	
Strictures	
Edema	
Debris, blood clots, sloughed papillae, fungal ball	
Malignancy	

may not always be possible and individual patients may transition between etiologic categories.

The prior absence of a uniform operational definition of AKI impeded epidemiologic studies and hampered clinical evaluations of preventative and therapeutic interventions. Older literature was characterized by multiple definitions using varying absolute and/or relative changes in the serum creatinine concentration with or without associated decrements in urine output, which made it difficult to compare findings across studies. In 2002, the Acute Dialysis Quality Improvement Initiative (ADQI) proposed the first consensus definition of AKI. The ADQI workgroup

proposed a classification scheme with three strata based on the magnitude of increase in serum creatinine and/or duration of oliguria (Table 28.4).

As proposed, the first stratum provides the greatest sensitivity for diagnosing AKI while higher strata provide increasing specificity of diagnosis. These three strata were combined with two outcome stages defined by the need for and duration of kidney replacement therapy (KRT), resulting in the five-tiered RIFLE classification (*R*isk of renal dysfunction, *I*njury to the Kidney, *F*ailure of kidney function and the two outcome stages, *L*oss of kidney function and *E*nd-stage kidney disease).¹¹ Subsequently, the Acute Kidney Injury Network (AKIN) modified the RIFLE classification by adding an absolute increase in serum creatinine of ≥ 0.3 mg/dL to the 50% relative increase in the serum creatinine to the definition of AKI and specifying that these increments develop over no more than 48 hours² (see Table 28.4). This definition has been further modified in the Kidney Disease: Improving Global Outcomes (KDIGO) Clinical Practice Guideline for Acute Kidney Injury, which clarifies that while the 0.3 mg/dL increment in serum creatinine needs to occur over 48 hours, from a known baseline value, the 50% increase may occur over a longer 7-day interval.³

The KDIGO Clinical Practice Guideline for AKI recognized a gap in the nosology of acute and CKD.³ On the basis of the definitions mentioned earlier, AKI has an onset of fewer than 7 days while CKD is defined by the presence of impaired kidney function or structural kidney damage for more than 3 months with implications for health.¹² Recognizing that some patients develop kidney disease with a more subacute onset than that of AKI but of less than 3 months' duration, the KDIGO Acute Kidney Injury Workgroup proposed the

Table 28.4 RIFLE, Acute Kidney Injury Network (AKIN) and Kidney Disease: Improving Global Outcomes (KDIGO) Definitions and Staging of Acute Kidney Injury

Definitions						
		RIFLE		AKIN		KDIGO
Serum creatinine		An increase of >50% developing over <7 days		An increase of >0.3 mg/dL or of >50% developing over <48 h		An increase of >0.3 mg/dL developing over <48 h; or an increase of >50% developing over <7 days
Urine output ^a		<0.5 mL/kg/h for >6 h				
Staging criteria						
Increase in serum creatinine						Urine output ^a
RIFLE		AKIN		KDIGO		
Risk	≥50%	Stage 1	≥0.3 mg/dL; or ≥50%	Stage 1	≥0.3 mg/dL; or ≥50%	<0.5 mL/kg/h for >6 h
Injury	≥100%	Stage 2	≥100%	Stage 2	≥100%	<0.5 mL/kg/h for >12 h
Failure	≥200%	Stage 3	≥200%	Stage 3	≥200%	<0.3 mL/kg/h for >24 h or anuria for >12 h
Loss	Need for kidney replacement therapy for >4 weeks					
End-stage	Need for kidney replacement therapy for >3 months					

^aThe urine output criteria for both definition and staging of AKI are the same for the RIFLE, AKIN, and KDIGO criteria.

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concept of acute kidney disease (AKD), defined as AKI or a reduction in GFR to <60 mL/min/1.73 m², a decrease in GFR by ≥35%, an increase in serum creatinine by >50%, or the presence of structural kidney damage of <3 months' duration. The concept of AKD has been further expanded by ADQI to include AKI that persists for more than 7 days.¹³

Several limitations to these criteria for diagnosis and staging of AKI have been recognized.^{14,15} First, while validation studies have demonstrated that AKI stage correlates with mortality risk, it is not clear that this is the appropriate metric for assessing their validity as a definition of kidney disease. Second, there is poor correlation between AKI stage and GFR. Since the magnitude of change in serum creatinine is time dependent, a patient may demonstrate progression over time from less severe (RIFLE-R or AKIN or KDIGO stage 1) to more severe AKI stage (RIFLE-F or AKIN or KDIGO stage 3) despite an improving GFR. Third, the definition of AKI by serum creatinine criteria relies on a referent baseline serum creatinine level, which is often unavailable. Furthermore, variations in specifications for this referent value (e.g., admission serum creatinine vs. most recent outpatient serum creatinine before admission vs. other definitions) can alter the classification of patients.¹⁶ Fourth, both RIFLE and AKIN definitions employ relative changes in serum creatinine to stage AKI. Analyses of creatinine kinetics demonstrate that the

time required to attain a fixed percent change in serum creatinine in the setting of severe AKI is dependent on the baseline level of kidney function, while the initial rate of change in serum creatinine is relatively independent of kidney function.¹⁴ Thus early in the course of AKI, absolute changes in serum creatinine may be detected more readily than relative changes. Fifth, concordance between the serum creatinine and urine output criterion is poor, even with regard to mortality risk.¹⁷ Transient changes in urine output may reflect variation in volume status or be due to the administration of medications and do not necessarily correlate with other parameters of kidney function. Finally, it must be noted that these classification systems are independent of the various etiologies of AKI (i.e., prerenal, intrinsic, and obstructive). Despite these shortcomings, the use of standardized classification schemes has enhanced the interpretation of epidemiologic studies and design of clinical trials.

Conceptually, AKI comprises a spectrum of structural and functional kidney disease in which there may be an evolution from injury to organ dysfunction and finally overt organ failure. Reliance solely on changes in serum creatinine and/or urine output to diagnose AKI has resulted in the inability to identify the incipient stages of intrinsic kidney damage, which may be the most opportune time for pharmacologic intervention.¹⁸ In order to facilitate the early diagnosis of

intrinsic injury, multiple biomarkers of tubular injury have been evaluated including N-acetyl-*B*-D-glucosaminidase (NAG), kidney injury molecule 1 (KIM-1), neutrophil gelatinase-associated lipocalin (NGAL), interleukin 18 (IL-18), liver fatty-acid binding protein (L-FABP), tissue inhibitor of metalloproteinase 2 (TIMP-2), and insulin-like growth factor-binding protein 7 (IGFBP7).^{19,24} In addition, serum cystatin C has been proposed as more sensitive (and in some settings more specific) than serum creatinine for detecting changes in GFR, and urinary cystatin C has been proposed as a marker of tubular injury.^{19,25,26} While most of these biomarkers have yet to be adequately validated for routine clinical use, they have the potential to provide an earlier diagnosis of intrinsic AKI than serum creatinine, to differentiate volume-responsive (prerenal) AKI from intrinsic disease, to diminish the confounding effect related to creatinine generation, and to provide prognostic information regarding the clinical course of an episode of AKI. One or more of these biomarkers may provide a means by which patients could be identified at the incipient stage of AKI to guide the implementation of specific therapy to ameliorate kidney damage or promote recovery of kidney function. For a detailed discussion on the pathophysiology of AKI, see Chapter 27.

INCIDENCE OF ACUTE KIDNEY INJURY

Estimates of the incidence of AKI are highly dependent on the case definition used, with rates among hospitalized patients ranging from as high as 44% when defined based on a change in serum creatinine of at least 0.3 mg/dL to as low as 1% using an increase in serum creatinine of at least 2.0 mg/dL.²⁷⁻³¹ Approximately 3% to 7% of hospitalized patients and 25% to 60% of intensive care unit (ICU) patients develop AKI, with 5% to 6% of the ICU population requiring KRT after developing AKI.²⁷⁻³¹ In a single-center analysis conducted in 1996 at an urban tertiary care hospital, AKI, defined as an increase in serum creatinine of 0.5 mg/dL for patients with a baseline serum creatinine of ≤ 1.9 mg/dL, of 1.0 mg/dL for patients with a baseline serum creatinine of 2.0 to 4.9 mg/dL, and of 1.5 mg/dL for patients with a baseline serum creatinine > 5 mg/dL, developed in 7.2% of 4622 consecutive patients.²⁷ The overall incidence of AKI is approximately 21.6% for all hospitalized adults worldwide,³² with known associations for accelerating CKD to end-stage kidney failure (ESKF).³³⁻³⁵ The more recent estimate of AKI incidence is considerably higher than the 4.9% investigators had observed in a similar study in 1979.³⁶ In the late 1970s, the most frequent etiology for AKI was decreased renal perfusion, observed in 39% of episodes, followed by medication-associated (16%), contrast-associated (11%), postoperative (9%), and sepsis-associated (6.5%). Overall mortality was 19.4%, with higher mortality rates associated with larger maximal increments in serum creatinine concentration.

Although definition is less of an issue regarding rates of AKI requiring KRT, reported rates vary considerably because of differences in characteristics of patient populations and variability in criteria for the initiation of KRT. In a multinational, multicenter observational study of 29,269 critically ill patients, 5.7% developed severe AKI

and 4.3% received KRT.³⁷ In a later, smaller multinational observational study, 13.5% of 1802 critically ill patients developed KRT-requiring AKI.³¹

Many epidemiologic studies of AKI have relied on data from large administrative databases. Such data need to be interpreted with caution, however, as administrative coding for AKI is incomplete and may only capture 20% to 30% of all episodes of AKI.^{38,39} Ascertainment of AKI requiring KRT using administrative data is substantially more complete. In an analysis of data from the National Hospital Discharge Survey in the United States, the Centers for Disease Control and Prevention observed an increase in hospital discharges with a diagnosis of AKI from 18 per 100,000 population in 1980 to 365 per 100,000 in 2005.⁴⁰ Similar trends have been observed in analyses of the U.S. Nationwide (more recently referred to as “National”) Inpatient Sample (NIS) and a 5% sample of U.S. hospitalized Medicare beneficiaries. In an analysis that combined administrative and clinical data from a single integrated health care delivery system, the incidence of AKI that did not require the use of KRT increased from 322.7 to 522.4 per 100,000 person-years from 1996 to 2003.⁴¹ Over the same period, AKI requiring KRT increased from 19.5 to 29.5 per 100,000 person-years.⁴¹ AKI was more common among men and older persons. In a more recent analysis using data from the U.S. National Inpatient Sample, a nationally representative administrative database of hospitalizations, the incidence of dialysis-requiring AKI increased from 222 per million person-years in 2000 to 533 per million person-years in 2009 with the largest rise in incidence occurring in patients 65 to 74 and 75 years of age and older.⁴² While a component of these temporal trends may be attributable to earlier initiation of dialysis and more frequent utilization of KRT in older patients, these changes are unlikely to account for the majority of the increase in incidence of severe AKI.

Preexisting kidney disease is one of the major risk factors for the development of dialysis requiring AKI.⁴³ More severe baseline CKD is associated with higher levels of risk. Compared with patients with baseline eGFR > 60 mL/min/1.73 m², patients with eGFR values of 45 to 59 mL/min/1.73 m² have a nearly twofold increased risk of developing dialysis-requiring AKI. This risk increases to more than 40-fold among patients with baseline eGFR values < 15 mL/min/1.73 m².⁴³ Underlying diabetes mellitus, hypertension, and the presence of proteinuria are also associated with the risk for hospital-acquired AKI.

CATEGORIZATION OF ACUTE KIDNEY INJURY

Although in the clinical setting AKI is often multifactorial, the etiology is commonly evaluated on the basis of three major pathophysiologic categories, namely prerenal, intrinsic, and postrenal (obstructive) AKI. Prerenal azotemia is the most common cause of AKI, accounting for about 40% to 55% of all cases.^{27,29,44} Prerenal AKI is caused by a reduction in blood flow to the kidneys that can result from absolute and/or effective reduction in intravascular volume. Common causes of absolute intravascular volume depletion leading to prerenal AKI include gastrointestinal losses (e.g., diarrhea, vomiting, and nasogastric suction), renal losses (e.g.,

over-diuresis and diabetes insipidus), and sequestration of extracellular fluid (e.g., “third spacing” as seen in acute pancreatitis) (see Table 28.1). Common conditions that cause effective intravascular volume contraction include heart failure and liver failure. In these conditions the absolute blood volume is increased, yet arterial blood flow to the kidneys is reduced, leading to prerenal azotemia. A more detailed narrative on prerenal AKI is contained in Chapter 27 on the pathophysiology of AKI.

Intrinsic AKI is frequently categorized on the basis of the anatomic location of primary renal injury including the vasculature, glomeruli, interstitium, and tubules (see Table 28.2). Diseases that affect the renal vasculature range from complete thrombosis of the renal artery or vein, which are uncommon events, to atheroembolic disease typically following instrumentation of renal or extrarenal vasculature, to vasculitides that affect small renal vessels including glomeruli. Such vasculitides may be confined to the kidney but are commonly systemic conditions with extrarenal manifestations. Acute interstitial nephritis (AIN) most often results from an idiosyncratic allergic response to a myriad of different pharmacologic agents, most commonly to antibiotics (e.g., methicillin and other penicillins, cephalosporins, sulfonamides, and quinolones) or nonsteroidal antiinflammatory drugs (NSAIDs) (e.g., ibuprofen).⁴⁵ Proton pump inhibitors have more recently been associated with impaired kidney function, hypothesized to be related, at least in part, to AIN.⁴⁶⁻⁴⁸ A multitude of other clinical conditions including bacterial infection can also lead to AIN. Systemic allergic signs such as fever, rash, and eosinophilia are often present in antibiotic-associated AIN but not usually present in NSAID-related AIN. The most common form of intrinsic AKI is ATI, also known as ATN. We use the term ATI preferentially because evidence of injury to renal tubular epithelial cells often predominates over overt epithelial cell necrosis.⁶ The etiologies of ATI are broadly categorized as ischemic, septic, and nephrotoxic. Prolonged prerenal azotemia and hypotension, even if short-lived, can lead to necrosis of tubular epithelial cells and ATI. Nephrotoxins leading to ATI can be extrinsic, such as iodinated contrast, antibiotics, and chemotherapeutic agents, or intrinsic, such as myoglobin, hemoglobin, and intratubular crystals (Table 28.5). Sepsis-associated AKI, although previously attributed to ischemia-reperfusion injury, is now believed to have a more complex pathogenesis and may develop in the absence of overt hypotension. These conditions including their pathogenesis and pathophysiology are discussed in detail in Chapter 27.

Postrenal (obstructive) AKI is characterized by blockage to the flow of urine. Obstruction, intrinsic or extrinsic, can occur at any point from the renal collecting system to the urethra (see Table 28.3). The development of postrenal AKI requires bilateral obstruction or unilateral obstruction in a patient with a solitary kidney or in the setting of CKD. The most common causes of postrenal AKI are functional or structural obstruction of the bladder neck due to prostatic conditions, anticholinergic agents, and neurogenic bladder. Additional etiologies of obstructive AKI include bilateral ureteral calculi, fibrosis, or blood clots; sloughed renal papillae (i.e.,

Table 28.5 Major Causes of Endogenous and Exogenous Toxins Causing Acute Tubular Injury

Endogenous toxins	Exogenous toxins
Myoglobinuria	Antibiotics
Muscle breakdown— trauma, compression, electric shock, hypothermia, hyperthermia, seizures, exercise, burns	Aminoglycosides Amphotericin B Antiviral agents—acyclovir, cidofovir, indinavir, foscarnet, tenofovir Pentamidine Vancomycin
Metabolic—hypokalemia, hypophosphatemia	
Infections—tetanus, influenza	
Toxins— <i>isopropyl alcohol</i> , ethanol, ethylene glycol, toluene, snake and insect bites, cocaine, heroin	
Drugs—HMG-CoA reductase inhibitors (statins), amphetamines, fibrates	
Inherited disease— deficiency of myophosphorylase, phosphofructokinase, carnitine palmitoyltransferase	
Autoimmune—polymyositis, dermatomyositis	
Hemoglobinuria	Chemotherapy
Mechanical—prosthetic valves, microangiopathic hemolytic anemia, extracorporeal circulation	Cisplatin Ifosfamide Placimycin 5-fluorouracil Cytarabine 6-thioguanine Methotrexate
Drugs—hydralazine, methyldopa	
Chemicals—benzene, arsine, fava beans, glycerol, phenol	
Immunologic—transfusion reaction	
Genetic—G6PD deficiency, PNH	
Intratubular obstruction from crystalluria or paraproteins	Calcineurin inhibitors
Tumor lysis syndrome HGPT deficiency Multiple myeloma Oxalate (ethylene glycol)	Cyclosporine Tacrolimus
	Organic solvents
	Toluene Ethylene glycol Poisons
	Miscellaneous
	Radiocontrast agents Intravenous immune globulin Nonsteroidal antiinflammatory drugs Oral phosphate bowel preps

G6PD, Glucose-6-phosphate dehydrogenase; PNH, paroxysmal nocturnal hematuria.

renal papillary necrosis); genitourinary malignancies; external compression from tumors or hemorrhage; and retroperitoneal fibrosis. A detailed description of renal obstruction is provided in [Chapter 39](#).

EVALUATION OF ACUTE KIDNEY INJURY

CLINICAL ASSESSMENT, URINARY FINDINGS, AND BLOOD AND RADIOGRAPHIC EVALUATION

The assessment of the patient with AKI requires a meticulous history and physical examination, comprehensive review of medical records, evaluation of urinary findings including the urinary sediment, review of laboratory tests, renal imaging, and, when appropriate, kidney biopsy⁴⁹ ([Table 28.6](#)). Analysis of serum creatinine concentration over time is invaluable for differentiating acute from CKD and identifying the timing of events that precipitated the acute decline in kidney function. The presence of an acute process is easily confirmed if a review of laboratory records reveals a sudden rise in blood urea nitrogen (BUN) and serum creatinine concentrations from previously stable baseline values. Spurious causes of elevated BUN and serum creatinine must be excluded before a diagnosis of AKI is made. When prior BUN and serum creatinine measurements are not available, key findings that suggest a chronic process is present include physical manifestations of hyperparathyroidism (resorption of distal phalangeal tufts or lateral aspect of clavicles), band keratopathy, “half-and-half” nails, and small echogenic kidneys on radiographic imaging. Enlarged kidneys do not necessarily rule out chronic processes such as diabetic nephropathy, HIV-associated nephropathy, amyloidosis, and polycystic kidney disease, all of which are characterized by enlarged kidney size even with moderate to advanced CKD. Anemia is a less useful differentiating feature as it is often present in both AKI and CKD. Once the presence of AKI has been confirmed, attention should focus on patient, urine, laboratory, and radiographic assessments to help distinguish among prerenal, intrinsic, and postrenal processes, to permit identification of the etiology of AKI, and to guide treatment.

CLINICAL ASSESSMENT OF THE PATIENT

Prerenal AKI should be suspected in clinical settings associated with intravascular volume depletion, including hemorrhage, excessive gastrointestinal, urinary, or insensible fluid losses and severe burns, or with reduced effective arterial blood volume due to congestive heart failure, liver disease, or nephrotic syndrome. The risk of intravascular volume depletion is increased in comatose, sedated, or obtunded patients and in patients with restricted access to salt and water. Clinical clues to a prerenal etiology of AKI on history include patient report of excessive thirst; orthostatic lightheadedness or dizziness; significant diarrhea and/or vomiting; diuretic use; recent use of medications that alter intrarenal hemodynamics including NSAIDs; and inhibitors of the renin-angiotensin axis including direct renin inhibitors, angiotensin-converting enzyme (ACE) inhibitors, and angiotensin receptor blockers (ARBs). Findings suggestive of volume depletion on

physical examination may include orthostatic hypotension (postural fall in diastolic blood pressure >10 mm Hg) and tachycardia (postural increase in heart rate >10 beats/minute), reduced jugular venous pressure, diminished skin turgor, dry mucous membranes, and the absence of axillary sweat. However, overt signs and symptoms of hypovolemia do not usually manifest until extracellular fluid volume has fallen by more than 10% to 20%. In addition, in patients with heart failure, liver disease, or nephrotic syndrome, renal hypoperfusion may be present despite total body volume overload. Findings on physical examination of peripheral edema, pulmonary vascular congestion, pleural effusion, cardiomegaly, the presence of an S3 heart sound, elevated jugular venous pressure, or hepatic congestion may point to a state of reduced cardiac output and decreased effective intravascular volume. The presence of acute or chronic liver disease is suggested by evidence of icterus, ascites, splenomegaly, palmar erythema, telangiectasia, and caput medusa. In select critically ill patients, invasive hemodynamic monitoring using central venous or pulmonary artery catheters or ultrasonography of the heart and central veins may assist in assessing intravascular volume status. A definitive diagnosis of prerenal AKI is usually based on prompt resolution of AKI after restoration of renal perfusion. In patients with underlying systolic heart failure, restoration of renal perfusion may be challenging and often requires the use of inotropic support.

There is a high likelihood of ischemic ATI if AKI follows a period of severe renal hypoperfusion and the impairment in kidney function persists or worsens despite restoration of renal perfusion. It should be noted, however, that significant hypotension is evident in less than 50% of patients with postsurgical ATI.⁵⁰ Although septic shock is a common cause of ATI, ATI may also develop in sepsis in the absence of overt hypotension.^{50,51} The diagnosis of nephrotoxic ATI requires a comprehensive review of all clinical, pharmacy, nursing, radiographic, and procedural notes for evidence of administration of nephrotoxic agents. Pigment-induced ATI may be suspected if the clinical assessment reveals risk factors for rhabdomyolysis (e.g., seizures, excessive exercise, alcohol or drug abuse, treatment with statins, prolonged immobilization, limb ischemia, and crush injury) or hemolysis, as well as selected signs and symptoms of the former (e.g., muscle tenderness, weakness, evidence of trauma, or prolonged immobilization).⁵²⁻⁵⁵

While most AKI is prerenal or ischemic, nephrotoxic, or septic ATI, patients should be carefully evaluated for other intrinsic renal parenchymal processes as their management and prognosis may differ substantially. Flank pain may be a prominent symptom of acute renal artery or renal vein occlusion, renal infarction (from systemic emboli), acute pyelonephritis, and rarely necrotizing glomerulonephritis.⁵⁶⁻⁵⁸ Interstitial edema leading to distention of the renal capsule and flank pain may be seen in up to one-third of patients with AIN.⁵⁹ Dermatologic examination can be highly informative. A maculopapular rash may accompany allergic interstitial nephritis. Subcutaneous nodules, livedo reticularis, digital ischemia, and palpable purpura may suggest atheroembolism or vasculitis. A malar (“butterfly”) rash may be associated with systemic lupus erythematosus, and impetigo or needle tracks from intravenous (IV) drug use

Table 28.6 Useful Clinical Features, Urinary Findings, and Confirmatory Tests in the Differential Diagnosis of Acute Kidney Injury

Cause of acute kidney injury	Some suggestive clinical features	Typical urinalysis	Some confirmatory tests
Prerenal azotemia	Evidence of true volume depletion (thirst, postural or absolute hypotension and tachycardia, low jugular venous pressure, dry mucous membranes and axillae, weight loss, fluid output > input) or decreased effective circulatory volume (e.g., heart failure and liver failure), treatment with NSAID, diuretic or ACE inhibitor/ARB	Hyaline casts $FE_{Na} < 1\%$, $FE_{urea} < 35\%$ $U_{Na} < 10$ mmol/L $SG > 1.018$	Occasionally requires invasive hemodynamic monitoring; rapid resolution of AKI with restoration of renal perfusion
Diseases involving large renal vessels			
Renal artery thrombosis	History of atrial fibrillation or recent myocardial infarction, nausea, vomiting, flank or abdominal pain	Mild proteinuria Occasionally RBCs	Elevated LDH with normal transaminase levels, renal arteriogram, MAG-3 renal scan, MRA ^a
Atheroembolism	Usually age >50 years, recent manipulation of aorta, retinal plaques, subcutaneous nodules, palpable purpura, livedo reticularis	Often normal Eosinophiluria Rarely casts	Eosinophilia, hypocomplementemia, skin biopsy, renal biopsy
Renal vein thrombosis	Evidence of nephrotic syndrome or pulmonary embolism, flank pain	Proteinuria, hematuria	Inferior venocavogram, Doppler flow studies, MRV ^a
Diseases of small renal vessels and glomeruli			
Glomerulonephritis or vasculitis	Compatible clinical history (e.g., recent infection) sinusitis, lung hemorrhage, rash or skin ulcers, arthralgias, hypertension, and edema	Red blood cell or granular casts, red blood cells, white blood cells, proteinuria	Low complement levels; positive antineutrophil cytoplasmic antibodies, antglomerular basement membrane antibodies, antistreptolysin O antibodies, anti-DNAse, cryoglobulins; renal biopsy
HUS/TTP	Compatible clinical history (e.g., recent gastrointestinal infection, cyclosporine, and anovulants), pallor, ecchymoses, and neurologic findings	May be normal, red blood cells, mild proteinuria, rarely red blood cell or granular casts	Anemia, thrombocytopenia, schistocytes on peripheral blood smear, low haptoglobin, increased LDH, renal biopsy
Malignant hypertension	Severe hypertension with headaches, cardiac failure, retinopathy, neurological dysfunction, papilledema	May be normal, red blood cells, mild proteinuria, rarely red blood cell casts	LVH by echocardiography or electrocardiogram, resolution of AKI with BP control
Ischemic or nephrotoxic acute tubular injury			
Ischemia	Recent hemorrhage, hypotension, surgery often in combination with vasoactive medication (e.g., ACE inhibitor and NSAID)	Muddy-brown granular or tubular epithelial cell casts, $FE_{Na} > 1\%$, $FE_{urea} > 60\%$ $U_{Na} > 20$ mmol/L, $SG \sim 1.010$	Clinical assessment and urinalysis usually inform diagnosis

Continued on following page

Table 28.6 Useful Clinical Features, Urinary Findings, and Confirmatory Tests in the Differential Diagnosis of Acute Kidney Injury (Continued)

Cause of acute kidney injury	Some suggestive clinical features	Typical urinalysis	Some confirmatory tests
Exogenous toxin	Recent contrast-enhanced procedure; nephrotoxic medications; certain chemotherapeutic agents often with coexistent volume depletion, sepsis, or chronic kidney disease	Muddy-brown granular or tubular epithelial cell cases, $FE_{Na} > 1\%$, $U_{Na} > 20$ mmol/L, $SG \sim 1.010$	Clinical assessment and urinalysis usually inform diagnosis
Endogenous toxin	History suggestive of rhabdomyolysis (coma, seizures, drug abuse, trauma) History suggestive of hemolysis (recent blood transfusion) History suggestive of tumor lysis (recent chemotherapy) myeloma (bone pain) or ethylene glycol ingestion	Urine supernatant tests positive for heme in absence of red cells Urine supernatant pink and tests positive for heme in absence of red cells Urate crystals, dipstick negative proteinuria, oxalate crystals, respectively	Hyperkalemia, hyperphosphatemia, hypocalcemia, increased CK, myoglobin Hyperkalemia, hyperphosphatemia, hypocalcemia, hyperuricemia, and free-circulating hemoglobin Hyperuricemia, hyperkalemia, hyperphosphatemia (for tumor lysis); circulating or urinary monoclonal protein (for myeloma); toxicology screen, acidosis, osmolal gap (for ethylene glycol) Systemic eosinophilia, renal biopsy
Diseases of the tubulointerstitium—allergic interstitial nephritis	Recent ingestion of drug and fever, rash, loin pain, or arthralgias	White blood cell casts, white blood cells (frequently eosinophiluria) red blood cells, rarely red blood cell casts, proteinuria (occasionally nephritic)	
Acute bilateral pyelonephritis	Fever, flank pain and tenderness, toxic state	Leukocytes, occasionally white blood cell casts, red blood cells, bacteria	Urine and blood cultures

may underlie infection-associated glomerulonephritis. Ophthalmologic examination is useful to assess for signs of atheroembolism; hypertensive or diabetic retinopathy; the keratitis, uveitis, and iritis of autoimmune vasculitides; icterus associated with chronic liver disease; and the rare but nevertheless pathognomonic band keratopathy of hypercalcemia and flecked retina of hyperoxalemia. Uveitis may also be an indicator of coexistent allergic interstitial nephritis, sarcoidosis, and tubulointerstitial nephritis and uveitis (TINU) syndrome.⁶⁰ Examination of the ears, nose, and throat may reveal conductive deafness and mucosal inflammation or ulceration suggestive of necrotizing granulomatous vasculitis or neural deafness caused by aminoglycoside toxicity. Respiratory failure, particularly if associated with hemoptysis, suggests the presence of a pulmonary-renal syndrome, and ascites and other stigmata of severe chronic liver disease suggest the possibility of hepatorenal syndrome (HRS). Cardiovascular assessment may reveal marked elevation in systemic blood pressure

suggesting malignant hypertension or scleroderma or demonstrate a new arrhythmia or murmur suggesting a potential source of thromboemboli or subacute bacterial endocarditis (acute glomerulonephritis), respectively. Chest or abdominal pain and reduced pulses in the lower limbs should suggest aortic dissection or, rarely, Takayasu arteritis. Abdominal pain and nausea are frequent clinical correlates of atheroembolic disease, commonly in patients who have recently undergone angiographic evaluation, particularly in the presence of widespread atherosclerotic disease. The presence of a tensely distended abdomen may indicate the presence of abdominal compartment syndrome and should prompt transduction of bladder pressure.⁶¹ Pallor and recent bruising are important clues to the thrombotic microangiopathies, and the combination of bleeding and fever should raise the possibility of AKI resulting from viral hemorrhagic fevers. A recent jejunoileal bypass may be a vital clue to oxalosis, a rare but reversible cause of AKI following bariatric surgery.⁶² Hyperreflexia and asterixis

often portend the development of uremic encephalopathy or may, in the presence of focal neurologic signs, suggest a diagnosis of thrombotic microangiopathy (i.e., hemolytic uremic syndrome [HUS] or thrombotic thrombocytopenic purpura [TTP], see Chapter 36).

Postrenal AKI may be asymptomatic if obstruction to the drainage of urine develops gradually. While anuria will be seen in complete obstruction, urine volume may be normal or even increased in the setting of partial obstruction. A pattern of fluctuating urine output may also be seen in some patients with partial obstruction. Suprapubic or flank pain may be the presenting complaint if there is acute distention of the bladder or renal collecting system and capsule, respectively. Colicky flank pain radiating to the groin suggests acute ureteral obstruction, most commonly from urinary stone disease. Prostatic disease should be suspected in older men with a history of nocturia, urinary frequency, urgency or hesitancy, and an enlarged prostate on rectal examination. Urinary retention may be exacerbated acutely in such patients by medications with anticholinergic properties, such as antihistamines or antidepressants. Rectal or pelvic examination may reveal obstructing tumors in female patients. Neurogenic bladder is a likely diagnosis in patients with spinal cord injury or autonomic insufficiency and should be suspected in patients with long-standing diabetes mellitus. Bladder distention may be evident on abdominal percussion and palpation in patients with bladder neck or urethral obstruction. Definitive diagnosis of postrenal AKI usually relies on examination of the postvoid bladder volume and radiographic evaluation of the upper urinary tract and is confirmed by improvement in kidney function following relief of the obstruction.

URINE ASSESSMENT

Urine volume is rarely helpful in distinguishing various forms and causes of AKI. Anuria can be seen with complete urinary tract obstruction but can also be seen with severe prerenal or intrinsic renal disease (e.g., renal artery occlusion, severe proliferative glomerulonephritis or vasculitis, and bilateral cortical necrosis). Patients with partial urinary tract obstruction may present with polyuria caused by secondary impairment of urinary concentrating mechanisms.

Assessment of the urine is essential in patients with AKI and is an inexpensive and useful diagnostic tool.⁶³⁻⁶⁵ Measured urine-specific gravity above 1.015 to 1.020 often accompanies prerenal AKI, although impaired urinary concentration may be present in patients with underlying CKD or as a result of diuretic therapy. Acute glomerulonephritis may also present with concentrated urine. Isosthenuria (a urine-specific gravity of 1.010, similar to that of plasma) is characteristic of ATI. Hematuria on dipstick may result from urologic trauma from catheterization, urologic disease, interstitial nephritis, acute glomerulonephritis, atheroembolic disease, renal infarction, or pigment (hemoglobinuric or myoglobinuric) nephropathy. The latter are suggested when the dipstick test for blood is strongly positive but there are few or no red blood cells seen on microscopic examination of the urinary sediment.

Examination of the urinary sediment of a centrifuged urine specimen complements the dipstick analysis and is

highly valuable for distinguishing among the various forms of AKI (Table 28.7). The sediment should be inspected for the presence of cells, casts, and crystals. In prerenal AKI, the urine sediment is typically bland (i.e., devoid of cells or casts) but may contain transparent hyaline casts. Hyaline casts are formed in concentrated urine from normal urinary constituents, principally Tamm-Horsfall protein secreted by epithelial cells of the loop of Henle. Postrenal AKI may also present with bland urine sediment, although hematuria is common in patients with intraluminal obstruction (e.g., stones, sloughed papilla, and blood clot) or prostatic disease. Renal tubular epithelial cells, epithelial cell casts, and pigmented “muddy-brown” granular casts are characteristic of ischemic or nephrotoxic ATI. These characteristic findings of ATI may be found in association with microscopic hematuria and mild “tubular” proteinuria (<1 g/day). Casts may be absent in approximately 20% to 30% of patients with ischemic or nephrotoxic ATI and are not a requisite for diagnosis^{63,66}; however, semiquantitative scoring systems have been developed to assess the presence of epithelial cells and casts in patients with AKI to assist in the

Table 28.7 Urine Sediment in the Differential Diagnosis of Acute Kidney Injury

Normal or few red blood cells or white blood cells

Prerenal azotemia
Arterial thrombosis or embolism
Preglomerular vasculitis
HUS/TTP
Scleroderma crisis
Postrenal AKI

Renal tubular epithelial cells and granular casts

Acute tubular injury

Dysmorphic red blood cells and red blood cell casts

Glomerulonephritis or vasculitis
Malignant hypertension
Rarely interstitial nephritis

White blood cell and white blood cell casts

Acute interstitial nephritis or exudative glomerulonephritis
Severe pyelonephritis
Marked leukemic or lymphomatous infiltration

Eosinophiluria (>5%)

Allergic interstitial nephritis (antibiotics > NSAIDs)
Atheroembolism

Crystalluria

Acute urate nephropathy
Calcium oxalate (ethylene glycol intoxication)
Acyclovir
Indinavir
Sulfonamides
Methotrexate

HUS, Hemolytic uremic syndrome; *NSAIDs*, nonsteroidal antiinflammatory drugs; *TTP*, thrombotic thrombocytopenic purpura.

diagnosis of ATI and correlate with the clinical course.^{64,67,68} Red blood cell (RBC) casts are almost always indicative of acute glomerular disease but may be observed, albeit rarely, in AIN. Dysmorphic RBCs, best seen using phase contrast microscopy, are a more common urinary finding in patients with glomerular injury but are a less specific finding than RBC casts. Urine sediment abnormalities vary in diseases involving preglomerular blood vessels, such as HUS, TTP, atheroembolic disease, and vasculitis involving medium-sized or large vessels, and range from benign to overtly nephritic. White blood cell casts and nonpigmented granular casts suggest interstitial nephritis, while broad granular casts are characteristic of CKD and probably reflect interstitial fibrosis and dilatation of tubules. Eosinophiluria (between 1% and 50% of urine leukocytes) is a common finding (90%) in drug-induced allergic interstitial nephritis.^{69,70} However, eosinophiluria has poor sensitivity and specificity for the diagnosis of AIN, with eosinophiluria of 1% to >5% occurring in a variety of other diseases including atheroembolization, ischemic and nephrotoxic AKI, proliferative glomerulonephritis, pyelonephritis, cystitis, and prostatitis. In a series of 566 patients who had urinary eosinophil testing and renal histology from kidney biopsy, eosinophiluria only had 31% sensitivity and 68% specificity for the diagnosis of interstitial nephritis.⁷¹ Uric acid crystals (pleomorphic) may be seen in the urine in prerenal AKI but should raise the possibility of acute urate nephropathy if seen in abundance. Oxalate crystalluria (either needle- or dumbbell-shaped monohydrate crystals or envelope-shaped dihydrate crystals) may suggest a diagnosis of ethylene glycol toxicity.⁷² A variety of other crystals may be seen in other medication-associated crystal nephropathies.⁷³⁻⁷⁵

Increased urinary protein excretion, characteristically <1 g/day, is a common finding in ischemic or nephrotoxic ATI and reflects both failure of injured proximal tubule cells to reabsorb normally filtered protein and excretion of cellular debris (tubular proteinuria). Proteinuria >1 g/day suggests injury to the glomerular ultrafiltration barrier (glomerular proteinuria) or excretion of light chains.^{76,77} The latter are not detected by conventional dipsticks (which detect albumin) and must be sought by other means (e.g., sulfosalicylic acid test). Heavy proteinuria is also a frequent finding (80%) in patients with allergic interstitial nephritis

triggered by NSAIDs. In addition to acute interstitial inflammation, these patients have a glomerular lesion that is almost identical to minimal-change disease.⁷⁸ A similar syndrome has been reported in patients receiving other agents such as ampicillin, rifampin, and interferon alfa.^{79,80} Hemolysis and rhabdomyolysis may often be differentiated by inspection of plasma, which is characteristically pink in hemolysis but clear in rhabdomyolysis.

Analysis of urine biochemical parameters may be helpful in differentiating between prerenal and intrinsic ischemic or nephrotoxic ATI (Table 28.8). Sodium is usually avidly reabsorbed from the glomerular filtrate in patients with prerenal AKI as a consequence of renal adrenergic activation, stimulation of the renin-angiotensin-aldosterone axis, suppression of atrial natriuretic peptide (ANP) secretion, and local changes in peritubular hemodynamics. In contrast, Na⁺ reabsorption is impaired in ATI as a result of injury to the renal tubular epithelium. Renal sodium handling can be assessed on the basis of urinary sodium concentration (U_{Na}) with values of <10 mmol/L commonly seen in prerenal disease compared with >20 mmol/L in ATI. Normalizing sodium excretion to creatinine provides a more sensitive index. The fractional excretion of sodium (FE_{Na}) is the ratio between urine sodium excretion (U_{Na} × V, where U_{Na} is the urine sodium concentration and V is the urine volume) and the filtered load of sodium (calculated as P_{Na} × CrCl, where P_{Na} is the plasma sodium concentration and CrCl is the creatinine clearance, which can be calculated as [(U_{Cr} × V)/P_{Cr}], where V is the urine volume and [U_{Cr}] and [P_{Cr}] are the urine and plasma creatinine concentrations, respectively). Since urine volume is in both the numerator and denominator of this ratio, the FE_{Na} can be calculated as [(U_{Na} ÷ P_{Na}) / (U_{Cr} ÷ P_{Cr})] × 100 using an untimed (spot) urine sample and simultaneous serum sodium and creatinine measurements and is usually <1% (frequently <0.5%), in the setting of prerenal azotemia, whereas it is typically >2% in patients with ischemic or nephrotoxic AKI. The utility of the FE_{Na} is limited in a variety of clinical settings. Values >1% are not uncommon in the setting of prerenal AKI in patients receiving diuretics, those with metabolic alkalosis and bicarbonaturia (in whom Na⁺ is excreted with HCO₃⁻ to maintain electroneutrality), in the presence of adrenal insufficiency, and in the setting of underlying CKD.⁸¹⁻⁸³

Table 28.8 Urine Indices used in the Differential Diagnosis of Prerenal Acute Kidney Injury (AKI) and Acute Tubular Injury (ATI)

Diagnostic index	Prerenal AKI	ATI
Fractional excretion of urea (%)	<35%	>60%
Fractional excretion of sodium (%)	<1 ^a	>2 ^a
Urine sodium concentration (mmol/L)	<20	>40
Urine creatinine/plasma creatinine ratio	>40	<20
Urine urea nitrogen/plasma urea nitrogen ratio	>8	<3
Urine-specific gravity	>1.018	~1.010
Urine osmolality (mOsm/kg H ₂ O)	>500	~300
Plasma BUN/creatinine ratio	>20	<10-15
Renal failure index, U _{Na} /(U _{Cr} /P _{Cr})	<1	>1
Urine sediment	Hyaline casts	Muddy-brown granular casts

^aFE_{Na} may be >1% in prerenal AKI associated with diuretic use and/or the setting of bicarbonaturia or chronic kidney disease; FE_{Na} is often

Conversely, a $FE_{Na} < 1\%$ may be observed in the setting of ATI, particularly in the settings of iodinated contrast administration, rhabdomyolysis, and sepsis, although it has been reported in 15% or more of patients with ATI from a variety of other causes including ischemia, burns, and exposure to selected nephrotoxins.^{81,82,84-86} It has been postulated that low FE_{Na} reflects a milder degree of intrinsic renal injury in which epithelial cell damage is probably localized to the corticomedullary junction and outer medulla with relative preservation of function in other Na^+ -transporting segments and may represent a transition state between prerenal azotemia and ATI. It should be recognized that a FE_{Na} of $< 1\%$ is not abnormal and reflects normal sodium homeostasis in patients on a moderate- to low-sodium diet. The FE_{Na} is also often $< 1\%$ in AKI caused by urinary tract obstruction, glomerulonephritis, and diseases of the renal vasculature; other parameters must be employed to distinguish these conditions from prerenal AKI.

A variety of other indices have also been proposed to differentiate between etiologies of AKI. The renal failure index, calculated as $U_{Na}/(U_{Cr} \div P_{Cr})$, provides comparable information to the FE_{Na} because clinical variations in serum Na^+ concentration are relatively small. The fractional excretion of urea (FE_{urea}) has been proposed as an alternative to the FE_{Na} , with particular utility in patients on diuretic therapy. Values of FE_{urea} calculated as $([U_{urea} \div P_{urea}]/[U_{Cr} \div P_{Cr}] \times 100) < 35\%$ are suggestive of a prerenal state.⁸⁷⁻⁸⁹ Similarly, indices of urinary concentrating ability such as urine-specific gravity, urine osmolality, urine/plasma creatinine or urea ratios, and serum urea nitrogen/creatinine ratio are of limited value in differentiating between prerenal and intrinsic AKI. This is particularly true for elderly patients, in whom urine-concentrating mechanisms are frequently impaired while mechanisms for Na^+ reabsorption are typically preserved.

LABORATORY EVALUATION

The pattern and timing of change of BUN and serum creatinine often provide clues to the cause of AKI. Enhanced tubular reabsorption of filtered urea in parallel with sodium and water reabsorption in prerenal states commonly leads to a disproportionate elevation in BUN relative to serum creatinine (ratio $> 20:1$). Conversely, with intrinsic AKI, the increase in BUN usually parallels the rise in serum creatinine, maintaining a ratio of $\sim 10:1$. However, severe malnutrition and low dietary protein intake blunt the rise in BUN and creatinine, while gastrointestinal bleeding, steroid therapy, and hypercatabolic states may lead to increases in the BUN that do not reflect prerenal physiology. In addition, aggressive volume resuscitation may rapidly expand the volume of distribution of urea and creatinine and may also obscure the acute rise in serum creatinine. Sepsis and other forms of critical illness have also been associated with decreased creatinine generation.^{90,91} The serum creatinine typically begins to rise within 24 to 48 hours when ATI results from an ischemic insult. Although the clinical course can be highly variable, the serum creatinine will generally peak within 7 to 10 days and, depending on the severity of the insult and underlying comorbid illnesses, AKI will resolve to varying degrees over the ensuing 1 to 2 weeks. Following

iodinated contrast exposure, the peak in serum creatinine generally occurs within 5 to 7 days. The time course of nephrotoxic ATI caused by aminoglycoside antibiotics or cisplatin is more variable, often with delayed onset of AKI (7–10 days).

Additional clues to the diagnosis can be obtained from biochemical and hematologic tests. The presence of marked hyperkalemia, hyperuricemia, and hyperphosphatemia point to cell lysis, which, in the setting of elevated creatine kinase levels and hypocalcemia, strongly suggests rhabdomyolysis.^{92,93} Biochemical signs of cell lysis with high levels of uric acid, normal or mildly elevated creatine kinase levels, and a urine uric acid-to-creatinine ratio > 1.0 are suggestive of acute urate nephropathy and tumor lysis syndrome.^{94,95} Severe hypercalcemia can precipitate AKI, commonly in the form of prerenal AKI from concomitant hypovolemia and renal vasoconstriction. AKI is associated with widening of both the serum anion ($Na^+ - [HCO_3^- + Cl^-]$) and osmolal (measured serum osmolality minus calculated osmolality) gaps. This suggests a diagnosis of ethylene glycol toxicity and should prompt a search for urine oxalate crystals. Severe anemia in the absence of hemorrhage may reflect the presence of hemolysis, multiple myeloma, or thrombotic microangiopathy (e.g., HUS, TTP, toxemia, disseminated intravascular coagulation, accelerated hypertension, SLE, scleroderma, and radiation injury). Other laboratory findings suggestive of thrombotic microangiopathy include thrombocytopenia, dysmorphic RBCs on peripheral blood smear, low circulating haptoglobin concentrations, and elevated circulating concentrations of lactate dehydrogenase. Systemic eosinophilia suggests allergic interstitial nephritis but may also be a prominent feature in other diseases such as atheroembolic disease and eosinophilic granulomatosis with polyangiitis (Churg-Strauss disease). Depressed serum complement and high titers of antiglomerular basement membrane antibodies, antineutrophil cytoplasmic antibodies, antinuclear antibodies, circulating immune complexes, or cryoglobulins are useful diagnostic tools in patients with suspected glomerulonephritis or systemic vasculitis.

NOVEL BIOMARKERS OF KIDNEY INJURY

A number of novel biomarkers of kidney injury have been evaluated for potential roles in the early identification, differential diagnosis, and prognosis of AKI.¹⁹⁻²⁶ While most of these biomarkers are not available for routine clinical use, they have the potential to provide an earlier diagnosis of intrinsic AKI, to differentiate volume-responsive (prerenal) AKI from intrinsic disease, and to provide prognostic information regarding the clinical course of an episode of AKI. Their role in clinical practice, however, remains the subject of debate.^{96,97}

CYSTATIN C

Cystatin C is a 13 kDa protein that is filtered by the glomerulus and completely reabsorbed and degraded by the proximal tubule. Cystatin C has been validated as an alternative marker of glomerular filtration.^{98,99} As the result of its shorter serum half-life, serum cystatin C concentrations change more rapidly than serum creatinine in response to changes in kidney function, allowing changes in serum

cystatin C to be detected sooner than changes in serum creatinine following the onset of AKI.¹⁰⁰ Under normal circumstances, urinary cystatin C is virtually undetectable; however, after tubular injury, tubular reabsorption of filtered cystatin is diminished and urinary cystatin C can be detected, raising the possibility of its use as an early marker of tubular injury.¹⁰¹

NEUTROPHIL GELATINASE-ASSOCIATED LIPOCALIN

Neutrophil gelatinase-associated lipocalin (NGAL) is a 25 kDa protein whose expression by renal tubular epithelial cells is markedly upregulated after ischemic or nephrotoxic kidney injury.^{102,103} NGAL is believed to enhance the trafficking of iron-siderophore complexes, enhancing the delivery of iron, upregulating hemeoxygenase-1, reducing apoptosis, and increasing the normal proliferation of renal tubule epithelial cells.¹⁰⁴ Urine and plasma NGAL have been evaluated in numerous clinical settings as an early biomarker of tubular injury.¹⁰⁵⁻¹¹⁵ Initial studies in children undergoing cardiac surgery demonstrated extremely high sensitivity and specificity for identification of AKI, with an area under the receiver operating characteristic (ROC) curve of >0.99; however, these early results have not been reproduced across other clinical settings.

KIDNEY INJURY MOLECULE 1

Kidney injury molecule 1 (KIM-1) is a transmembrane protein whose expression is markedly upregulated in the proximal tubule after tubular injury.¹¹⁶⁻¹¹⁹ The extracellular component of the KIM-1 protein is shed into the urine post tubular injury, permitting its potential use as a marker of tubular damage; however, the time course of peak KIM-1 expression in the urine is later than seen with NGAL.^{112,120-122} Moreover, studies have demonstrated elevated levels of KIM-1 in non-AKI settings, including CKD and renal cell carcinoma, which reduce its specificity for AKI.^{123,124}

INTERLEUKIN 18

Interleukin 18 (IL-18) is a proinflammatory cytokine whose expression is increased in the kidney after ischemic and nephrotoxic injury.¹²⁵ Urinary IL-18 levels have been shown to rise within 6 hours after tubular injury and cardiac surgery, as well as in critically ill patients.^{114,115,126,127}

LIVER FATTY ACID-BINDING PROTEIN

Despite its name, liver fatty acid-binding protein (L-FABP) is expressed in the proximal tubule.^{128,129} Elevated L-FABP levels may be detected in the urine within 6 hours of ischemic or nephrotoxic injury, permitting its potential use as a marker of tubular injury.^{122,130-132} In a meta-analysis of published studies, the sensitivity and specificity of urinary L-FABP for diagnosis of AKI were each approximately 75%.¹³³

TISSUE INHIBITOR OF METALLOPROTEINASE 2 (TIMP-2) AND INSULIN-LIKE GROWTH FACTOR-BINDING PROTEIN 7 (IGFBP7)

Tissue inhibitor of metalloproteinase 2 (TIMP-2) and insulin-like growth factor-binding protein 7 (IGFBP7) are expressed in epithelial cells and act in an autocrine and paracrine manner to arrest cell cycle in AKI.¹³⁴⁻¹³⁶ In

three discovery cohorts comprising 522 patients, these two biomarkers were identified as having the highest discriminant ability among 340 candidate biomarkers of AKI.²⁴ In a subsequent validation study of 728 patients, this pair of biomarkers had an area under the ROC curve of 0.80, which was significantly better than the performance of other candidate biomarkers including NGAL, KIM-1, IL-18, and L-FABP.²⁴ A combination test that includes TIMP-2 and IGFBP7 is available for commercial use.

RADIOLOGIC EVALUATION

Imaging of the abdomen is a highly useful adjunct to laboratory testing to determine the etiology of AKI. In cases of suspected obstructive uropathy, postvoid residual volumes of >100 to 150 mL suggest a diagnosis of bladder outlet obstruction. While plain films rarely provide definitive evidence of postrenal AKI, they may identify the presence of calcium-containing stones that can cause obstructive disease. Renal ultrasonography is the screening test of choice to assess cortical thickness, differences in cortical and medullary density, the integrity of the collecting system, and kidney size.¹³⁷ Although pelvicalyceal dilatation is usual in cases of urinary tract obstruction (98% sensitivity), dilatation may not be observed in the volume-depleted patient, during the initial 1 to 3 days after obstruction, when the collecting system is relatively noncompliant, or in patients with obstruction caused by ureteric encasement or infiltration (e.g., retroperitoneal fibrosis and neoplasia).¹³⁸ Alternatively, computed tomography may be used to visualize the kidneys and collecting system, although contrast administration should ideally be avoided in patients with AKI. Visualization of the collecting system may be suboptimal in the absence of contrast enhancement; however, unenhanced computed tomographic scans are useful for the identification of obstructing ureteral stones.^{139,140} Ultrasonography and computed tomography have essentially replaced the use of IV pyelography, which now has little or no role in the evaluation of AKI. Cytoscopic retrograde or percutaneous antegrade pyelography are useful tests for the precise localization of the site of obstruction and can be combined with placement of ureteral stents or percutaneous nephrostomy tubes to allow therapeutic decompression of the urinary tract. Radionuclide scans have been proposed as useful for assessing renal blood flow, glomerular filtration, tubule function, and infiltration by inflammatory cells in AKI; however, these tests lack specificity and yield conflicting or poor results in controlled studies.^{141,142} Magnetic resonance angiography (MRA) of the kidneys is extremely useful for detecting renal artery stenosis and has been used in the evaluation of acute renovascular crises.¹⁴³ However, given the association of gadolinium-based contrast administration with the development of nephrogenic systemic fibrosis, contrast-enhanced MRA is contraindicated in the majority of patients with AKI.^{144,145} Doppler ultrasonography and spiral CT are also useful in patients with suspected vascular obstruction; however, contrast angiography remains the gold standard for definitive diagnosis. For more discussion on the use of point-of-care ultrasound (POCUS) at the bedside to evaluate volume status, hemodynamics, and structural concerns in AKI, see [Chapter 26](#).

KIDNEY BIOPSY

Kidney biopsy in AKI is usually reserved for patients in whom prerenal and postrenal AKI have been excluded and the cause of intrinsic AKI is unclear.^{146,147} Kidney biopsy is particularly useful when clinical assessment, urinalysis, and laboratory investigation suggest diagnoses other than ischemic or nephrotoxic injury that may respond to specific therapy. Examples include antiglomerular basement membrane disease and other forms of necrotizing glomerulonephritis, vasculitis, HUS and TTP, allergic interstitial nephritis, and myeloma cast nephropathy.

ETIOLOGY OF ACUTE KIDNEY INJURY IN SPECIFIC CLINICAL SETTINGS

The differential diagnosis of AKI in several common clinical settings warrants special mention (Table 28.9).

ACUTE KIDNEY INJURY IN THE SETTING OF CANCER

There are several potential etiologies of AKI in patients with cancer. Prerenal AKI is common in the setting of underlying malignancy and may be related to tumor- or chemotherapy-induced vomiting or diarrhea, reduced oral intake secondary to anorexia, the use of NSAIDs for pain management, and malignancy-associated hypercalcemia.^{148,149} Intrinsic AKI can be triggered by a variety of chemotherapeutic agents. Cisplatin is the classical chemotherapeutic medication associated with AKI.^{150,151} The principal site of renal damage with cisplatin is the proximal tubule. The nephrotoxicity of cisplatin is dose dependent, yet AKI can result from a single exposure. Electrolyte disturbances including hypomagnesemia and hypokalemia are common following cisplatin administration. Other platinum-containing chemotherapy agents, such as carboplatin and oxaliplatin, are less nephrotoxic than cisplatin but are not risk free, particularly when high cumulative doses are administered. Ifosfamide, which has been used to treat germ cell tumors, sarcomas, other solid tumors, and occasionally lymphoma, is also associated with AKI in a dose-dependent fashion.¹⁵²⁻¹⁵⁴ Methotrexate nephrotoxicity occurs following IV administration of high doses (>1 g/m²), primarily as the result of precipitation of the drug and metabolites within the tubular lumen.^{73,155,156} Risk factors for methotrexate nephrotoxicity include volume depletion and the presence of acidic urine. Direct tubular toxicity may also contribute to the development of AKI. Chemotherapy agents targeting vascular endothelial growth factor (VEGF) or the VEGF receptor, such as bevacizumab and the tyrosine kinase inhibitor sunitinib, are associated with hypertension, proteinuria, thrombotic microangiopathy, and AKI.^{157,158} Checkpoint inhibitors, such as nivolumab, may cause a variety of immune-related adverse events, including AIN.^{159,160}

Renal parenchymal invasion by solid and hematologic cancers is reported in 5% to 10% of autopsy studies but is an uncommon cause of AKI.^{161,162} Infiltration of leukemic cells into the renal parenchyma can precipitate AKI and typically

presents with hematuria, proteinuria, and enlarged kidneys on ultrasound imaging. Prompt diagnosis is important as the AKI may respond to chemotherapeutic intervention.

The tumor lysis syndrome, which is associated with hyperuricemia, hyperphosphatemia, and hypocalcemia, is a well-recognized etiology of AKI in patients with cancers.^{163,164} Tumor lysis syndrome occurs most commonly after the initiation of chemotherapy for patients with poorly differentiated, rapidly growing lymphoproliferative malignancies (e.g., Burkitt lymphoma, acute lymphoblastic, and promyelocytic leukemia). Yet it can occur spontaneously and in the setting of certain solid tumors that are highly sensitive to radiation and/or chemotherapy (e.g., testicular carcinoma). The Cairo-Bishop criteria, which include both laboratory and clinical criteria, have been used to provide a standard definition for the diagnosis of tumor lysis syndrome.¹⁶⁴ (Table 28.10). AKI associated with the tumor lysis syndrome is thought to be triggered by direct tubular injury and luminal obstruction by uric acid and calcium phosphate crystals. Prophylactic therapy with aggressive volume administration and either xanthine oxidase inhibitors to inhibit uric acid synthesis or recombinant uricase to convert uric acid to allantoin has markedly reduced the incidence of this form of AKI.¹⁶⁵⁻¹⁶⁸ Less common causes of AKI include tumor-associated glomerulonephritis and thrombotic microangiopathy induced by medications or irradiation. Chemotherapy-associated thrombotic microangiopathy is a well-recognized complication of several agents including mitomycin C and gemcitabine.¹⁶⁹⁻¹⁷¹

AKI is a common complication of multiple myeloma.^{77,172} Causes of AKI in this setting include intravascular volume depletion, myeloma cast nephropathy, sepsis, hypercalcemia, ATI induced by drugs or tumor lysis during therapy, cryoglobulinemia, hyperviscosity syndrome, and plasma cell infiltration. Multiple myeloma may also result in impaired kidney function as the result of amyloidosis or light chain deposition disease; however, these most commonly present with proteinuria and a more subacute decline in kidney function. Myeloma cast nephropathy results from the binding of filtered immunoglobulin Bence Jones proteins to Tamm-Horsfall glycoprotein forming casts that obstruct the tubular lumen. Higher excretion rates of free light chains, volume depletion, and hypercalcemia are associated with higher risks for development of myeloma cast nephropathy. Prompt treatment to lower free light chain burden may result in recovery of kidney function. Studies of the effectiveness of plasmapheresis in the treatment of myeloma cast nephropathy have yielded conflicting results.¹⁷³⁻¹⁷⁶ The use of dialysis membranes that are permeable to light chains and other proteins with molecular weights lower than albumin (high cutoff membranes) has also been proposed as a potential therapeutic strategy; however, data from clinical trials evaluating the efficacy of this strategy are similarly conflicting.¹⁷⁷⁻¹⁷⁹ For more on kidney disease and cancer see Chapter 57.

ACUTE KIDNEY INJURY IN PREGNANCY

In the industrialized world, the incidence of dialysis-requiring AKI in the setting of pregnancy is approximately 1 in 20,000 births.^{180,181} The marked decline in this

Table 28.9 Major Causes of Acute Kidney Injury (AKI) in Specific Clinical Settings**AKI in the cancer patient**

- Prerenal azotemia
 - Hypovolemia (e.g., poor intake, vomiting, and diarrhea)
- Intrinsic AKI
 - Exogenous nephrotoxins: chemotherapy, antibiotics, contrast agents
 - Endogenous toxins: hyperuricemia, hypercalcemia, tumor lysis, paraproteins
 - Other: radiation, HUS/TTP, glomerulonephritis, amyloid, malignant infiltration
- Postrenal AKI
 - Ureteric or bladder neck obstruction

AKI after cardiac surgery

- Prerenal azotemia
 - Hypovolemia (surgical losses, diuretics), cardiac failure, vasodilators
- Intrinsic AKI
 - Ischemic ATI (even in absence of hypotension)
 - Atheroembolic disease after aortic manipulation/intraaortic balloon pump
 - Preoperative or perioperative administration of contrast agent
 - Allergic interstitial nephritis induced by perioperative antibiotics
- Postrenal AKI
 - Obstructed urinary catheter, exacerbation of voiding dysfunction

AKI in pregnancy

- Prerenal azotemia
 - Acute fatty liver of pregnancy with fulminate hepatic failure
- Intrinsic AKI
 - Preeclampsia or eclampsia
 - Postpartum HUS/TTP
 - HELLP syndrome
 - Ischemia: postpartum hemorrhage, abruptio placentae, amniotic fluid embolus
 - Direct toxicity of illegal abortifacients
- Postrenal AKI
 - Obstruction with pyelonephritis

AKI after solid organ or bone marrow transplantation (BMT)

- Prerenal azotemia
 - Intravascular volume depletion (e.g., diuretic therapy)
 - Vasoactive drugs (e.g., calcineurin inhibitors and amphotericin B)
 - Hepatorenal syndrome, venoocclusive disease of liver (BMT)
- Intrinsic AKI
 - Postoperative ischemic ATI (even in absence of hypotension)
 - Sepsis
 - Exogenous nephrotoxins: aminoglycosides, amphotericin B, radiocontrast
 - HUS/TTP (e.g., cyclosporine or myeloablative radiotherapy related)
 - Allergic tubulointerstitial nephritis
- Postrenal AKI
 - Obstructed urinary catheter

AKI and pulmonary disease (pulmonary-renal syndrome)

- Prerenal azotemia
 - Diminished cardiac output complicating pulmonary embolism, severe pulmonary hypertension, or positive pressure mechanical ventilation
- Intrinsic AKI
 - Vasculitis:
 - Goodpasture syndrome, ANCA-associated vasculitis, SLE, Churg-Strauss syndrome, polyarteritis nodosa, cryoglobulinemia, right-sided endocarditis, lymphomatoid granulomatosis, sarcoidosis, scleroderma
 - Toxins:
 - Ingestion of paraquat or diquat
 - Infections:
 - Legionnaire disease, *Mycoplasma* infection, tuberculosis, disseminated viral or fungal infection
- AKI from any cause with hypervolemia and pulmonary edema
- Lung cancer with hypercalcemia, tumor lysis, or glomerulonephritis

Continued on following page

Table 28.9 Major Causes of Acute Kidney Injury (AKI) in Specific Clinical Settings (Continued)**AKI and liver disease**

Prerenal azotemia
 Reduced true (gastrointestinal hemorrhage, GI losses from lactulose, diuretics, large-volume paracentesis) circulatory volume or effective (hypoalbuminemia, splanchnic vasodilatation)
 Hepatorenal syndrome type 1 or 2
 Tense ascites with abdominal compartment syndrome
 Intrinsic AKI
 Ischemic (severe hypoperfusion—see earlier) or direct nephrotoxicity and hepatotoxicity of drugs or toxins (e.g., carbon tetrachloride, acetaminophen, tetracyclines, and methoxyflurane)
 Tubulointerstitial nephritis + hepatitis caused by drugs (e.g., sulfonamides, rifampin, phenytoin, allopurinol, and phenindione), infections (leptospirosis, brucellosis, Epstein-Barr virus infection and cytomegalovirus infection), malignant infiltration (leukemia and lymphoma) or sarcoidosis
 Glomerulonephritis or vasculitis (e.g., polyarteritis nodosa, ANCA-associated GN, cryoglobulinemia, SLE, postinfectious hepatitis, or liver abscess)

AKI and nephrotic syndrome

Prerenal azotemia
 Intravascular volume depletion (diuretic therapy, hypoalbuminemia)
 Intrinsic AKI
 Manifestation of primary glomerular disease
 Collapsing glomerulopathy (e.g., HIV and pamidronate)
 Associated ATI (elderly hypertensive males)
 Associated interstitial nephritis (NSAIDs, rifampin, interferon alfa)
 Other—amyloid or light chain deposition disease, renal vein thrombosis, severe interstitial edema

ANCA, anti-neutrophil cytoplasmic antibody; ATI, acute tubular injury; GN, glomerulonephritis; HELLP, Hemolysis, Elevated Liver enzymes, Low Platelets; HIV, human immunodeficiency virus; HUS, hemolytic uremic syndrome; NSAID, nonsteroidal antiinflammatory drug; SLE, systemic lupus erythematosus.

Table 28.10 Cairo-Bishop Definition of Tumor Lysis Syndrome**Diagnosis of laboratory tumor lysis syndrome**

Requires at least 2 of the following criteria achieved in the same 24-h interval from 3 days before to 7 days after chemotherapy initiation:

- Uric acid: ≥ 8.0 mg/dL or $\geq 25\%$ increase from baseline
- Potassium: ≥ 6.0 mmol/L or $\geq 25\%$ increase from baseline
- Phosphorus: ≥ 4.6 mg/dL (≥ 6.5 mg/dL in children) or $\geq 25\%$ increase from baseline
- Calcium: ≤ 7.0 mg/dL or $\geq 25\%$ decrease from baseline

Diagnosis of clinical tumor lysis syndrome

Laboratory tumor lysis syndrome plus at least 1 of the following:

- Serum creatinine ≥ 1.5 times the age-adjusted upper limit of normal
- Cardiac arrhythmia/sudden death
- Seizure

Adapted from Cairo MS, Bishop M. Tumour lysis syndrome: new therapeutic strategies and classification. *Br J Haematol*. 2004;127:3–11.

complication over the past 50 years is a result of improved prenatal care and advancements in obstetrics practice. In early pregnancy, ATI induced by nephrotoxic abortifacients remains a relatively common cause of AKI

in developing countries but is rare in the developed world. Ischemic ATI, severe toxemia of pregnancy, and postpartum HUS and TTP are the most common causes of AKI in late-term pregnancy.^{180,182,183} Ischemic ATI is usually precipitated by placental abruption or postpartum hemorrhage and less commonly by amniotic fluid embolism or sepsis. Glomerular filtration is usually normal in mild or moderate preeclampsia; however, AKI may complicate severe preeclampsia.^{183,184} In this setting, AKI is typically transient and found in association with intrarenal vasospasm, marked hypertension, and neurologic abnormalities. A variant of preeclampsia, the HELLP syndrome (Hemolysis, Elevated Liver enzymes, Low Platelets), is characterized by a benign initial course that can rapidly deteriorate with the development of thrombotic microangiopathy with hemolysis, coagulation abnormalities, derangement in hepatic function, and AKI.^{184–186} Immediate delivery of the fetus is indicated in this setting. Thrombotic microangiopathy can also develop in the postpartum setting and typically occurs in patients who have had a normal pregnancy.¹⁸⁷ Postpartum thrombotic microangiopathy is characterized by thrombocytopenia, microangiopathic anemia, and normal prothrombin and partial thromboplastin times and frequently results in long-term impairment of renal function.

Acute fatty liver of pregnancy (AFLP) occurs in approximately 1 in 7000 pregnancies and is associated with AKI, likely as a result of intrarenal vasoconstriction, as occurs in the HRS. Although the exact origin of AFLP

is unknown, the incidence is increased in women who carry a fetus with a defect in fatty-acid oxidation and who are themselves carriers of a genetic mutation that compromises intramitochondrial fatty-acid oxidation.¹⁸⁶ Acute bilateral pyelonephritis may also precipitate AKI in pregnancy and should be obvious from the patient's presentation (fever, flank pain); findings on urinalysis (bacteria, leukocytes); and laboratory tests (leukocytosis, elevated serum creatinine).^{182,185,188,189} The diagnosis of postrenal AKI in the pregnant patient is particularly challenging due to the physiologic dilatation of the collecting system that normally occurs in the second and third trimesters. As a result, determining the presence of abnormal findings on renal ultrasound is more difficult. For further discussion on pregnancy-associated AKI (PrAKI) see Chapter 58.

ACUTE KIDNEY INJURY IN THE SETTING OF CARDIAC SURGERY

An acute deterioration in kidney function is a relatively common complication following cardiac surgery with an incidence of 7.7% to 42% depending on the criteria used to define AKI.¹⁹⁰⁻¹⁹⁴ AKI requiring dialytic support occurs in up to 5% of patients post cardiac surgery.¹⁹⁰⁻¹⁹⁴ AKI in the perioperative period is most commonly attributed to prerenal azotemia associated with decreased cardiac function or to ATI. Risk factors for cardiac surgery-associated AKI can be broadly categorized into presurgical patient-related factors; surgical factors; and postoperative events. The principal patient-related risk factors include underlying CKD, advanced age, left ventricular dysfunction, previous myocardial revascularization, diabetes mellitus, and peripheral vascular disease.¹⁹³⁻¹⁹⁶ Operative factors include the need for emergent surgery, prolonged time on cardiopulmonary bypass, insertion of intraaortic balloon pump, the performance of concomitant valvular surgery, and redo coronary artery bypass grafting (CABG). Several studies have compared the incidence of AKI following on-pump versus off-pump CABG, with some data suggesting that off-pump CABG is associated with a lower incidence of AKI.¹⁹⁷⁻²⁰¹ Postoperative factors associated with an increased risk for AKI include reduced cardiac output, bleeding, vasodilatory shock, and overzealous use of diuretics and afterload reducing agents.

Additional potential etiologies of AKI after CABG include the administration of iodinated contrast media in the preoperative, perioperative, and/or postoperative periods; antibiotic-associated AIN; and atheroembolic disease.²⁰² Whereas prerenal azotemia and ATI typically occur within days of the surgical procedure, atheroembolic AKI may take longer to develop and can be distinguished by the characteristic clinical features of livedo reticularis, cyanosis, and gangrenous digital lesions, as well as the findings of eosinophilia, eosinophiluria, and hypocomplementemia.

ACUTE KIDNEY INJURY AFTER SOLID ORGAN OR BONE MARROW TRANSPLANT

Nonrenal solid organ transplant recipients have a particularly high risk of AKI from cardiopulmonary and hepatic failure, sepsis, and the nephrotoxic effects of antimicrobial

and immunosuppressive agents. In a large retrospective multicenter study, 25% of all nonrenal solid organ transplant recipients developed AKI, with 8% requiring KRT.²⁰³ The development of AKI requiring dialysis was associated with a 9- to 12-fold increase in mortality. AKI developed in 35% of heart transplant recipients and 15% of lung transplant recipients. Up to 30% of liver transplant recipients develop AKI, many of whom had CKD before transplantation.^{204,205} There are conflicting data as to whether impaired kidney function pretransplant predicts outcomes in patients undergoing orthotopic liver transplantation; however, patients with impaired kidney function preoperatively have longer hospital and ICU stays and are more likely to need dialysis compared with patients with intact preoperative kidney function.²⁰⁶⁻²⁰⁸

AKI is a well-recognized complication of hematopoietic cell transplantation.^{148,209,210} The three types of hematopoietic cell transplantation are myeloablative autologous, myeloablative allogeneic, and nonmyeloablative allogeneic, and the incidence, severity, and outcomes of AKI following these forms of hematopoietic cell transplantation vary considerably.^{209,211,212} In a study of 272 patients who underwent myeloablative hematopoietic cell transplantation (predominantly allogeneic), 53% developed AKI and 24% required dialysis.²¹³ Of patients with dialysis-requiring AKI, the mortality rate was 84%. A recent study found an incidence of severe AKI in this patient population of 73%.²¹⁴ AKI following nonmyeloablative allogeneic hematopoietic cell transplantation is less common.^{214,215} A study of 253 patients demonstrated an incidence of AKI of 40% within 3 months of hematopoietic cell transplantation with just 4.4% of patients requiring dialysis.²¹⁵ The incidence of AKI following myeloablative autologous hematopoietic cell transplantation is considerably lower.^{216,217} A study of 173 patients following autologous hematopoietic cell transplantation reported an incidence of AKI of 21%, with 5% of patients requiring dialysis.²¹⁷ The absence of graft-versus-host disease and more rapid engraftment likely account for the lower incidence of AKI in this setting. Causes of hematopoietic cell transplantation-associated AKI include hypovolemia, sepsis, tumor lysis syndrome, direct tubular toxicity from cytoreductive therapy, thrombotic microangiopathy, graft-versus-host disease, antibiotics, immunosuppressive agents, and hepatic venoocclusive disease (VOD). VOD results from acute radiochemotherapy-induced endothelial cell injury of hepatic venules.^{213,218-220} This condition occurs most commonly in conditioning regimens that include total body irradiation and cyclophosphamide and/or busulfan and in the setting of myeloablative allogeneic hematopoietic cell transplantation. The syndrome is characterized clinically by profound jaundice and avid salt retention with edema and ascites within the first month after engraftment and subsequently the development of AKI. Oliguric AKI is common in moderate VOD and certain in severe cases. The mortality rate for patients with severe VOD approaches 100%. BK virus is a human polyoma virus that is a common opportunistic infection in solid organ transplant recipients and in patients following hematopoietic cell transplantation.²²¹ Detectable BK viruria may be seen in up to 50% of patients undergoing hematopoietic cell transplantation.²²² Reactivation of

latent BK virus infection in immunosuppressed patients is associated with both hemorrhagic cystitis and renal involvement with tubular atrophy and fibrosis with an inflammatory lymphocytic infiltrate with intranuclear BK virus inclusion bodies.²²³ The diagnosis is suggested by rising viral titers in blood and/or urine, and the mainstay of treatment is minimization of immunosuppression.

ACUTE KIDNEY INJURY ASSOCIATED WITH PULMONARY DISEASE

The coexistence of AKI and pulmonary disease (pulmonary-renal syndrome) classically suggests the presence of Goodpasture syndrome, antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis, or other vasculitides.²²⁴⁻²²⁶ The detection of antiglomerular basement membrane antibodies, antineutrophil cytoplasmic antibodies, or low serum complement concentrations can be helpful in differentiating among the various causes of pulmonary-renal syndrome, although the urgent need for definitive diagnosis and treatment may mandate lung or renal biopsy. Several toxic ingestions and infections may also precipitate simultaneous pulmonary and kidney injury that mimics vasculitis-associated pulmonary-renal syndrome. Furthermore, AKI of any cause may be complicated by secondary hypervolemia and pulmonary edema. Severe lung disease and ventilator support with increased intrathoracic pressure may compromise cardiac output and induce prerenal AKI.

ACUTE KIDNEY INJURY IN ASSOCIATION WITH LIVER DISEASE

The differential diagnosis of AKI in patients with liver disease is broad. Common causes of AKI in this setting include intravascular volume depletion, gastrointestinal bleeding, sepsis, and nephrotoxins. Most cases of AKI in advanced liver disease are due to prerenal azotemia, ATI, or the HRS, and differentiating these conditions can be clinically challenging.²²⁷⁻²²⁹ Although a urine sodium concentration <20 mmol/L and fractional excretion rate of sodium $<1\%$ are typical of prerenal AKI and HRS, high-dose diuretics that are commonly prescribed in patients with advanced liver disease may lead to higher sodium excretion rates. Differentiating ATI from other forms of AKI is further confounded by the fact that bile-stained casts, which can be seen in prerenal AKI and HRS, have a similar appearance to the classical “muddy-brown” granular casts of ATI.²³⁰ Kidney disease in patients with liver disease may also result from acute glomerular disease including IgA nephropathy, hepatitis B virus–associated membranous nephropathy, and hepatitis C virus–associated membranoproliferative glomerulonephritis with cryoglobulinemia. Acetaminophen toxicity may cause nephrotoxic ATI in addition to being one of the most common etiologies of acute hepatotoxicity.

The term “hepatorenal syndrome” (HRS) is typically reserved for a clinical syndrome marked by irreversible AKI that develops in patients with advanced cirrhosis, although it has also been described in the setting of fulminant viral and alcoholic hepatitis. HRS almost certainly represents the terminal stage of a state of hypoperfusion that begins early

during chronic liver disease. The precise pathophysiologic mechanisms underlying the hemodynamic alterations in HRS are incompletely understood. In the early stages of HRS, increased vascular capacitance as the result of splanchnic and systemic vasodilatation is thought to trigger activation of the renin-angiotensin and sympathetic nervous systems.²²⁷ Renal perfusion is preserved in this stage by the local release of renal vasodilatory factors; however, these compensatory mechanisms are eventually overwhelmed and progressive renal hypoperfusion ensues. An inadequate increase in cardiac output relative to the fall in vascular resistance is thought to also contribute to the development of HRS.

Clinically, the presentation of HRS closely resembles that of prerenal AKI. However, unlike prerenal AKI, HRS does not improve with aggressive expansion of the intravascular space. Criteria for the diagnosis of HRS have undergone revision (Table 28.11).²³¹ Previous criteria were based on an increase in serum creatinine to >1.5 mg/dL in the setting of cirrhosis with ascites,²³² whereas the updated criteria have been harmonized with the creatinine component of the KDIGO AKI definition.²³¹ Other criteria include failure of kidney function to improve after at least 2 days with diuretic withdrawal and volume expansion with albumin, the absence of shock or concurrent or recent treatment with nephrotoxic drugs, and the absence of parenchymal kidney disease (defined by proteinuria >500 mg/day, hematuria [>50 RBC/hpf] and/or abnormal renal ultrasonography).²³¹

Two subtypes of HRS have been described. Under the prior criteria, type 1 HRS was characterized by a rapid onset of AKI defined by at least a doubling of the serum creatinine concentration to a level of at least 2.5 mg/dL or a reduction in glomerular filtration of $\geq 50\%$ to a level <20 mL/min over a 2-week period.^{233,234} In the revised criteria, the diagnosis of type 1 HRS is based on meeting KDIGO criteria for stage 2 or higher AKI (i.e., doubling of serum creatinine from baseline).²³¹ Type 1 HRS typically develops in hospitalized patients and may be precipitated by variceal bleeding, overly rapid diuresis, the performance of paracentesis, or most commonly, the development of spontaneous bacterial peritonitis. Other postulated triggers include infections, minor surgery, or the use of NSAIDs or other drugs. However, caution must be exerted in these cases to exclude reversible causes of AKI. Type 1 HRS is generally characterized by a fulminant course with oliguria, encephalopathy, marked hyperbilirubinemia, and death within 1 month of clinical presentation. However, recent advances in the management of HRS discussed later suggest that there may be a trend toward better survival in those patients who respond to therapy.^{235,236} Type 2 HRS is typified by a more gradual decline in renal function that develops in the setting of diuretic-resistant ascites and avid sodium retention. The prognosis of type 2 HRS is considerably better than that of type 1 HRS, with a reported median survival of 6 months and a 1-year survival as high as 30%.^{237,238} The development of sudden deterioration in kidney function after a prolonged stable period may occur in patients with type 2 HRS, leading to outcomes similar to that of patients with type 1 HRS. To better align the HRS nomenclature with the nosology for AKI, AKD, and CKD, it has been proposed that type 1 HRS now be called HRS-AKI and that type 2 HRS

Table 28.11 Diagnostic Criteria for Hepatorenal Syndrome

- Cirrhosis with ascites
- AKI defined as
 - ≥ 0.3 mg/dL increase in serum creatinine over <48 h
 - $\geq 50\%$ increase in serum creatinine known or presumed to have occurred within prior 7 days
- No improvement of serum creatinine (decrease to a level of ≤ 1.5 mg/dL) after at least 2 days of diuretic withdrawal and volume expansion with albumin (1 g/kg body weight per day to a maximum of 100 g/day)
- Absence of shock
- Absence of parenchymal kidney disease as indicated by:
 - Proteinuria >500 mg/day
 - Microhematuria (>50 red blood cells per high-power field); and/or
 - Abnormal renal ultrasonography

Type 1 hepatorenal syndrome

Rapid progressive AKI defined based on increase in serum creatinine by $>2\times$ baseline

Type 2 hepatorenal syndrome

Moderate renal dysfunction with a steady or slowly progressive course

AKI, Acute kidney injury.

Adapted from Angeli P, Gines P, Wong F, et al. Diagnosis and management of acute kidney injury in patients with cirrhosis: revised consensus recommendations of the International Club of Ascites. *Gut*. 2015;64:531–537.

be subdivided into subacute (HRS-AKD) and chronic (HRS-CKD) based on duration less than or more than 3 months (Table 28.12).²³⁹

Definitive treatment of HRS is dependent on recovery of hepatic function or successful liver transplantation.

However, the use of vasoconstrictive agents combined with volume expansion with colloid has shown promise for improving kidney function.^{240–243} It is postulated that by reversing the splanchnic and peripheral vasodilatation, more normal renal perfusion can be restored. Vasoconstrictive regimens that have been used include norepinephrine, combination therapy with midodrine and octreotide, and the vasopressin-agonist terlipressin.^{240–248} Although vasoconstrictive therapy is associated with improvement in kidney function and patients who respond have an improved prognosis, the use of vasoconstrictive therapy has not been shown to improve overall prognosis in patients with AKI, suggesting that survival remains limited by the underlying severity of liver disease.

ACUTE KIDNEY INJURY AND NEPHROTIC SYNDROME

AKI in the context of nephrotic syndrome presents a unique array of potential diagnoses. Epithelial injury, if severe, can trigger both nephrotic range proteinuria and acute or subacute kidney injury.^{249,250} The epithelial injury typically occurs as a manifestation of primary glomerular diseases, such as collapsing glomerulopathy or crescentic membranous nephropathy. Less dramatic visceral epithelial cell injury, in combination with proximal tubular injury (e.g., pan-epithelial cell injury induced by NSAIDs or possible undiagnosed viral illness) or interstitial nephritis (e.g., rifampicin or ampicillin induced) can also present as AKI complicating the nephrotic syndrome.^{251–253} Massive excretion of light chain proteins in patients with multiple myeloma may also present in this fashion.^{254,255} ATI in association with nephrotic syndrome is seen in a subpopulation of older patients with minimal-change disease and in other patients with nephrosis and severe hypoalbuminemia, particularly with overzealous diuresis. In general, patients with AKI complicating nephrotic syndrome have higher blood pressure and urinary protein excretion

Table 28.12 Updated Classification of Hepatorenal Syndrome (HRS)

Old terminology	New classification	Criteria
Type 1 HRS	HRS-AKI	• Absolute increase in serum creatinine of ≥ 0.3 mg/dL within 48 h and/or • Urine output ≤ 0.5 mL/kg body weight per hour for ≥ 6 h
Type 2 HRS	HRS-AKD	• Relative increase in serum creatinine $\geq 50\%$ using the last available outpatient serum creatinine within 3 months as baseline • eGFR <60 mL/min/1.73 m² for <3 months in the absence of other causes
	HRS-CKD	• Relative increase in serum creatinine $<50\%$ using the last available outpatient serum creatinine within 3 months as baseline • eGFR <60 mL/min/1.73 m² for ≥ 3 months in the absence of other causes

AKD, acute kidney disease; AKI, acute kidney injury; CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate.

Adapted from Angeli P, Garcia-Tsao G, Nadim MK, Parikh CR. News in pathophysiology, definition and classification of hepatorenal syndrome: a step beyond the International Club of Ascites (ICA) consensus document. *J Hepatol*. 2019;71:811–822.

than patients without AKI.²⁴⁹ The higher incidence of arteriosclerosis in biopsy samples from these patients may point to preexisting hypertensive nephrosclerosis as a risk factor for the development of this complication. Renal vein thrombosis must always be considered in the differential diagnosis of the nephrotic syndrome and AKI, particularly in the pediatric population and in adults with membranous nephropathy in association with high-grade proteinuria and hypoalbuminemia. For further discussion on nephrotic syndrome, see [Chapter 30](#).

COMPLICATIONS OF ACUTE KIDNEY INJURY

The acute loss of kidney function in AKI results in multiple derangements in fluid, electrolyte and acid-base homeostasis, and in hematologic, gastroenterologic, and immunologic function ([Table 28.13](#)).

POTASSIUM HOMEOSTASIS IN ACUTE KIDNEY INJURY

Hyperkalemia is a common and potentially life-threatening complication of AKI.^{256,257} Serum K⁺ typically rises by 0.5 mmol/L/day in oligoanuric patients and reflects impaired excretion of K⁺ derived from a patient's diet, the administration of K⁺-containing solutions and drugs administered as potassium salts, as well as the release of K⁺ from the injured tubular epithelium. Hyperkalemia may be compounded by coexistent metabolic acidosis and/or hyperglycemia or other hyperosmolar states that promote K⁺ efflux from cells. Hyperkalemia present at the time of diagnosis of AKI or the rapid development of severe hyperkalemia suggests massive tissue destruction, as might be seen with rhabdomyolysis, hemolysis, or tumor lysis.^{52,92,258} Hyperuricemia and hyperphosphatemia may accompany hyperkalemia in these settings. Mild hyperkalemia (<6.0 mmol/L) is usually asymptomatic. Higher levels are frequently associated with electrocardiographic abnormalities including peaked T-waves, prolongation of the PR-interval, flattening of P-waves, widening of the QRS

complex, and intraventricular conduction defects.²⁵⁹⁻²⁶¹ These electrocardiographic findings may precede the onset of life-threatening cardiac arrhythmias such as bradycardia, heart block, ventricular tachycardia, ventricular fibrillation, and asystole. In addition, hyperkalemia may induce neuromuscular abnormalities such as paresthesias, hyporeflexia, weakness, ascending flaccid paralysis, and respiratory failure.

Hypokalemia is unusual in AKI but may complicate nonoliguric ATI caused by aminoglycosides, cisplatin, or amphotericin B, presumably because of impaired K⁺ reabsorption resulting from epithelial cell injury in the thick ascending limb of the loop of Henle.^{262,263} For more discussion on potassium homeostasis, see [Chapter 16](#).

ACID-BASE HOMEOSTASIS IN ACUTE KIDNEY INJURY

Normal metabolism of dietary protein yields between 50 and 100 mmol/day of fixed nonvolatile acids (principally sulfuric and phosphoric acid) that are excreted by the kidneys to maintain acid-base homeostasis. Predictably, AKI is commonly complicated by metabolic acidosis, typically with a widening of the serum anion gap due to retention of phosphates, sulfates, and organic anions.²⁶⁴ Acidosis may be severe (daily fall in plasma HCO₃⁻ >2 mmol/L) when the generation of H⁺ is increased by additional mechanisms (e.g., diabetic or fasting ketoacidosis; lactic acidosis complicating generalized tissue hypoperfusion, liver disease, or sepsis; metabolism of ethylene glycol).^{72,229,265} In contrast, metabolic alkalosis is an infrequent finding but may complicate overly aggressive correction of acidosis with HCO₃⁻, overzealous use of combination loop and thiazide diuretics, or loss of gastric acid by vomiting or nasogastric aspiration. For further discussion on acid-base homeostasis, see [Chapter 15](#).

MINERAL AND URIC ACID HOMEOSTASIS IN ACUTE KIDNEY INJURY

Mild to moderate hyperphosphatemia (5–10 mg/dL) is a common consequence of AKI, and hyperphosphatemia

Table 28.13 Complications of Acute Kidney Injury

Metabolic	Cardiovascular	Gastrointestinal	Neurologic	Hematologic	Infectious	Other
Hyperkalemia	Pulmonary edema	Nausea	Neuromuscular irritability	Anemia	Pneumonia	Hiccups
Metabolic acidosis	Arrhythmias	Vomiting	Asterixis	Bleeding	Septicemia	Elevated parathyroid hormone
Hyponatremia	Pericarditis	Malnutrition	Seizures		Urinary tract infection	Low total triiodothyronine and thyroxine
Hypocalcemia	Pericardial effusion	Hemorrhage	Mental status changes			Normal thyroxine
Hyperphosphatemia	Pulmonary embolism					
Hypermagnesemia	Hypertension					
Hyperuricemia	Myocardial infarction					

may be severe (10 to 20 mg/dL) in highly catabolic patients or when AKI is associated with rapid cell death as in rhabdomyolysis, severe burns, hemolysis, or tumor lysis.²⁶⁶⁻²⁶⁹ Factors that potentially contribute to hypocalcemia include skeletal resistance to the actions of parathyroid hormone, reduced levels of 1, 25-dihydroxyvitamin D, Ca^{2+} sequestration in injured tissues such as muscle in the setting of rhabdomyolysis, and metastatic deposition of calcium phosphate salts in the setting of severe hyperphosphatemia.²⁷⁰⁻²⁷² Hypocalcemia is usually asymptomatic, possibly because of the counterbalancing effects of acidosis on neuromuscular excitability. However, symptomatic hypocalcemia can occur in patients with rhabdomyolysis or acute pancreatitis or after treatment of acidosis with HCO_3^- .²⁷⁰ Clinical manifestations of hypocalcemia include perioral paresthesias, muscle cramps, seizures, hallucinations, and confusion, as well as prolongation of the QT-interval and nonspecific T-wave changes on electrocardiogram. The Chvostek (contraction of facial muscles on tapping of the jaw over the facial nerve) and Trousseau (carpopedal spasm after occlusion of arterial blood supply to the arm for 3 minutes with a blood pressure cuff) signs are useful indicators of latent tetany in high-risk patients.

Mild asymptomatic hypermagnesemia is common in oliguric AKI and reflects impaired excretion of ingested magnesium (dietary magnesium, magnesium-containing laxatives, or antacids).^{273,274} More significant hypermagnesemia is usually the result of overzealous parenteral magnesium administration, as in the management of AKI associated with preeclampsia. Hypomagnesemia occasionally complicates nonoliguric ATI associated with cisplatin or amphotericin B and, as with hypokalemia, likely reflects injury to the thick ascending limb of loop of Henle, a principal site for Mg^{2+} reabsorption.^{263,275,276} Hypomagnesemia is usually asymptomatic but may occasionally manifest as neuromuscular instability, cramps, seizures, cardiac arrhythmias, or resistant hypokalemia or hypocalcemia.^{273,277}

Uric acid is cleared from blood by glomerular filtration and secretion by proximal tubule cells, and asymptomatic hyperuricemia (10–15 mg/dL) is not atypical in established AKI. Higher levels suggest increased production of uric acid and may point to a diagnosis of acute urate nephropathy.²⁷⁸⁻²⁸⁰ The urinary uric acid-to-creatinine ratio on a random specimen has been proposed as a means to distinguish between hyperuricemia caused by overproduction and impaired excretion. In a small series of patients, this ratio was >1 in 5 patients with acute uric acid nephropathy and was <1 in 27 patients with acute kidney injury due to other etiologies.²⁸¹ In a subsequent case series, elevations in the uric acid-to-creatinine ratio to values of >1 were described in other etiologies of AKI, most notably in patients with infections who were markedly hypercatabolic.²⁸²

VOLUME OVERLOAD AND CARDIAC COMPLICATIONS IN AKI

Extracellular volume overload is an almost inevitable consequence of diminished salt and water excretion in AKI and may present clinically as mild hypertension, increased jugular venous pressure, pulmonary vascular

congestion, pleural effusion, ascites, peripheral edema, increased body weight, and life-threatening pulmonary edema. Hypervolemia may be particularly troublesome in patients receiving multiple IV medications, high volumes of enteral or parenteral nutrition, and/or excessive volumes of maintenance IV fluids. Moderate or severe hypertension is unusual in ATI and should suggest other diagnoses such as hypertensive nephrosclerosis, glomerulonephritis, preeclampsia, renal artery stenosis, and other diseases of the renal vasculature.^{182,283-285} Excessive water ingestion or administration of hypotonic saline or dextrose solutions can trigger hyponatremia, which, if severe, may cause cerebral edema, seizures, and other neurologic abnormalities.²⁸⁶ Cardiac complications include arrhythmias and myocardial infarction. Although these events may reflect primary cardiac disease, abnormalities in myocardial contractility and excitability may be triggered or compounded by hypervolemia, acidosis, hyperkalemia, and other metabolic sequelae of AKI.²⁸⁷ For further discussion on cardiorenal syndromes, see Chapter 42.

HEMATOLOGIC COMPLICATIONS IN ACUTE KIDNEY INJURY

Anemia develops rapidly in AKI and is usually multifactorial in origin. Contributing factors include inhibition of erythropoiesis, hemolysis, bleeding, hemodilution, and reduced RBC survival time.²⁸⁸⁻²⁹⁰ Prolongation of the bleeding time is also common, resulting from mild thrombocytopenia, platelet dysfunction, and clotting factor abnormalities (e.g., factor VIII dysfunction).

NUTRITIONAL AND GASTROINTESTINAL COMPLICATIONS OF ACUTE KIDNEY INJURY

Malnutrition remains one of the most frustrating and troublesome complications of AKI. The majority of patients have net protein breakdown, which may exceed 200 g/day in catabolic patients.²⁹¹⁻²⁹³ Malnutrition is usually multifactorial in origin and may reflect inability to eat, loss of appetite, and/or inadequate nutritional support; the catabolic nature of the underlying medical disorder (e.g., sepsis, rhabdomyolysis, and trauma); nutrient losses in drainage fluids or dialysate; and increased breakdown and reduced synthesis of muscle protein and increased hepatic gluconeogenesis, probably through the actions of toxins, hormones (e.g., glucagon and parathyroid hormone), or other substances (e.g., proteases) that accumulate in AKI.²⁹⁴⁻²⁹⁸ Nutrition may also be compromised by the high incidence of acute gastrointestinal hemorrhage, which complicates up to 15% of cases of AKI. Mild gastrointestinal bleeding is common (10% to 30%) and usually due to stress ulceration of gastric or small intestinal mucosa.^{299,300}

INFECTIOUS COMPLICATIONS OF ACUTE KIDNEY INJURY

Infection is the most common and serious complication of AKI, occurring in 50% to 90% of cases and accounting for up to 75% of deaths.^{27,256,301-303} It is unclear whether this high incidence of infection is due to a defect in host

immune responses or repeated breaches of mucocutaneous barriers (e.g., intravenous cannulas, mechanical ventilation and bladder catheterization) resulting from therapeutic interventions.

OTHER SEQUELAE OF ACUTE KIDNEY INJURY

Protracted periods of severe AKI or short intervals of catabolic anuric AKI often lead to development of uremic syndrome. Clinical manifestations of uremic syndrome, in addition to those already listed, include pericarditis; pericardial effusion; cardiac tamponade; gastrointestinal complications such as anorexia, nausea, vomiting, and ileus; and neuropsychiatric disturbances including lethargy, confusion, stupor, coma, agitation, psychosis, asterixis, myoclonus, hyperreflexia, restless leg syndrome, focal neurologic deficit, and/or seizures. The uremic toxin(s) responsible for this syndrome has yet to be defined. Candidate molecules include urea, other products of nitrogen metabolism such as guanidine compounds, products of bacterial metabolism such as aromatic amines and indoles, and other compounds that are inappropriately retained in the circulation in AKI or are underproduced, such as nitric oxide (NO).³⁰⁴

COMPLICATIONS DURING RECOVERY FROM ACUTE KIDNEY INJURY

A vigorous diuresis may complicate the recovery phase of AKI, precipitate intravascular volume depletion, and result in delayed recovery of kidney function. This diuretic response probably reflects the combined effects of an osmotic diuresis induced by retained urea and other byproducts of protein metabolism, excretion of retained salt and water accumulated during AKI, and delayed recovery of tubular resorptive function relative to glomerular filtration leading to salt-wasting.³⁰⁵⁻³⁰⁸ Hyponatremia may also complicate this recovery phase if free water losses are not replenished or are inappropriately replaced by relatively hypertonic electrolyte solutions. Hypokalemia, hypomagnesemia, hypophosphatemia, and hypocalcemia are rarer metabolic complications during recovery from AKI. Mild transient hypercalcemia is relatively frequent during recovery and appears to be a consequence of delayed resolution of secondary hyperparathyroidism. In addition, hypercalcemia may complicate recovery from rhabdomyolysis because of mobilization of sequestered Ca^{2+} from injured muscle.³⁰⁹

MANAGEMENT OF ACUTE KIDNEY INJURY

The treatment of AKI varies considerably on the basis of its etiology and clinical presentation. There is no effective pharmacologic therapy to counteract the pathophysiologic processes and ameliorate the renal parenchymal damage in many forms of AKI, most notably ATI. As a result, the management of AKI generally focuses on implementing, when possible, interventions to prevent its development, providing supportive care to ameliorate derangements of fluid and electrolyte homeostasis, and instituting treatment to prevent and

mitigate uremic complications (Table 28.14). In cases of severe AKI, KRT is often required. The ultimate goals of management are to prevent death, facilitate recovery of kidney function, and minimize the risk for de novo and/or progressive CKD.

MANAGEMENT OF PRERENAL ACUTE KIDNEY INJURY

INTRAVASCULAR VOLUME DEPLETION

Prerenal AKI is defined as hemodynamically mediated kidney dysfunction that is rapidly reversible following

Table 28.14 Supportive Management Of Acute Kidney Injury

Management issue	Treatment
Intravascular volume overload	Restriction of salt (<1-2 g/day) and water (<1 L/day) intake Diuretic therapy (if nonoliguric) Ultrafiltration
Hyponatremia	Restriction of oral and intravenous free water
Hyperkalemia	Calcium gluconate (10 mL of 10% solution over 5 min) if ECG changes are present Glucose (50 mL of 50%) + insulin (10-15 U regular) IV Albuterol (10-20 mg by nebulizer or MDI) Kidney replacement therapy Loop diuretics (if nonoliguric) K ⁺ binding resin Discontinue K ⁺ supplements or K ⁺ -sparing diuretics
Metabolic acidosis	Restriction of dietary potassium Restriction of dietary protein Sodium bicarbonate (if HCO_3^- <15 mmol/L)
Hyperphosphatemia	Kidney replacement therapy Restriction of dietary phosphate intake Phosphate binding agents (aluminum hydroxide, calcium carbonate, calcium acetate, sevelamer, lanthanum)
Hypocalcemia	Oral or intravenous replacement (if symptomatic or sodium bicarbonate to be administered)
Hypermagnesemia	Discontinue magnesium-containing antacids
Nutrition	Caloric intake: 20-30 kcal/day Protein intake: Nondialysis requiring: 0.8-1.0 g/kg/day Dialysis requiring: 1.0-1.5 g/kg/day Continuous kidney replacement therapy: up to 1.7 g/kg/day Enteral route of nutrition preferred
Drug dosage	Adjust all doses for GFR and kidney replacement modality

ECG, Electrocardiogram; GFR, glomerular filtration rate; MDI, metered-dose inhaler.

normalization of renal perfusion.³¹⁰ Prevention of prerenal AKI from intravascular volume depletion pertains to early recognition and treatment of conditions that involve the loss of extracellular fluid including vomiting, diarrhea, excessive diuresis, and bleeding before hypoperfusion of the kidneys occurs. In patients in whom intravascular volume depletion leads to prerenal AKI, treatment consists of restoration of normal circulating blood volume. The optimal composition of administered fluids in patients with hypovolemic prerenal AKI depends on the source of fluid loss and associated electrolyte and acid-base disturbances. The initial management commonly consists of intravascular volume resuscitation with an isotonic crystalloid solution. Recent data demonstrate that balanced crystalloids reduce major adverse kidney events compared with isotonic saline in hospitalized patients.^{311,312} Red blood cell transfusion should be used for hemorrhagic hypovolemia when there is ongoing bleeding, particularly if the patient is hemodynamically unstable or the blood hemoglobin concentration is dangerously low.

The relative merits of colloid and crystalloid resuscitation fluids in the management of nonhemorrhagic renal, extrarenal, and third-space fluid losses are controversial with advocates for the use of colloids positing that they are more effective at restoring circulating blood volume due to greater retention in the intravascular compartment. However, randomized controlled trials and meta-analyses comparing colloid with crystalloid replacement for resuscitation in critically ill patients have not confirmed this theoretical benefit and have demonstrated an increased need for KRT and other adverse outcomes associated with colloid formulations containing hydroxyethyl starch.³¹³⁻³²⁰ In a meta-analysis of 55 trials involving 3504 patients randomly assigned to treatment with albumin or crystalloid, there was no evidence of either improved outcomes, decreased mortality or other complications associated with albumin administration.³²¹ These results were subsequently confirmed in a nearly 7000-patient multicenter, randomized, controlled trial of fluid resuscitation in hypovolemic medical and surgical ICU patients in which 28-day survival, development of single or multiple organ failure, and duration of hospitalization were similar in both groups.³²² Although specific data on the development of AKI were not described, the need for KRT was similar with saline compared with albumin resuscitation. However, in a post hoc analysis of patients with traumatic brain injury, albumin resuscitation was associated with increased mortality risk.³²³ The use of synthetic colloid solutions has been proposed as an alternative to albumin administration; however, hydroxyethyl starch preparations have been associated with an increased risk of AKI. In a multicenter, randomized, controlled trial comparing fluid resuscitation with hydroxyethyl starch with a 3% gelatin solution in 129 patients with sepsis, hydroxyethyl starch was associated with a more than twofold increased risk of AKI.³¹⁶ A subsequent meta-analysis confirmed the increased risk of AKI associated with hydroxyethyl starch across 34 studies that included 2604 individuals.³¹⁷ In an ensuing randomized controlled trial that included 7000 critically ill patients who were assigned to receive either 6% hydroxyethyl starch or isotonic saline, there was an approximately 20% increased risk of AKI

treated with KRT with use of hydroxyethyl starch.³²⁰ On the basis of these data demonstrating no benefit and potential increased risk of AKI, along with the higher costs associated with colloid administration, their routine use for volume resuscitation in hypovolemia and sepsis is not advisable. In particular, hydroxyethyl starch solutions should be used sparingly. If utilized, there should be regular monitoring of kidney function. In such instances, the risk of hyperoncotic renal failure should be minimized by the concomitant use of appropriate crystalloid solutions.^{3,316,317,319,320}

Experimental data have suggested that volume resuscitation with isotonic sodium chloride solutions, which contain supraphysiologic concentrations of chloride, may exacerbate renal vasoconstriction and diminish GFR as compared with isotonic crystalloid solutions with lower chloride content.³²⁴⁻³²⁶ In healthy patients, magnetic resonance imaging demonstrated that infusion of isotonic saline was associated with reduced renal blood flow velocity and renal cortical tissue perfusion as compared with administration of a reduced-chloride isotonic crystalloid solution.³²⁷ In a subsequent open-label, sequential period study conducted in a single ICU, replacing use of high-chloride IV solutions with fluids containing lower chloride content was associated with a reduction in the incidence of KDIGO stage 3 AKI from 14% to 8.4% and in the use of KRT from 10% to 6.3%.³²⁸ Subsequent prospective randomized clinical trials have yielded divergent results.^{311,312,329,330} Two of these were large pragmatic, cluster-randomized clinical trials conducted in parallel at the same institution. They compared the administration of isotonic saline with balanced crystalloid solution (i.e., fluid with electrolyte composition that more closely resembles plasma).^{311,312} In the Saline Against Lactated Ringer's of Plasma-Lyte in the Emergency Department (SALT-ED) trial of >13,000 non-critically ill patients, balanced crystalloid was associated with a decrease in the incidence of 30-day major adverse kidney events (i.e., death, new KRT, persistent kidney impairment defined by a >200% increase in serum creatinine at time of hospital discharge, within 30 days) compared with saline (4.7% vs. 5.6%, $P = 0.01$), but not with the primary outcome of hospital-free days (days alive after discharge to day 28).³¹¹ The Isotonic Solutions and Major Adverse Renal Events Trial (SMART) that included 15,802 critically ill patients found that compared with isotonic saline, balanced crystalloids were associated with a decrease in 30-day major adverse kidney events (14.3% vs. 15.4%, $P = 0.04$).³¹² In the SALT-ED trial, the benefit in the composite outcome was predominantly due to a lower rate of persistent kidney impairment while in SMART, 30-day mortality predominated. Collectively, these findings support a benefit of balanced crystalloid compared with isotonic sodium chloride, yet the benefits were not homogenous. The greatest benefit was present among the critically ill patients with sepsis and among non-critically ill patients who had a baseline serum creatinine >1.5 mg/dL or hyperchloremia (Cl >110 mmol/L) on initial presentation. Although the balanced fluids were more physiologic regarding their chloride content, they were hypotonic and associated with higher rates of hyponatremia.

The Balanced Solutions in Intensive Care Study (BaSICS) was a multicenter trial in Brazil that randomized 11,052 critically ill patients in a 2 × 2 factorial design to receive either isotonic saline or a balanced electrolyte solution

infused rapidly or at a slower rate.^{329,331} In this study, neither electrolyte composition nor rate of fluid administration was associated with mortality at 90 days or with the development of AKI. Similarly, the Plasma-Lyte 148 versus Saline (PLUS) study randomized 5037 critically ill patients in Australia and New Zealand to receive Plasma-Lyte, a balanced electrolyte solution, or isotonic saline as intravenous fluids in the ICU.³³⁰ As in BaSICS, the use of balanced electrolyte solutions was not associated with either a reduction in mortality, AKI, or need for acute KRT as compared with isotonic saline. A subsequent systematic review and meta-analysis that included all four of these studies and included data from 34,450 patients found risk ratios for 90-day mortality of 0.96 (95% confidence interval [CI]: 0.91–1.01), AKI of 0.96 (95% CI: 0.89–1.02), and need for acute KRT of 0.95 (95% CI: 0.81–1.11).³³² Using Bayesian techniques, the authors estimated that the effect of using balanced crystalloid solutions in place of isotonic saline ranged from a 9% relative reduction to a 1% relative increase in the risk of death at 90 days and from an 11% reduction to a 2% relative increase in the risk of developing AKI.³³²

The volume and electrolyte content of urinary and gastrointestinal losses, as well as patients' serum electrolytes and acid-base status, should be closely monitored to guide adjustments in the composition of the replacement fluids. Although the potassium content in gastric juices tends to be low, concomitant urinary potassium losses may be quite high as the result of metabolic alkalosis.

HEART FAILURE

Management of AKI in the setting of heart failure depends on the clinical setting and etiology of the heart failure. In patients with heart failure in whom AKI has developed in the setting of excessive diuresis, withholding of diuretics and administering cautious volume replacement may be sufficient to restore kidney function. In acute decompensated heart failure (ADHF), AKI may develop despite worsening volume overload; intensification of diuretic therapy is often required for treatment of pulmonary vascular congestion. Although diuretic therapy may exacerbate prerenal AKI, it can also result in improvement in kidney function via several postulated mechanisms: 1) decreasing ventricular distention resulting in a shift from the descending limb to the ascending limb of the Starling curve and improvement in myocardial contractility; 2) decreasing venous congestion^{325,333-336}; and 3) diminishing intraabdominal pressure.³³⁷ Additional therapies for ADHF in the setting of AKI include inotropic support, vasodilators for afterload reduction, and mechanical support including intraaortic balloon pumps and ventricular assist devices. The use of invasive hemodynamic monitoring in ADHF has been controversial; although it is often used to guide pharmacologic management, clinical data have not demonstrated improved renal outcomes when management is guided by pulmonary artery catheters.³³⁸ The role of isolated ultrafiltration in ADHF is also controversial. Although negative fluid balance can be achieved more readily using extracorporeal ultrafiltration than with conventional diuretic therapy, studies have not demonstrated differences in kidney function or survival.³³⁹⁻³⁴¹ In the Ultrafiltration versus Intravenous Diuretics for Patients Hospitalized for Acute Decompensated Heart Failure (UNLOAD) trial, hypervolemic patients with

heart failure who were randomized to isolated ultrafiltration had more rapid fluid loss and decreased rehospitalizations within 90 days as compared with patients randomized to diuretic therapy, with no differences in kidney function.³⁴⁰ In contrast, in the subsequent Cardiorenal Rescue Study in Acute Decompensated Heart Failure (CARRESS-HF) trial, ultrafiltration was inferior to diuretic therapy with respect to the bivariate endpoint of change in serum creatinine level and body weight 96 hours after enrollment ($P = 0.003$), owing primarily to worsening of kidney function in the ultrafiltration group.³⁴¹ On the basis of these data, extracorporeal ultrafiltration cannot be recommended for primary management of patients with decompensated heart failure.

LIVER FAILURE AND HEPATORENAL SYNDROME

Although volume-responsive prerenal azotemia is common in patients with advanced liver disease, differentiation from HRS and intrinsic AKI may be difficult.²²⁷ While patients with liver failure are typically total body sodium overloaded with peripheral edema and ascites, true hypovolemia or reduced effective systemic arterial blood volume is often an important contributory factor to the development of AKI. The underlying pathophysiology of salt and water retention in cirrhosis involves multiple pathways. Portal hypertension leads directly to ascites formation, while splanchnic and peripheral vasodilatation results in a state of relative arterial underfilling, which activates neurohumoral vasoconstrictors that produce intrarenal vasoconstriction, salt and water retention, and decreased GFR.³⁴² Volume-responsive AKI may develop in the setting of excessive diuresis, increased gastrointestinal losses (often as the result of therapy for hepatic encephalopathy), rapid drainage of ascites, or spontaneous bacterial peritonitis. Worsening hepatic function is often associated with diuretic resistance and progressive or precipitous worsening of kidney function. It has been postulated that an inadequate increase in cardiac output in response to the fall in peripheral vascular resistance may be central to development of HRS.³⁴³

Differentiation between volume-responsive prerenal AKI and the HRS is based on the clinical response to volume loading. The optimal fluid for volume expansion in this setting has been controversial. Most recent expert opinion advocated the use of hyperoncotic (20% or 25%) albumin at a dose of 1 g/kg per day.^{232,344} However, there is an absence of rigorous data supporting this regimen as compared with volume expansion with isotonic crystalloid solutions. There are more data regarding the use of albumin infusion to prevent AKI in patients undergoing large-volume (>5 L) paracentesis^{238,345,346} and in the treatment of spontaneous bacterial peritonitis.³⁴⁷ In a randomized controlled trial, patients undergoing paracentesis who received an infusion of approximately 10 g of albumin per liter of drained ascites experienced less activation of the renin-angiotensin system and a significantly lower rate of worsening kidney function than patients who did not receive albumin infusion.³⁴⁵ In a subsequent study, albumin infusion was superior to administration of either dextran or gelatin solutions in preventing AKI after large-volume paracentesis.³⁴⁶ Current recommendations are to infuse 6 to 8 g of albumin per liter of ascites drained when paracentesis volume exceeds 5 L. In a randomized controlled trial comparing antibiotics

alone with antibiotics plus albumin for the treatment of spontaneous bacterial peritonitis, infusion of 1.5 g/kg of albumin at the initiation of treatment and an additional 1 g/kg on the third day of treatment was associated with reduced rates of both AKI and mortality,³⁴⁷ although the benefit appears to be restricted to patients in whom the serum creatinine is >1 mg/dL, the BUN is >30 mg/dL, or the total bilirubin is >4 mg/dL.³⁴⁸

Definitive therapy of HRS requires restoration of hepatic function, usually achieved through liver transplantation.^{229,344} The role of peritoneovenous shunting (e.g., LeVeen and Denver shunts) in HRS has been inadequately studied. In a subset of 33 patients with HRS included in a randomized trial comparing peritoneovenous shunts with medical therapy, shunting was not associated with improved survival.³⁴⁹ These data need to be interpreted with caution due to the small sample size and because data on improvement in kidney function were not reported. In addition, because of poor long-term patency rates and high rates of complications, particularly encephalopathy, the use of peritoneovenous shunts has largely been supplanted by transjugular portosystemic shunts (TIPS). TIPS has been demonstrated to provide better control of ascites than sequential paracentesis,³⁵⁰⁻³⁵³ and in one series lower rates of HRS,³⁵¹ albeit with a higher risk of encephalopathy.³⁵⁴ In a small case series, TIPS was reported to be effective as primary therapy for HRS³⁵⁵ but has not been evaluated in a randomized trial.³⁴⁴ Pharmacologic therapy with vasoconstrictors, when combined with albumin infusion, has been associated with improvement in kidney function in patients with HRS.^{344,356} Agents that have shown benefit include norepinephrine,³⁵⁷ the combination of octreotide and midodrine,^{235,358-360} and the V₁-vasopressin receptor agonist terlipressin,^{240-243,361} although only terlipressin has been evaluated in randomized controlled trials. In the Reversal of Hepatorenal Syndrome Type 1 with Terlipressin (REVERSE) trial, similar rates of confirmed reversal of HRS were found with terlipressin³⁶² and placebo (19.6% vs. 13.1%, $P = 0.22$), although the mean decrease in serum creatinine was more pronounced with terlipressin (1.1 mg/dL vs. 0.6 mg/dL, $P = 0.001$).²⁴² In the Study to Confirm Efficacy and Safety of Terlipressin in Hepatorenal Syndrome Type 1 (CONFIRM), verified reversal of HRS was observed in 32% of patients in the terlipressin group and 17% in the placebo group.²⁴³ However, mortality was higher among terlipressin-treated patients (51% vs. 45%), fewer terlipressin-treated patients received liver transplants (23% among terlipressin-treated patients vs. 29% among placebo-treated patients), and terlipressin was associated with more adverse events, particularly respiratory failure, with death due to respiratory disorders in 11% of patients treated with terlipressin as compared with 2% of patients in the placebo arm.²⁴³ In a subsequent meta-analysis that included 974 patients, the risk ratio for reversal of HRS was 2.1 (95% CI: 1.5–2.9) with a 40% reduction in the need for KRT within 30 days but no benefit in 90-day survival.²⁴⁸ In a meta-analysis of studies comparing terlipressin to norepinephrine, both agents were associated with similar efficacy in reversal of HRS but adverse events were more common with terlipressin.³⁶² On the basis of these data, terlipressin has been approved for use in the United States. However, given the risk of respiratory failure,

it should not be used in patients with hypoxia, it should not be used in patients with grade 3 acute-on-chronic liver failure, and given a low probability of benefit, it should not be used in patients whose serum creatinine is >5 mg/dL.

ABDOMINAL COMPARTMENT SYNDROME

Acute kidney injury can result from elevations in intraabdominal pressure, resulting in a clinical presentation with similar features to prerenal AKI. The abdominal compartment syndrome is defined by an intraabdominal pressure ≥ 20 mm Hg associated with dysfunction of one or more organ systems.³⁶³ However, intraabdominal pressures lower than 20 mm Hg may be associated with abdominal compartment syndrome, while values higher than this threshold do not universally lead to abdominal compartment syndrome.³⁶⁴⁻³⁶⁷ Abdominal compartment syndrome typically develops in critically ill patients, most commonly in the setting of trauma with abdominal hemorrhage, abdominal surgery, massive fluid resuscitation, liver transplantation, and gastrointestinal conditions including peritonitis and pancreatitis. Mechanisms underlying the development of AKI in abdominal compartment syndrome are believed to involve renal vein compression and constriction of the renal artery from sympathetic and renin-angiotensin system activation and reduced cardiac output.³⁶⁸⁻³⁷⁰ Oliguria, which can lead to anuria, often develops and, as is true for other forms of AKI associated with impaired renal perfusion, urine sodium concentration is commonly reduced.

The diagnosis of abdominal compartment syndrome, which should be suspected in patients with acute abdominal distension and/or rapidly accumulating ascites or abdominal trauma, can be made by simple transduction of bladder pressure.^{61,363,364} Treatment is prompt abdominal decompression; if ascites is present, decompression may be achieved by performing large-volume paracentesis and in patients with severe ileus or colonic distension, bowel decompression may be sufficient, yet surgical laparotomy is often required for definitive therapy.

MANAGEMENT OF POSTRENAL ACUTE KIDNEY INJURY

The principle underlying the management of postrenal AKI is prompt relief of urinary tract obstruction. This topic is reviewed extensively in [Chapter 39](#). Urethral or bladder neck obstruction may be relieved with the placement of a transurethral or suprapubic bladder catheter. Similarly, ureteric obstruction may be acutely relieved by placement of percutaneous nephrostomy tubes or by cystoscopically placed ureteral stents. After the initial relief of obstruction, most patients experience a physiologic diuresis that resolves after several days as the result of excretion of volume and solutes retained during the period of renal obstruction. However, approximately 5% of patients may have a more prolonged diuretic phase after relief of obstruction because of delayed recovery of tubule function relative to GFR resulting in salt-wasting syndrome, which may require IV fluid replacement to maintain blood pressure.^{307,308,371} After initial relief of obstruction, urologic evaluation is required for definitive evaluation and management of the underlying cause of obstruction.

PREVENTION OF INTRINSIC ACUTE KIDNEY INJURY

GENERAL PRINCIPLES

Strategies to prevent intrinsic AKI vary on the basis of the specific etiology of kidney injury. Optimization of cardiovascular function and restoration of intravascular volume status are key interventions to minimize the risk that prerenal AKI evolves into ischemic ATI. There is compelling evidence that aggressive intravascular volume expansion reduces the incidence of ATI after major surgery or trauma, burns, and cholera.^{55,313,372,373} AKI due to sepsis is common and is associated with mortality rates as high as 80%.^{37,303,374} The role of early goal-directed therapy (EGDT) using resuscitation to defined hemodynamic targets (MAP >65 mm Hg, CVP 10–12 mm Hg, urine output >0.5 mL/kg per hour, ScvO₂ >70%) using a combination of crystalloid solutions, red cell transfusion, and vasopressors guided by invasive hemodynamic monitoring in improving overall outcomes and decreasing the risk of AKI has been controversial. In a seminal single-center randomized controlled trial, EGDT resulted in a significant reduction in overall organ dysfunction and mortality in patients presenting with severe sepsis or septic shock, although specific data on the incidence of AKI were not reported.³⁷⁵ However, EGDT was not associated with a reduction in dialysis-requiring AKI in the Protocolized Care for Early Septic Shock (ProCESS), the Australasian Resuscitation in Sepsis Evaluation (ARISE), or the Protocolized Management in Sepsis (PRoMiSe) trials or in a patient-level meta-analysis that combined data from these trials (Fig. 28.1).³⁷⁶⁻³⁷⁹ Although the benefits of EGDT were not confirmed in these three later trials, early recognition of sepsis, prompt initiation of antibiotic therapy, and rapid volume resuscitation and hemodynamic stabilization improve outcomes and are likely to minimize the risk of AKI.³⁸⁰ The role of maintenance of normoglycemia in critically ill patients in minimizing the risk of AKI has also been controversial. Two single-center randomized controlled trials that used intensive insulin management to maintain blood glucose levels of 80 to 110 mg/dL as compared with conventional management maintaining the glucose concentration between 180 and 220 mg/dL each resulted in decreased rates of AKI, defined on either

the basis of change in serum creatinine or the need for KRT.³⁸¹⁻³⁸³ However, the benefits of tight glycemic control were not confirmed in the Normoglycemia in Intensive Care Evaluation–Survival Using Glucose Algorithm Regulation (NICE-SUGAR) trial, a 6104-patient multicenter trial that compared intensive therapy to achieve a target glucose of approximately 80 to 110 mg/dL with more conventional therapy designed to maintain the blood glucose <180 mg/dL.³⁸⁴ In the NICE-SUGAR trial, intensive glycemic control was associated with an increased risk of hypoglycemia, an increased mortality risk (27.5% vs. 24.9%, $P = 0.02$), and no reduction in the need for KRT.

In surgical patients, avoidance of hypotension has been associated with a decreased risk of AKI. In a retrospective analysis of more than 33,000 patients who underwent noncardiac surgery, episodes of intraoperative hypotension with a mean arterial blood pressure <55 mm Hg were associated with a marked increase in the probability of AKI.³⁸⁵ The adjusted odds of AKI with intraoperative hypotension increased with duration of hypotension, from an odds ratio of 1.2 with <10 minutes of hypotension, to 1.3 with 10 to 20 minutes of hypotension and 1.5 with more than 20 minutes of hypotension. Volume overload is typical after surgical procedures and small trials suggested better patient outcomes following abdominal surgery using a restrictive fluid management strategy. However, a large randomized controlled trial comparing restrictive and liberal fluid management strategies in patients undergoing major abdominal surgery at increased risk of complications found no benefit with regard to the primary outcome of 1-year disability-free survival but found a higher rate of acute kidney injury associated with the restrictive fluid strategy (8.6% vs. 5.0%, $P < 0.001$).³⁸⁶

Intravascular volume depletion has been identified as a risk factor for ATI resulting from iodinated contrast, rhabdomyolysis, hemolysis, cisplatin, amphotericin B, multiple myeloma, aminoglycosides, and other nephrotoxins; crystal-associated AKI related to acyclovir and acute urate nephropathy; and AKI stemming from hypercalcemia.^{76,77,164,173,258,280,373,387-389} Restoration of adequate intravascular volume prevents the development of experimental and human ATI in many of these clinical

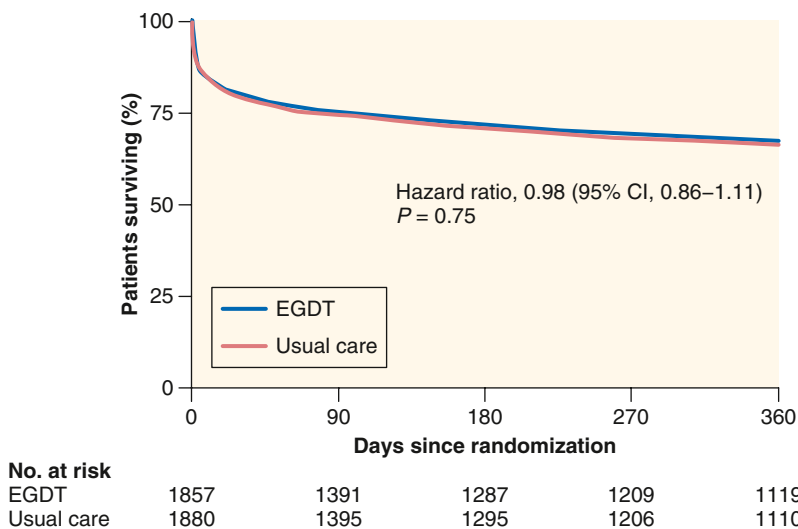


Fig. 28.1 One-year survival comparing early goal-directed therapy (EGDT) to usual care in patients with sepsis. Kaplan-Meier survival curves comparing EGDT to usual care in patients with sepsis from pooled patient-level data from the ProCESS, ARISE, and PRoMiSe trials. Kidney replacement therapy was required in 11.0% of patients who were randomized to EGDT as compared with 10.6% of patients randomized to usual care (OR: 1.02; 95% CI: 0.81–1.28; $P = 0.88$). (From The PRISM Investigators. Goal-directed therapy for septic shock—a patient-level meta-analysis. *N Engl J Med*. 2017;376:2223–2234.)

settings. Avoidance of potentially nephrotoxic medications or insults in high-risk patients and settings is also important to reduce the risk for ATI. Specifically, among patients with advanced cardiac and/or liver disease in whom renal perfusion may be diminished, the use of selective or nonselective NSAIDs that inhibit the production of vasodilatory prostaglandins may exacerbate intrarenal vasoconstriction and precipitate AKI.³⁹⁰⁻³⁹⁴ Diuretics, NSAIDs (including selective cyclo-oxygenase-II [COX-II] inhibitors), ACE inhibitors, angiotensin receptor blockers, and other inhibitors of the renin-angiotensin-aldosterone system (RAAS) should be used with caution in patients with suspected absolute or effective intravascular volume depletion or in patients with renovascular disease as these agents may convert reversible prerenal AKI to ischemic ATI. The combined use of agents that block the RAAS, diuretics, and NSAIDs has been identified as a risk factor for AKI, particularly among patients with heart failure, liver failure, or other conditions with reduced baseline renal perfusion.^{395,396}

Careful monitoring of circulating drug levels appears to reduce the incidence of AKI associated with aminoglycoside antibiotics and calcineurin inhibitors.³⁹⁷⁻³⁹⁹ The observation that the antimicrobial efficacy of aminoglycosides persists in tissues even after the drug has been cleared from the circulation (postantibiotic killing) has led to the use of once-daily dosing with these agents. Dosing regimens that provide higher peak drug levels but less frequent administration appear to provide comparable antimicrobial activity and less nephrotoxicity than older conventional dosing regimens.³⁹⁹⁻⁴⁰² Nephrotoxicity of drugs may also be reduced through changes in formulation. For example, the use of lipid-encapsulated formulations of amphotericin B may decrease the risk of amphotericin-induced AKI.⁴⁰³

CONTRAST-ASSOCIATED ACUTE KIDNEY INJURY

Contrast-associated AKI (CA-AKI) has historically been defined by and commonly manifests clinically as small absolute (>0.5 mg/dL) and/or relative ($>25\%$) increases in serum creatinine that occur within 2 to 4 days post iodinated contrast administration. While severe AKI is relatively uncommon due to iodinated contrast alone, the incidence of CA-AKI defined by relatively minor increments in serum creatinine has been shown in past studies to occur in up to 15% or more of high-risk patients.⁴⁰⁴ A series of recent observational retrospective studies have questioned the true incidence of CA-AKI and challenged the concept that iodinated contrast is nephrotoxic, particularly among individuals receiving intravenous (IV) contrast media for computed tomography.⁴⁰⁵⁻⁴⁰⁸ A meta-analysis of 13 nonrandomized studies that included 25,950 patients found a similar risk of AKI in patients who received IV iodinated contrast compared with patients who underwent radiographic procedures without intravascular contrast (relative risk [RR] = 0.79; 95% CI: 0.62–1.02).⁴⁰⁶ Similarly, an observational study of more than 29 million patients hospitalizations found that the risk for AKI among patients who received intravascular iodinated contrast during their hospitalization was comparable with the risk observed among patients who had not received contrast (RR 0.93; 95% CI 0.88–0.97).⁴⁰⁹ While the results of these retrospective analyses are adjusted for medical comorbidity and other risk

factors for renal injury, such adjustment cannot fully account for all factors that influence the decision to administer contrast, rendering it likely that patients in these studies who did not receive contrast may have been at higher risk for AKI than patients who received contrast.

The apparent reduction in risk of kidney toxicity from iodinated contrast over time likely reflects advancement in technology. Over the past 25 years, there has been considerable progress in developing less nephrotoxic contrast agents.⁴¹⁰ The use of lower-osmolal contrast agents in place of the older and more nephrotoxic high-osmolal agents resulted in a decreased incidence of CA-AKI.^{411,412} Data regarding the added benefit associated with the iso-osmolal radiocontrast agent iodixanol have been less consistent⁴¹³⁻⁴¹⁹ and may reflect heterogeneity in the risk of CA-AKI associated with specific lower-osmolal agents.⁴²⁰ In addition, imaging technology has improved with use of higher-resolution computed tomography scanners and improved imaging technology for angiography, allowing reduction in the volume of administered contrast media, further reducing the risk of toxicity.

On the basis of these data, the American College of Radiology and National Kidney Foundation have issued consensus statements on the use of intravenous iodinated contrast media.⁴²¹ On the basis of an extensive review of published data, they estimated the risk of AKI following contrast administration to be similar to that of the general population for individuals with an eGFR of 45 mL/min/1.73 m² or higher, to be 0% to 2% for individuals with an eGFR of 30 to 44 mL/min/1.73 m² and to be 0% to 17% for individuals with an eGFR <30 mL/min/1.73 m² and recommend prophylaxis for prevention of AKI in individuals undergoing imaging with intravenous contrast who have an eGFR <30 mL/min/1.73 m² and are not undergoing maintenance dialysis and in high-risk patients with an eGFR of 30 to 44 mL/min/1.73 m².⁴²¹ Similar guidance was developed for the management of patients with kidney disease requiring cardiovascular catheterization or intervention in a workshop jointly sponsored by the National Kidney Foundation and Society of Cardiovascular Angiography and Interventions (SCAI).⁴²²

Although is prudent to consider intravascular iodinated contrast as potentially nephrotoxic and to implement evidence-based preventive care in patients at risk for CA-AKI, excessive concern regarding the risk of CA-AKI should not prevent performance of clinically indicated diagnostic and therapeutic procedures. Two decades ago, Chertow and colleagues⁴²³ observed that elderly individuals with CKD who presented with acute myocardial infarction were less likely to undergo coronary angiography than propensity-matched individuals without CKD, a finding that they described as “renalism.” They further estimated that the failure to perform clinically indicated angiography was associated with an increase in mortality risk. In an analysis of U.S. veterans hospitalized with acute coronary syndrome between 2013 and 2017, individuals with CKD were again less likely to undergo angiography despite matching for clinical indication for angiography based on the Global Registry of Acute Coronary Events (GRACE) 2.0 risk score, with nonperformance of angiography associated with an adjusted risk ratio for all-cause mortality at 6 months of 1.4 (95% CI: 1.3–1.5).⁴²⁴ Thus while CA-AKI is associated

with increased morbidity and mortality, failure to perform clinically indicated diagnostic and therapeutic interventions because of excessive concern regarding the risk of AKI also has untoward consequences.

Interventions to ameliorate risk of contrast-associated acute kidney injury Periprocedural intravenous crystalloid

Periprocedural Intravenous Crystalloid. Given the known timing of administration of iodinated contrast, prevention of CA-AKI has been the focus of numerous clinical trials (Table 28.15). Past studies have demonstrated that the administration of IV fluids to high-risk patients before and after exposure to intravascular iodinated contrast diminished the risk for contrast-associated AKI (CA-AKI), although the optimal regimen for fluid administration is unknown.^{3,425,426} In a small clinical trial that was stopped early due to safety concerns, periprocedural IV isotonic saline was associated with a markedly lower rate of AKI after contrast exposure than oral fluid administration.⁴²⁵ In a larger randomized trial, isotonic IV saline significantly reduced the incidence of CA-AKI following coronary angiography compared with IV half-normal saline with a particular benefit noted in diabetic patients and those receiving large volumes of contrast.⁴²⁶ One study challenged the principle that IV fluid administration reduces the risk for CA-AKI.⁴²⁷ This noninferiority clinical trial randomized 660 patients with baseline CKD undergoing procedures with iodinated contrast to receive IV isotonic saline or no IV fluids and found a similar rate of CA-AKI (2.7% with saline vs. 2.6% without fluid), concluding that withholding IV fluid was not inferior to administering IV saline with regard to the prevention of small increments in serum creatinine. A subsequent study randomized 523 patients with stage 3 CKD undergoing contrast-enhanced computed tomography to no preprocedural volume administration or preprocedural isotonic sodium bicarbonate.⁴²⁸ Overall, AKI developed in 2.1% of participants, 2.7% in the group receiving no preprocedural fluids, and 1.5% in

participants administered preprocedural intravenous sodium bicarbonate. Interpretation of both trials is limited by relatively low sample size and inclusion of a predominantly low-risk population who did not meet criteria for periprocedural prophylactic intervention proposed by the American College of Radiology–National Kidney Foundation consensus recommendations.⁴²¹

Intravenous Sodium Bicarbonate. A number of small clinical trials compared the effects of isotonic sodium bicarbonate compared with isotonic saline for prevention of contrast-associated AKI.⁴²⁹⁻⁴³⁸ These studies were generally underpowered and yielded conflicting results. Subsequent meta-analyses concluded that there was an overall benefit associated with bicarbonate administration with regard to AKI defined by small changes in serum creatinine, although there was no demonstrable benefit with regard to the need for dialysis,⁴³⁹⁻⁴⁴¹ leading several clinical practice guidelines to recommend the administration of either IV isotonic sodium chloride or sodium bicarbonate to high-risk patients receiving iodinated contrast.^{3,442,443} However, the Prevention of Serious Adverse Events Following Angiography (PRESERVE) trial provided more definitive findings regarding the comparative effects of sodium bicarbonate and sodium chloride for the prevention of CA-AKI.⁴⁴⁴ PRESERVE was a multinational randomized clinical trial that used a 2 × 2 factorial design to compare IV isotonic sodium bicarbonate with IV isotonic sodium chloride and oral N-acetylcysteine with placebo for the prevention of serious adverse outcomes and CA-AKI following angiographic procedures in 4993 patients with CKD. IV sodium bicarbonate did not reduce the incidence of a primary outcome composed of 90-day death, need for dialysis, or persistent decline in kidney function (OR = 0.93; 95% CI: 0.72–1.22). Similarly, bicarbonate did not reduce the incidence of CA-AKI assessed 3 to 5 days post angiography (OR = 1.16; 95% CI: 0.96–1.41). While the optimal rate and duration of IV saline administration is not known, it is reasonable to administer isotonic sodium chloride at a rate of 1 mL/kg per hour for 6 to 12 hours before and 6 to 12 hours after a contrast-enhanced procedure in at-risk hospitalized patients. For at-risk outpatients, an alternative regimen of 3 mL/kg over 1 hour before the procedure followed by 6 mL/kg administered over 2 to 6 hours following the procedure may be more feasible.

Table 28.15 Effectiveness of Preventive Interventions for Contrast-Associated Acute Kidney Injury

No benefit	Unclear benefit	Beneficial
Loop diuretics ^a	Statins	Isotonic intravenous fluids
Mannitol ^a	Natriuretic peptides	Low/iso-osmolal contrast media
Dopamine ^a	Theophylline/aminophylline	
Fenoldopam ^a	Ascorbic acid	
Hemodialysis ^a	Hemofiltration	
N-acetylcysteine		
Sodium bicarbonate ^b		

^aPotentially deleterious.

^bAs compared with isotonic saline.

N-acetylcysteine. N-acetylcysteine (NAC) is an antioxidant with vasodilatory properties that was postulated to potentially prevent CA-AKI based on its capacity to scavenge ROS, reduce the depletion of glutathione, and stimulate the production of vasodilatory mediators including nitric oxide.^{445,446} Clinical trials of oral and IV NAC have yielded conflicting findings.⁴⁴⁷⁻⁴⁵⁶ Although initially used at a dose of 600 mg twice daily,⁴⁴⁷ subsequent studies suggested greater efficacy with higher doses of up to 1200 mg twice daily.^{454,455} In the Acetylcysteine for Contrast-Induced Nephropathy (ACT) trial, 2308 patients were randomized to receive 1200 mg of NAC or placebo twice daily beginning before the procedure and continuing for three doses post procedure.⁴⁵⁶ No differences were observed in the incidence of CA-AKI at 48 to 96 hours post contrast administration or in the incidence of death or need for dialysis within 30 days. However, the

overall study population had relatively well-preserved kidney function, with a median serum creatinine of 1.1 mg/dL, and less than 16% of patients had a baseline serum creatinine >1.5 mg/dL. In the aforementioned PRESERVE trial, the administration of NAC in a dose of 1200 mg by mouth twice daily for 5 days beginning just before angiography was not associated with a reduction in 90-day death, need for dialysis, or persistent impairment in kidney function (OR = 1.02; 95% CI: 0.78–1.33) or in a reduction in CA-AKI (OR = 1.06; 95% CI: 0.87–1.28).⁴⁴⁴ On the basis of these findings, NAC should not be used to reduce the risk of CA-AKI.

Other Agents. Trials of other pharmacologic interventions including furosemide, dopamine, fenoldopam, calcium channel blockers, and mannitol have failed to demonstrate significant benefit and, in some cases, have been associated with an increased risk of CA-AKI (Table 28.15).⁴⁵⁷⁻⁴⁶³ Studies on the benefit of natriuretic peptides, aminophylline, theophylline, and ascorbic acid have also yielded conflicting results.⁴⁶⁴⁻⁴⁷³ Given the absence of convincing data on the efficacy of these interventions, as well as potential safety concerns with the use of natriuretic peptides, aminophylline, and theophylline in patients with cardiovascular disease, their routine use is not recommended.⁴⁷² There have been multiple trials and meta-analyses that investigated statins for the prevention of CA-AKI. While many, albeit not all, demonstrated a benefit of statins, particularly in high dose, with respect to the development of CA-AKI, the effect of this class of medication on more serious outcomes including need for dialysis and progressive CKD remains unclear.⁴⁷⁴⁻⁴⁸⁴ Furthermore, in the majority of patients requiring angiographic procedures, there are likely other indications for statin therapy. Kidney replacement therapies for the prevention of CA-AKI have been largely ineffective, and in some instances the use of “prophylactic” hemodialysis has been associated with harm.⁴⁸⁵⁻⁴⁸⁷ The interpretation of studies of hemofiltration for prevention of CA-AKI is confounded by their use of change in serum creatinine as an endpoint because hemofiltration lowers serum creatinine concentration.^{488,489} Given the risks associated with intravenous line placement and the kidney replacement procedures themselves, along with lack of definitive benefit, use of dialysis or hemofiltration to prevent CA-AKI is not currently recommended^{3,490} (Fig. 28.2).

PREVENTION OF OTHER FORMS OF INTRINSIC ACUTE KIDNEY INJURY

Allopurinol (10 mg/kg/day in three divided doses, or 100 mg/m² every 8 hours, max 800 mg per day) is useful for limiting uric acid generation in patients at high risk for acute urate nephropathy; however, AKI can develop despite the use of allopurinol, probably through the toxic actions of hypoxanthine crystals on tubule function.^{164,258,278,280,491,492} In the setting of high rates of uric acid generation such as tumor lysis syndrome, the use of recombinant urate oxidase (rasburicase, 0.05–0.2 mg/kg) may be more effective. Rasburicase catalyzes the degradation of uric acid to allantoin and has been shown to be effective both as prophylaxis and treatment for acute uric acid-mediated tumor lysis syndrome and to prevent the development of AKI due to tumor lysis syndrome associated hyperuricemia.^{168,258,492-495}

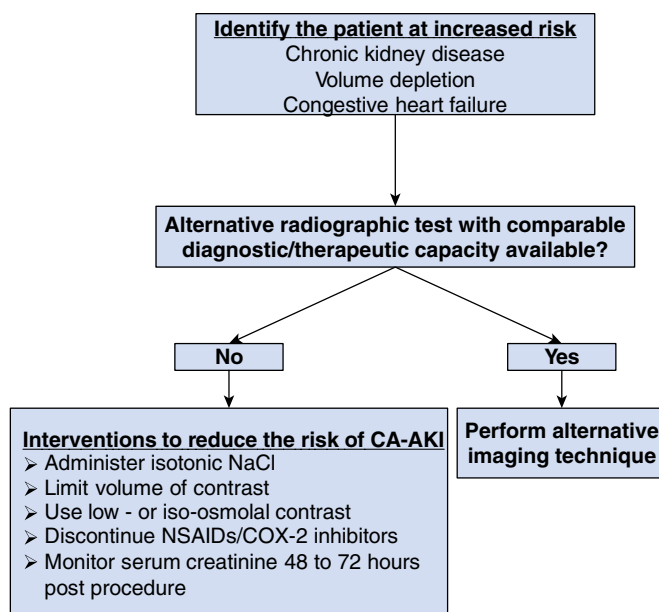


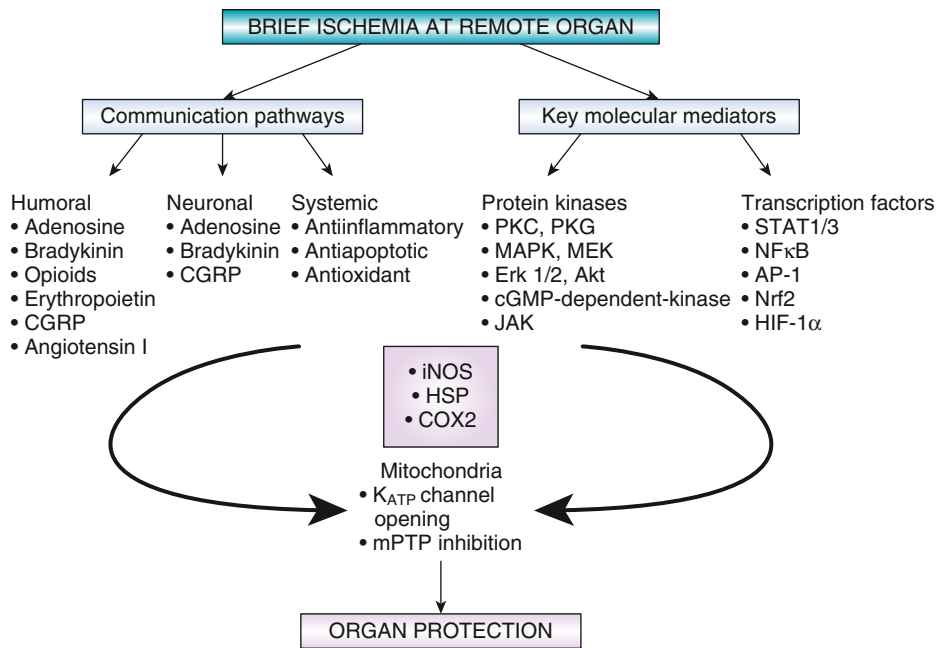
Fig. 28.2 Algorithm for mitigation of risk of contrast-associated acute kidney injury (CA-AKI). (COX-2, Cyclo-oxygenase-2; NSAIDs, nonsteroidal antiinflammatory drugs.)

In oligoanuric patients, prophylactic hemodialysis may be used to acutely lower uric acid levels.

Amifostine, an organic thiophosphate, has been demonstrated to ameliorate cisplatin nephrotoxicity in patients with solid organ or hematologic malignancies.⁴⁹⁶⁻⁴⁹⁹ N-acetylcysteine limits acetaminophen-induced renal injury if given within 24 hours of ingestion, and dimercaprol, a chelating agent, may prevent heavy metal nephrotoxicity.^{500,501} Ethanol inhibits ethylene glycol metabolism to oxalic acid and other toxic metabolites, but its use has been largely replaced by fomepizole, an inhibitor of alcohol dehydrogenase that decreases production of ethylene glycol metabolites and prevents the development of AKI.⁵⁰²⁻⁵⁰⁵

REMOTE ISCHEMIC PRECONDITIONING

Remote ischemic preconditioning (RIPC) has been investigated as a potential intervention for the prevention of AKI. RIPC involves the implementation of brief episode(s) of ischemia/reperfusion of distant tissue (e.g., with sequential brief inflation and deflation of a blood pressure tourniquet on a limb that is hypothesized to enhance the kidneys' resistance to a subsequent more prolonged period of ischemia by means of hormonal mediators, hormonal and neuronal signaling pathways, and antiinflammatory and molecular mediators (Fig. 28.3). Various trials and meta-analyses have examined the benefit of RIPC in the setting of cardiac surgery. A trial that randomized 240 high-risk patients undergoing cardiac surgery to RIPC or sham RIPC demonstrated a lower rate of AKI, defined by KDIGO criteria, with RIPC (37.5% vs. 52.5%, $P = 0.02$), with no effect on secondary endpoints including myocardial infarction, stroke, or death.⁵⁰⁶ A considerably larger trial that enrolled 1612 patients undergoing cardiac surgery found no difference



Gassnov – JASN 2014

Fig. 28.3 Postulated mechanisms for remote ischemic preconditioning. Postulated mechanisms for renal protective effects of remote ischemic preconditioning. *AP-1*, Activator protein-1; *cGMP*, cyclic guanosine monophosphate; *CGRP*, calcitonin gene-related peptide; *COX2*, cyclooxygenase 2; *HIF-1α*, hypoxia-inducible factor 1α; *HSP*, heat shock protein; *iNOS*, inducible nitric oxide synthase; *JAK*, Janus kinase; *MEK*, MAPK kinase; *mPTP*, mitochondrial permeability transition pore; *Nrf2*, nuclear factor (erythroid-derived 2)-like 2; *STAT1/3*, signal transducer and activator of transcription. (From Gassnov N, Nia AM, Caglayan E, Er F. Remote ischemic preconditioning and renoprotection: from myth to a novel therapeutic option? *J Am Soc Nephrol*. 2014;25(2):216–224.)

in the rate of AKI (a secondary end point) between RIPC and sham conditioning groups or in the incidence of the combined primary end point of death from cardiovascular causes, nonfatal myocardial infarction, coronary revascularization, or stroke within 12 months.⁵⁰⁷ A series of meta-analyses of studies that included patients undergoing cardiac and, in some instances, vascular surgery found lower rates of AKI with RIPC compared with control but failed to demonstrate a benefit on need for KRT or death.^{508–510} On the basis of these cumulative data, the use of RIPC for the prevention of adverse outcomes following cardiac and vascular procedures is not routinely recommended.

PHARMACOLOGIC THERAPY FOR ACUTE TUBULAR INJURY

During the past 2 decades there has been extensive investigation into the pathogenesis of AKI using experimental animal models and cultured cells. These studies have resulted in substantial advances in our understanding of the pathophysiology of ATI in humans and led to the discovery of an array of potentially novel targets for the treatment of this common and serious disease. However, multiple interventions shown to ameliorate AKI in animals have failed to be effective in humans with ATI (Table 28.16). There are many possible reasons for the lack of success in translating therapeutic successes for AKI from animal models to clinical practice. Differences in the etiology of ATI in animal models and human disease may contribute to differential responses to pharmacologic therapy. Another principal obstacle relates to the difficulty in identifying the incipient stage of ATI before elevations in the serum creatinine concentration or clinical evidence of decreased urine output. Over the past decade, several novel serum and urinary biomarkers have been investigated for their ability to identify AKI in its

Table 28.16 Pharmacologic Interventions found to be Ineffective for Acute Tubular Injury in Clinical Trials

Diuretics
Dopamine
Fenoldopam
Thyroid hormone
Alpha melanocyte stimulating hormone
Atrial natriuretic peptide
Alkaline phosphatase
Insulin growth factor
Erythropoietin
Prostaglandin A1

earliest stages and differentiate ATI from volume-responsive AKI.¹⁹ Work in this area may facilitate the identification of those patients most likely to respond to treatments that have been found to be effective in animal models.

DOPAMINE

Historically, low-dose dopamine (“renal-dose” dopamine; <2 mg/kg/min) was widely advocated for the management of oliguric AKI.^{511–513} In experimental studies of animals and healthy human volunteers, low-dose dopamine increased renal blood flow and to a lesser extent GFR. However, low-dose dopamine has not been demonstrated to prevent or alter the course of ischemic or nephrotoxic ATI in prospective clinical trials.^{514–518} This absence of clinical benefit may relate to differences in the hemodynamic response to low-dose dopamine in patients with renal disease as compared to healthy individuals. In contrast to the reduction in renal resistive index associated with low-dose dopamine in critically ill patients without kidney disease, dopamine infusion is associated with an increase

in renal resistance in patients with AKI.⁵¹⁹ Moreover, dopamine, even at low doses, is potentially toxic in critically ill patients and can induce tachyarrhythmias, myocardial ischemia, as well as extravasation necrosis.⁵¹⁹ Thus the routine administration of low-dose dopamine to ameliorate or reverse the course of AKI is not justified on the basis of the balance of experimental and clinical evidence.^{520,521}

FENOLDOPAM

Fenoldopam is a selective postsynaptic dopamine agonist that acts on D1 receptors and mediates more potent renal vasodilatation and natriuresis than dopamine.⁵²² However, fenoldopam is a potent antihypertensive agent and causes hypotension by decreasing peripheral vascular resistance. Several small studies suggested that fenoldopam could reduce the incidence of AKI in high-risk clinical situations^{523,524}; however, a subsequent larger randomized trial comparing fenoldopam to standard hydration in patients undergoing invasive angiographic procedures found no benefit in regard to decreasing the incidence of contrast-associated AKI.⁴⁶⁰ In another large RCT, fenoldopam administration failed to reduce mortality or the need for KRT in ICU patients with early ATI.⁵²⁵ Therefore there is currently no clinical role for fenoldopam in the prevention or treatment of AKI.

NATRIURETIC PEPTIDES

ANP is a 28-amino acid polypeptide synthesized in cardiac atrial muscle.^{526,527} ANP augments GFR by triggering afferent arteriolar vasodilatation and constriction of the efferent arteriole.^{528,529} In addition, ANP inhibits sodium transport and lowers oxygen requirements in several nephron segments.^{530,531} Synthetic analogs of ANP showed promise in the management of ATI in the laboratory setting; however, these benefits in animal models of AKI have failed to translate into clinical benefit in humans. A large multicenter, prospective, randomized placebo-controlled trial of anaritide, a synthetic analog of ANP, in patients with ATI failed to show clinically significant improvement in dialysis-free survival or overall mortality,⁵³² although there was an improvement in dialysis-free survival in oliguric patients. This benefit in oliguric patients was not confirmed in a subsequent prospective study.⁵³³ It has been suggested that the absence of benefit may be related to both the relatively late initiation of therapy and the effect of ANP on systemic blood pressure. In a subsequent pilot study, low-dose recombinant ANP administration in high-risk cardiac surgery patients was associated with a reduction in the requirement for postoperative KRT.⁵³⁴ Until these results are confirmed in a larger, multicenter trial, the use of ANP in this setting cannot be recommended. Trials of ANP for the prevention of contrast-associated AKI have generated mixed results.^{469,470} Ularitide (urodilantin) is a natriuretic pro-ANP fragment produced within the kidney. In a small randomized trial, ularitide did not reduce the need for dialysis in patients with AKI.⁵³⁵ A recent meta-analysis of ANP for the treatment of AKI concluded that the paucity of high-quality studies precluded a determination of the effects of this therapy.⁵³⁶

LOOP DIURETICS

High-dose IV diuretics are commonly prescribed to increase urine output in patients with oliguric AKI. Although this

strategy assists in volume management and minimizes the risk of progressive volume overload, there is no evidence that diuretic therapy alters the natural history of AKI or improves mortality or dialysis-free survival. In a retrospective analysis, diuretic therapy was associated with an increased risk of death and nonrecovery of renal function.⁵³⁷ These risks were restricted, however, to patients who did not respond to diuretic administration with increased urine volume; in diuretic-responsive patients, outcomes were similar to untreated patients. In a prospective randomized trial, high-dose IV furosemide augmented urine output but did not alter the outcome of established AKI.⁵³⁸ In a post-hoc analysis of data from the Fluid and Catheter Treatment Trial, a positive fluid balance after AKI in patients with acute lung injury was strongly associated with increased mortality while diuretic therapy was associated with improved 60-day patient survival.⁵³⁹ Given the risks of loop diuretics in AKI, including irreversible ototoxicity and exacerbation of prerenal AKI, these agents should be used solely to facilitate the management of extracellular volume overload (*vide infra*).⁵⁴⁰ Of note, a single administration of furosemide in a dose of 1.0 to 1.5 mg/kg has been shown to help characterize the risk for progressive AKI.⁵⁴¹ Specifically, a urine volume of <200 mL in the 2 hours following furosemide administration demonstrated sensitivity and specificity for progression to AKIN stage 3 of 87.1% and 84.1%, respectively.

MANNITOL

The osmotic diuretic mannitol, which also has renal vasodilatory and oxygen-free radical scavenging properties, has been investigated as a preventive treatment for AKI.^{542,543} No adequate data exist to support the routine administration of mannitol to oliguric patients. Moreover, when administered to severely oliguric or anuric patients, mannitol may trigger expansion of intravascular volume and pulmonary edema, as well as severe hyponatremia due to an osmotic shift of water from the intracellular to the intravascular space.^{543,544-547}

MANAGEMENT OF OTHER ETIOLOGIES OF INTRINSIC ACUTE KIDNEY INJURY

ACUTE VASCULITIS AND ACUTE GLOMERULAR DISEASE

The management of acute vasculitis involving the kidney and acute glomerular disease is covered in detail in Chapters 31 to 34. AKI caused by acute glomerulonephritis or vasculitis may respond to corticosteroids, alkylating agents, rituximab, and plasmapheresis depending on the primary etiology of the disease. Plasma exchange is useful in the treatment of sporadic TTP and possibly sporadic HUS in adults.^{548,549} The role of plasmapheresis in drug-induced thrombotic microangiopathies is less certain, and removal of the offending agent is the most important initial therapeutic maneuver.^{148,550,551} Postdiarrheal HUS in children is usually managed conservatively as evidence suggests that early antibiotic therapy may actually promote the development of HUS.⁵⁵² Treatment with eculizumab, a humanized monoclonal antibody that prevents cleavage of complement component C5 into C5a and C5b, inhibiting terminal complement activation, may be considered in patients with nondiarrheal (complement mediated) HUS unresponsive to plasma exchange.⁵⁵³

Hypertension and AKI associated with scleroderma may be exquisitely sensitive to treatment with ACE inhibitors.⁵⁵⁴⁻⁵⁵⁶ Management of vasculitis is discussed in detail in Chapter 31.

ACUTE KIDNEY INJURY IN MULTIPLE MYELOMA

Early studies suggested that plasmapheresis may be of benefit in AKI due to myeloma cast nephropathy.^{173,557,558} Clearance of circulating light chains with concomitant chemotherapy to decrease the rate of production had been postulated to reverse renal injury in patients with circulating light chains, heavy Bence Jones proteinuria, and AKI. A subsequent randomized controlled trial compared plasma exchange and standard chemotherapy with chemotherapy alone. Although the study did not demonstrate improvement with plasma exchange with regard to a composite outcome of death, dialysis dependence, or GFR <30 mL/min at 6 months, the study was inadequately powered to definitively exclude a clinical benefit and there was a trend toward improved outcomes with plasmapheresis.¹⁷⁴⁻¹⁷⁶ More recently, it has been suggested that the use of dialysis membranes that are permeable to light chains and other proteins with molecular weights lower than albumin (high cutoff membranes) may be an effective therapeutic strategy in patients with AKI due to light chain cast nephropathy; however, data from clinical trials have yielded inconclusive results.¹⁷⁷⁻¹⁷⁹ Thus primary management should focus on prompt initiation of highly effective chemotherapy. Other contributors to AKI in multiple myeloma, such as hypercalcemia, should also be promptly treated. Multiple myeloma is discussed in more detail in Chapter 35.

ACUTE INTERSTITIAL NEPHRITIS

Most cases of acute interstitial nephritis (AIN) are due to an allergic response to a medication.⁵⁵⁹ The initial therapeutic step in AIN is discontinuation of the offending medication or treatment of the probable inciting factor if not drug induced. Data on the efficacy of corticosteroids derive from small observational studies, which have yielded highly discordant results. While some studies suggest that early use of corticosteroids (i.e., before significant renal damage and within 7–14 days of discontinuation of the offending medication)⁵⁶⁰ may be beneficial, other studies demonstrate no clear evidence of efficacy.⁵⁹ There have been no large, prospective randomized clinical trials investigating the role of corticosteroids in the treatment of AIN. As corticosteroids are associated with a series of potentially serious side effects, their use should be considered on a case-by-case basis. If corticosteroid therapy is being considered and no patient-related contraindications exist, one potential regimen used in a recent study involves the IV administration of methylprednisolone (250–500 mg/day) for 3 to 4 days, followed by oral prednisone at a dose of 1 mg/kg/day tapered over 8 to 12 weeks.⁵⁶⁰ However, there are no data supporting the superiority of this specific approach over others. Mycophenolate mofetil has also been investigated as a therapeutic agent for AIN. In a study of eight patients with AIN, six experienced improvement and two experienced stabilization in kidney function with mycophenolate mofetil therapy.⁵⁶¹ While this small case series suggests a possible role for mycophenolate mofetil in the treatment of AIN, additional data are needed to confirm its safety and efficacy

for this indication. Tubulointerstitial diseases are discussed further in Chapter 37.

NONDIALYTIC SUPPORTIVE MANAGEMENT OF ACUTE KIDNEY INJURY–ASSOCIATED COMPLICATIONS

Metabolic complications such as intravascular volume overload, hyperkalemia, hyperphosphatemia, and metabolic acidosis are common in oliguric AKI, and preventive measures should be implemented beginning with initial diagnosis (Table 28.14). Adequate nutrition should be provided to meet caloric requirements and minimize catabolism; however, excessive protein feeding should be avoided as higher protein delivery is not associated with improved outcomes in critical illness and has been associated with increased mortality in critically ill patients with AKI.⁵⁶² In addition, all medications that are normally excreted by the kidney need to be adjusted on the basis of the severity of renal impairment.

EXTRACELLULAR VOLUME OVERLOAD

After correction of intravascular volume deficits, salt and water intake should be adjusted to match ongoing losses (urinary, gastrointestinal, drainage sites, insensible losses). Extracellular volume overload can usually be managed by restriction of salt and water intake and judicious use of diuretics. High doses of loop diuretics (e.g., the equivalent of 200 mg of furosemide administered as an IV bolus infusion or 20 mg per hour as a continuous infusion) or combination therapy with both thiazide and loop diuretics may be required. If adequate diuresis cannot be attained, further use of diuretics should be discontinued to minimize the risk of complications such as ototoxicity. Fluid administration should be closely monitored to avoid progressive volume overload. Although there is a strong association between progressive fluid overload and mortality risk in patients with AKI,⁵⁶³⁻⁵⁶⁵ a causal relationship has not been definitively established because volume overload may also be a surrogate for other determinants of mortality such as hemodynamic instability and capillary leak. Fluid conservative management has, however, been demonstrated to result in improved outcomes in critically ill patients with respiratory failure.⁵⁶⁶ Ultrafiltration or dialysis may be required for volume management when conservative measures fail.

HYPONATREMIA AND HYPERNATREMIA

Hyponatremia associated with a fall in effective serum osmolality can usually be corrected by restriction of water intake. Conversely, hypernatremia is treated by administration of water, hypotonic saline solutions, or hypotonic dextrose-containing solutions (the latter are effectively hypotonic because dextrose is rapidly metabolized).

HYPERKALEMIA

Mild hyperkalemia (<5.5 mmol/L) should be managed initially by restriction of dietary potassium intake and the discontinuation of potassium supplements and potassium-sparing diuretics. More severe degrees of hyperkalemia (5.5–6.5 mmol/L) can usually be controlled by combining these measures with the administration of exchange resins

to enhance gastrointestinal potassium losses. While sodium polystyrene sulfonate has been widely used for decades, concerns have been raised regarding its safety, particularly when administered in 70% sorbitol, due to reports of bowel necrosis.^{567,568} Newer exchange resins including sodium zirconium cyclosilicate and patiomer are also effective at decreasing serum potassium concentration, although patiomer is not labeled for acute management of hyperkalemia. Loop diuretics can also increase potassium excretion in diuretic-responsive patients. Emergency measures need to be employed in patients with more severe hyperkalemia and in patients with electrocardiographic manifestations of hyperkalemia. In patients with severe hyperkalemia with concomitant electrocardiographic manifestations, the IV administration of calcium will antagonize the cardiac and neuromuscular effects of hyperkalemia and is a valuable emergency temporizing measure, allowing time for the additional measures described later to be implemented. IV calcium must be used with caution, however, if there is concomitant severe hyperphosphatemia or evidence of digitalis toxicity. IV insulin (10–20 U of regular insulin) promotes potassium entry into cells and lowers extracellular potassium concentration within 15 to 30 minutes, with an effect that lasts for several hours.^{569,570} Concomitant administration of IV dextrose (25–50 g over 30 to 60 minutes) is required to prevent hypoglycemia in patients who do not have hyperglycemia. β -adrenergic agonists, such as inhaled albuterol (10–20 mg by nebulizer), also promote rapid potassium uptake into the intracellular compartment.⁵⁶⁹ Although sodium bicarbonate also stimulates potassium uptake into the intracellular compartment, this effect is not sufficiently rapid to be clinically useful for the emergent management of hyperkalemia.⁵⁷⁰ Emergent dialysis is indicated if hyperkalemia is resistant to these measures.

METABOLIC ACIDOSIS

Treatment of metabolic acidosis depends on the clinical setting and etiology. As a general rule, metabolic acidosis does not require emergent treatment unless the serum HCO_3^- concentration falls below 15 mmol/L or the pH is lower than 7.15 to 7.20. In patients with AKI in whom metabolic acidosis is due to the underlying renal failure, more severe acidosis can be corrected by either oral or IV bicarbonate administration. Initial rates of replacement should be based on estimates of HCO_3^- deficit and adjusted thereafter according to serum levels. In patients with underlying lactic acidosis, the role of bicarbonate therapy is controversial and the primary focus of therapy should be on correction of the underlying cause.^{571–574} Patients treated with IV bicarbonate need to be monitored for complications of therapy including metabolic alkalosis, hypocalcemia, hypokalemia, hyponatremia, and volume overload.

DISTURBANCES OF CALCIUM, PHOSPHATE, MAGNESIUM, AND URIC ACID

Hypocalcemia does not usually require treatment unless it is severe or symptomatic, as may occur in patients with rhabdomyolysis or pancreatitis, or after administration of bicarbonate. Hyperphosphatemia can often be controlled by restricting dietary phosphate intake and the use of oral phosphate binders (e.g., aluminum hydroxide, calcium salts,

sevelamer carbonate, and lanthanum carbonate). Caution should be employed when using aluminum-containing phosphate binders as prolonged use may result in aluminum intoxication, which can contribute to osteomalacia; short-term use is rarely associated with bone disease, and the feared neurologic complications of aluminum intoxication are restricted to patients with inadvertent parenteral exposure. Hypermagnesemia can be prevented through avoidance of magnesium-containing medications, such as antacids, and limiting magnesium content of parenteral nutrition. Hyperuricemia is usually mild in AKI (<15 mg/dL) and does not require specific intervention. Severe hyperuricemia secondary to cell lysis may be managed by blocking xanthine oxidase with allopurinol or by enhancing degradation with recombinant uricase as previously described.

NUTRITIONAL MANAGEMENT

Patients with AKI are clinically heterogeneous and individualized nutritional management is required, especially in critically ill patients on KRT in whom protein catabolic rates can exceed 1.5 g/kg body weight/day.^{3,292,293,295,296,575,576} The objective of nutritional management in AKI is to provide sufficient calories to preserve lean body mass, avoid starvation ketoacidosis, and promote healing and tissue repair while minimizing production of nitrogenous waste. If the duration of impaired kidney function is likely to be short and the patient is not extremely catabolic and does not require KRT, then dietary protein should be approximately 0.8 to 1.0 g/kg body weight/day.³ Protein intake should not be restricted in patients in whom AKI is likely to be prolonged, are hypercatabolic, or are receiving KRT. Protein intake in these patients should generally be 1.0 to 1.5 g/kg body weight/day.^{292,293,575,576} There is no evidence of improved outcomes with protein intake higher than 1.7 g/kg body weight/day, even in extremely hypercatabolic patients with a meta-analysis of data from clinical trials in critically ill patients demonstrating no overall benefit to higher protein delivery with increased mortality in the subgroup of individuals with AKI (risk ratio 1.4; 95% CI: 1.14–1.8).^{3,562} Total caloric intake should generally be 20 to 30 kcal/kg body weight/day and should not exceed 35 kcal/kg per day.^{3,292,293,575,576} Benefits of vigorous parenteral hyperalimentation have not been consistently demonstrated; enteral nutrition support is preferred as it avoids the morbidity associated with parenteral nutrition while providing support to intestinal function.²⁹⁵ Water-soluble vitamins and trace elements should be supplemented in patients receiving KRT.^{575,576}

ANEMIA

Severe anemia is generally managed with blood transfusion. Transfusion is usually not required for patients with a hemoglobin concentration above 7 g/dL.⁵⁷⁷ Whether there is a role for erythropoiesis stimulating agents in AKI has not been definitively determined.⁵⁷⁸ Patients with AKI or other acute illness are relatively resistant to the effects of these agents, and their onset of action is delayed. In randomized controlled trials in critically ill patients, recombinant human erythropoietin decreased transfusion requirement but had no effect on other outcomes.^{579,580} Uremic bleeding usually responds to desmopressin, correction of anemia, estrogens, or dialysis.

DRUG DOSING

Doses of drugs that are excreted by the kidney must be adjusted for impaired kidney function and the use of KRT.⁵⁸¹⁻⁵⁸³ Whenever possible, pharmacokinetic monitoring should be employed to ensure appropriate drug dosing, especially for agents with narrow therapeutic windows (see Chapter 56). In addition to careful monitoring for toxicity of agents that are normally excreted by the kidney, careful attention must be paid to dosing of antibiotics and other drugs removed by KRT to assure that therapeutic drug levels are achieved, particularly in patients receiving augmented intensity of KRT. Drug dosing in kidney disease is further discussed in Chapter 56.

KIDNEY REPLACEMENT THERAPY IN AKI

GENERAL PRINCIPLES

KRT is the generic term for the multiple modalities of dialysis and hemofiltration employed in the management of kidney failure. Although kidney transplantation is also a form of KRT for end-stage renal disease, transplantation does not play a role in the management of AKI. KRT facilitates the management of patients with AKI, allowing correction of acid-base and electrolyte disturbances, amelioration of volume overload, and removal of byproducts of nitrogen metabolism (so-called “uremic solutes”). Although KRT can forestall or reverse the life-threatening complications of uremia associated with severe and prolonged AKI, it does not hasten and can potentially delay the recovery of kidney function in patients with AKI,⁵⁸⁴ and it can be associated with potentially life-threatening complications.⁵⁸⁵ Despite more than 60 years of research and clinical experience,^{586,587} numerous questions regarding the optimal application of KRT in AKI remain.^{3,588-591}

INDICATIONS FOR AND TIMING OF INITIATION OF KIDNEY REPLACEMENT THERAPY

In clinical practice there are wide variations in the timing of initiation of KRT for patients with AKI.⁵⁹² Widely accepted indications for initiation of KRT include volume overload unresponsive to diuretic therapy, severe metabolic acidosis or hyperkalemia despite appropriate medical therapy, and overt uremic manifestations including encephalopathy, pericarditis, or uremic bleeding diathesis (Table 28.17). However, even these specific indications are subject to substantial clinical interpretation, and in many patients, KRT is initiated in the absence of these specific indications in response to a clinical course marked by progressive azotemia or sustained oliguria. The correlation between the BUN concentration and onset of uremic symptoms is relatively weak, although the longer the duration and the greater the severity of azotemia, the more likely it is that overt symptoms will develop. Observational series and small clinical trials dating from the 1950s through the 1980s suggested that initiating KRT when the BUN concentration approached 90 to 100 mg/dL was associated with improved survival as compared with more delayed initiation of therapy.⁵⁹³⁻⁵⁹⁷ More recent observational studies have suggested that initiation of KRT at even less severe degrees of azotemia may further improve survival.⁵⁹⁸⁻⁶⁰¹ These studies need to be interpreted with

Table 28.17 Indications for Kidney Replacement Therapy

Absolute indications	Volume overload unresponsive to diuretic therapy Persistent hyperkalemia despite medical therapy Severe metabolic acidosis Overt uremic symptoms Encephalopathy Pericarditis Uremic bleeding diathesis
Relative indications	Progressive azotemia without uremic manifestations Persistent oliguria

caution, as the outcomes associated with earlier initiation of KRT may reflect differences related to the reasons for initiation of therapy (e.g., volume overload or hyperkalemia vs. progressive azotemia) rather than a benefit due to the earlier therapy per se. In addition, these observational series only included patients in whom KRT was actually initiated rather than the broader population of patients with AKI, including patients who either recovered kidney function or died without receiving KRT.

Numerous prospective clinical trials have evaluated the timing of initiation of KRT in AKI. In a small, randomized, controlled trial of critically ill patients randomized to early, high-volume hemofiltration, early low-volume hemofiltration, or late low-volume hemofiltration, there was no benefit associated with earlier initiation of treatment.⁶⁰² In a subsequent trial comparing earlier to later initiation of dialysis in patients with community-acquired AKI, mortality was lower in patients initiated on dialysis later than in the group of patients started earlier with no difference in recovery of kidney function between groups.⁶⁰³ This latter trial needs to be interpreted with caution, however, as almost half of the patients admitted with community-acquired AKI were excluded due to urgent need for dialysis.

Four larger, randomized clinical trials have evaluated the timing of KRT among critically ill patients with AKI (Table 28.18).⁶⁰⁴⁻⁶⁰⁷ The Effect of Early versus Delayed Initiation of Renal Replacement Therapy on Mortality in Critically Ill Patients with Acute Kidney Injury (ELAIN) trial randomized 231 patients with AKI (based on KDIGO stage 2 and plasma NGAL level >150 ng/mL) at a single surgical ICU to early or delayed initiation of KRT and demonstrated reduced 90-day mortality (hazard ratio 0.66; 95% CI: 0.45–0.97) with earlier initiation of KRT.⁶⁰⁴ Patients in the early KRT group also demonstrated decreased duration of KRT (9 days vs. 25 days, $P = 0.04$) and reduced hospital length of stay (51 days vs. 82 days, $P < 0.001$). However, the separation in timing of initiation of KRT between early and late groups was less than 1 day, raising questions about the therapeutic mechanisms underlying the observed marked reductions in mortality, duration of KRT, and hospital length of stay and suggesting that unrecognized differences between the two treatment groups may have contributed to the surprisingly large effect size.

The Artificial Kidney Initiation in Kidney Injury (AKIKI) trial was a multicenter trial that randomized 620 patients with

Table 28.18 Clinical Trials: Timing of Initiation of Kidney Replacement Therapy (KRT) in Critically Ill Patients with Acute Kidney Injury (AKI)

	ELAIN ⁶⁰⁴	AKIKI ⁶⁰⁵	IDEAL-ICU ⁶⁰⁶	STARTR-AKI ⁶⁰⁷
Patients	231	620	488	2927
Centers	1	31	29	168
Population	Surgical ICU	Mixed ICU	Mixed ICU	Mixed ICU
Initial KRT modality	CVVHDF	IHD (55%)/CKRT (45%)	IHD (34%)/CKRT (66%)	IHD (26%)/PIKRT (4%)/CKRT (69%)
AKI entry criteria	KDIGO stage 2	KDIGO stage 3	RIFLE-F	KDIGO stage 2 or 3
Additional inclusion criteria	Fluid overload Severe sepsis Vasopressor dependent Nonrenal organ dysfunction	Mechanical ventilation Vasopressor dependent	Septic shock	Age ≥18 Receiving care in ICU Equipoise among treating physicians
Criteria for delayed KRT	KDIGO stage 3	[K ⁺] >6.0 mmol/L Pulmonary edema pH <7.15 BUN >112 mg/dL Oligo-anuria >72 hours	[K ⁺] >6.5 mmol/L Pulmonary edema pH <7.15 No recovery after 48 hours	[K ⁺] ≥6.0 mmol/L pH ≤7.20 or tCO ₂ ≤12 mmol/L Respiratory failure No recovery after 72 hours
Time from enrollment to KRT (hours)	6.0	4.3	7.6	4.4
early	25.5	57	51.5	29.1
late	21	53	44	25
difference				
Patients in late group without KRT	9%	49%	38%	38.2%
Mortality (early vs. late)	30.4% vs. 40.3% (<i>P</i> = 0.11)	41.6% vs. 43.5% (<i>P</i> = 0.79)	45.1% vs. 42.1% (<i>P</i> = 0.48)	43.9% vs. 43.7% (<i>P</i> = 0.92)
Day 28				
Day 60	38.4% vs. 50.4% (<i>P</i> = 0.07)	48.5% vs. 49.7% (<i>P</i> = 0.79)	57.7% vs. 53.7% (<i>P</i> = 0.38)	
Day 90	39.3% vs. 54.7% (<i>P</i> = 0.03)			

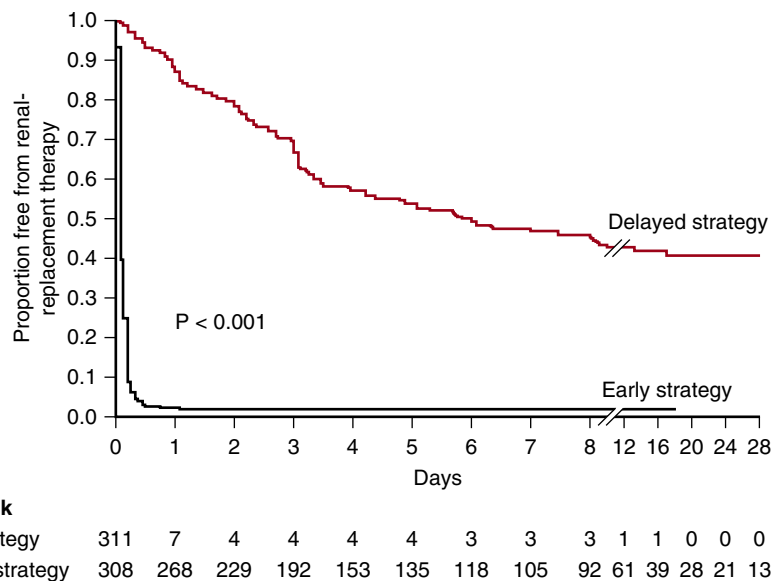
AKIKI, Artificial Kidney Initiation in Kidney Injury; CKRT, continuous kidney replacement therapy; ELAIN, Effect of Early vs. Delayed Initiation of Renal Replacement Therapy on Mortality in Critically Ill Patients with Acute Kidney Injury; ICU, intensive care unit; IDEAL-ICU, Initiation of Dialysis Early Versus Delayed in the Intensive Care Unit; IHD, intermittent hemodialysis; STARTR-AKI, Standard versus Accelerated Initiation of Renal Replacement Therapy in Acute Kidney Injury.

KDIGO stage 3 AKI who required mechanical ventilation and/or catecholamine support to early or delayed initiation of KRT.⁶⁰⁵ KRT was initiated as soon as possible in patients randomized to the early arm but was withheld unless hyperkalemia, metabolic acidosis, or volume overload resulting in pulmonary edema were present or oliguria persisted for more than 72 hours or the BUN exceeded 112 mg/dL. There was no difference in 60-day mortality between the groups (48.5% in early strategy group vs. 49.7% in delayed strategy group, *P* = 0.79). Notably, nearly half of patients assigned to the delayed strategy group never required initiation of KRT (see Fig. 28.4).

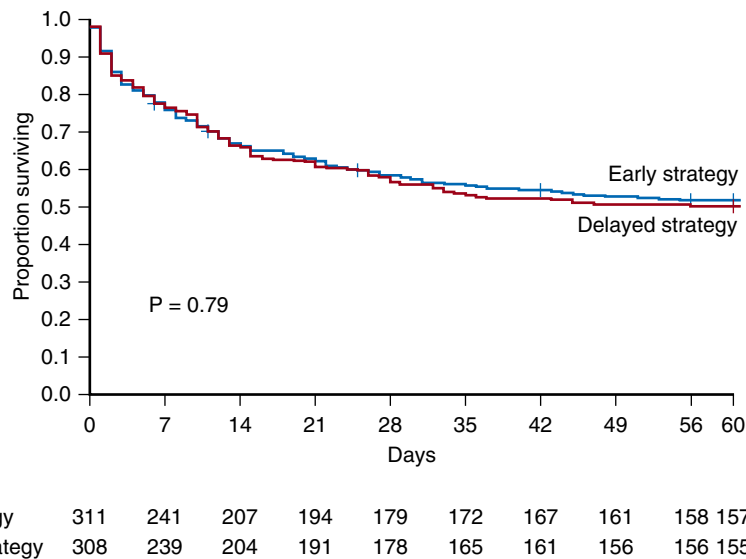
The Initiation of Dialysis Early versus Delayed in the Intensive Care Unit (IDEAL-ICU) trial also failed to demonstrate a benefit to earlier initiation of KRT in 488 patients with sepsis-associated AKI.⁶⁰⁶ In the IDEAL-ICU trial, patients were enrolled if they met the criteria for RIFLE-F, AKI, had sepsis, and did not have an emergent indication for KRT. In the early-treatment arm, patients initiated KRT within 12 hours of eligibility while in the delayed-treatment arm, KRT was initiated if a specific indication for KRT developed or if there was no recovery of kidney function

after 48 hours. In the early-treatment arm, mortality at 90 days was 58% as compared with 54% in the delayed-treatment arm (*P* = 0.38). Ninety-seven percent of patients in the early-treatment arm received KRT as compared with only 62% of patients in the delayed-treatment arm. Of those in the delayed arm who did not receive KRT, 75% had spontaneous recovery of kidney function while 23% died before specified criteria for initiation of KRT.

The Standard versus Accelerated Initiation of Renal Replacement Therapy in Acute Kidney Injury (STARTR-AKI) trial was a multinational trial that enrolled 3019 critically ill patients with KDIGO stage 2 or 3 AKI at 168 hospitals in 15 countries.⁶⁰⁷ Patients were eligible if they did not have advanced CKD, a treatable form of AKI, or an urgent indication for KRT, and their treating physicians had equipoise regarding the immediate need to start or defer initiation of KRT. Patients randomized to early initiation had KRT started as soon as possible and within 12 hours of meeting full eligibility while patients randomized to the standard (delayed) initiation arm had KRT withheld unless specific criteria were present (serum potassium ≥6.0 mmol/L, pH ≤7.20, serum bicarbonate ≤12 mmol/L, or the presence



A



B

Fig. 28.4 Timing of initiation of kidney replacement therapy (KRT) and survival in the Acute Kidney Initiation in Kidney Injury (AKIKI) Trial comparing early and delayed strategies for initiation of KRT. Timing of initiation of KRT (A) and Kaplan-Meier probability of survival (B) in the Acute Kidney Initiation in Kidney Injury (AKIKI) trial. Most (98%) of the patients in the early-treatment group initiated KRT at a median of 4.3 hours after reaching stage 3 AKI as compared with 51% of patients who initiated KRT at a median of 57 hours. Sixty-day mortality was 48.5% in the early-treatment group versus 49.7% in the delayed-treatment group (HR: 1.03; 95% CI: 0.81–1.29; $P = 0.84$). (From Gaudry S, et al. Initiation strategies for renal-replacement therapy in the intensive care unit. *N Engl J Med*. 2016;375:122–133.)

of respiratory failure due to volume overload) or persistent AKI for 72 hours after randomization. Ninety-seven percent of patients randomized to the early initiation arm received KRT with a median time from full eligibility to start of KRT of 6.1 hours (IQR: 3.9–8.8 hours) as compared with 62% of individuals randomized to the standard (delayed) arm, with a median time to initiation of KRT of 31.1 hours (IQR 19.0–17.8 hours). All-cause mortality at 90 days was not different between the two treatment arms (43.9% in the early arm vs. 43.7% in the standard arm) (Fig. 28.5); however, dialysis dependence at day 90 among surviving patients was higher in the early initiation arm (10.4% vs. 6.0%; HR 1.74, 95% CI: 1.24–2.43).

A meta-analysis that pooled data from these four large trials and several smaller studies found no benefit associated with earlier initiation of KRT, with a log odds ratio of -0.04 (95% CI: -0.16 – 0.07).⁶⁰⁸ There was also a trend to greater dialysis dependence associated with

earlier initiation of KRT, although this did not achieve statistical significance (log odds ratio 0.24; 95% CI: -0.03 – 0.51).

In a follow-up to their initial trial, the AKIKI investigators evaluated a strategy of even later initiation of KRT.⁶⁰⁹ Two-hundred seventy-eight critically ill patients with AKI who had oliguria for more than 72 hours or a BUN of more than 112 mg/dL without other indications for initiation of KRT were randomized to begin KRT immediately or wait until objective criteria (hyperkalemia, acidosis, or pulmonary edema) were present or the BUN exceeded 140 mg/dL. Greater delay in initiation of KRT did not result in further reduction in the number of KRT-free days but was associated with an increased hazard ratio for death (1.65; 95% CI: 1.09–2.50; $P = 0.018$) in a prespecified multivariable analysis. Thus the authors concluded that while early initiation of KRT was not beneficial, excessive delay was associated with an increased risk of preventable death.

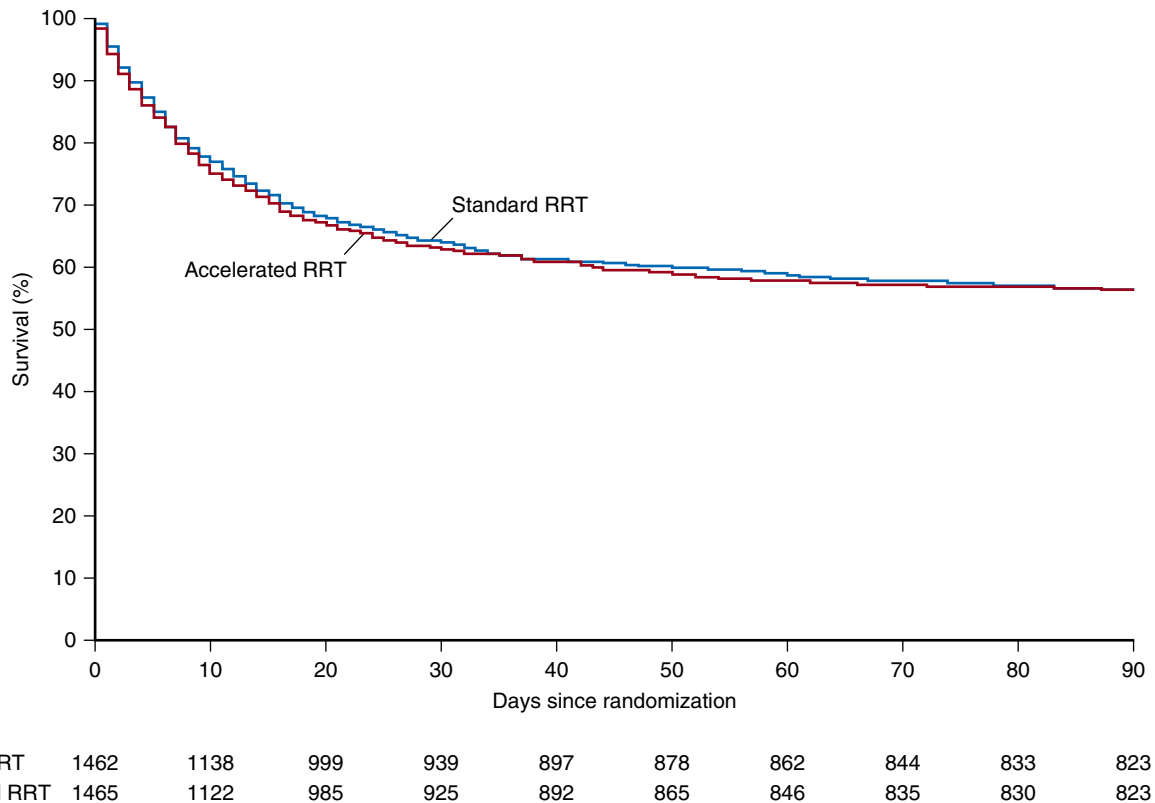


Fig. 28.5 Survival with standard (delayed) versus accelerated initiation of kidney replacement therapy (KRT) in the Standard versus Accelerated Initiation of Renal Replacement Therapy in the Acute Kidney Injury (STARRT-AKI) Trial. Kaplan-Meier plot of survival probability among 2927 evaluable patients in the STARRT-AKI Trial, which compared strategies of early and delayed KRT in critically ill patients with AKI. Death at 90 days occurred in 643 patients (43.9%) in the accelerated KRT arm as compared with 639 patients (43.7%) in the standard (delayed) KRT arm ($P = 0.92$). RRT, Renal replacement therapy. (From STARRT-AKI Investigators, Australian and New Zealand Intensive Care Society Clinical Trials Group, United Kingdom Critical Care Research Group, et al. Timing of initiation of renal-replacement therapy in acute kidney injury. *N Engl J Med*. 2020;383:240–251.)

Although volume overload unresponsive to diuretic therapy is a widely accepted indication for initiation of KRT, wide variations in the degree of volume overload at initiation of therapy exist.^{564,610,611} Observational studies have demonstrated a strong association between the degree of volume overload and mortality risk, leading to the suggestion that KRT should be initiated early, before the development of progressive volume overload.^{563,612} The association between volume overload and mortality risk does not, however, establish a causal relationship; disease processes that contribute to the development of volume overload may independently contribute to mortality risk in these patients. In a newer, single-center, prospective observational study of 820 critically ill patients treated with continuous KRT (CKRT), mortality was associated with greater positive fluid balance at time of CKRT initiation.⁶¹³ However, after adjusting for covariates including age, SOFA score, and vasopressor dose, cumulative fluid balance at time of CKRT initiation was no longer associated with ICU or hospital mortality (OR 1.01 per 1000 mL fluid accumulation; 95% CI: 0.97–1.05).⁶¹³ Similarly, in a secondary analysis of the STARRT-AKI trial, earlier initiation of KRT attenuated cumulative fluid balance; however, among patients with greater fluid accumulation at the time of randomization, earlier KRT initiation did not have an effect on all-cause mortality.⁶¹⁴

The KDOQI Clinical Practice Guidelines for Acute Kidney Injury, which predate the ELAIN, AKIKI, IDEAL-ICU, and STARRT-AKI trials, do not make strong recommendations on the timing of initiation of KRT.³ They suggest that KRT be “... initiated emergently when life-threatening changes in fluid, electrolyte, and acid-base balance exist”³ and further suggest that “... the broader clinical context, the presence of conditions that can be modified by KRT, and trends of laboratory tests—rather than single BUN and creatinine thresholds alone—[be considered] when making the decision to start KRT.”³ The accumulated data since publication of these guidelines demonstrate that early initiation of KRT before the presence of objective clinical indications is not associated with improved clinical outcomes; however, undue delay may be associated with increased mortality.⁶¹⁵

DISCONTINUATION OF KIDNEY REPLACEMENT THERAPY

KRT should be discontinued when kidney function recovers or because continued provision of dialytic support is no longer consistent with the patient’s overall goals of care.³ Recovery of kidney function is usually heralded by increased urine volume. Although no specific threshold of urine output correlates with sufficient recovery of kidney function, it is unlikely that a urine output of less than roughly 1 L/

day is sufficient to sustain dialysis independence. Although diuretics may increase daily urine volume, there is no evidence that diuretic therapy promotes recovery of kidney function.⁶¹⁶ Improved solute clearance is manifested by spontaneous fall in blood urea and creatinine concentrations or a persistent downward trend in predialysis values. The role of creatinine clearance measurement to assess recovery of kidney function is uncertain, with a paucity of data to define specific thresholds for recovery of kidney function. In the Acute Renal Failure Trial Network (ATN) Study, KRT was continued if measured creatinine clearance on a 6-hour timed urine collection was <12 mL/min; KRT was stopped if the clearance was >20 mL/min; and the decision was left to the discretion of the clinician if the creatinine clearance was between 12 and 20 mL/min.^{617,618} While specific criteria for discontinuation of KRT cannot be provided, patients should be carefully monitored for evidence of recovery.⁶¹⁹

CHOICE OF MODALITY OF KIDNEY REPLACEMENT THERAPY

Multiple modalities of KRT are available for the management of patients with AKI including conventional intermittent hemodialysis (IHD), peritoneal dialysis (PD), multiple forms of continuous KRT (CKRT), and prolonged intermittent kidney replacement therapies (PIKRTs), such as sustained low-efficiency dialysis (SLED; also known as extended duration dialysis). Detailed descriptions of the technical aspects of these modalities are provided in Chapters 62, 63, and 64 (CKRT and PIKRT). Objective data to guide the selection of modality for individual patients are limited, and the choice of modality is often guided by the resources of the health care institution and technical expertise of the physicians and nursing staff. The KDIGO Clinical Practice Guideline for Acute Kidney Injury suggests that for the majority of patients, the available modalities of KRT are complementary, with the caveats that CKRT and PIKRT be used in hemodynamically unstable patients and CKRT be used for patients with acute brain injury or other causes of increased intracranial pressure or generalized brain edema.³

Intermittent hemodialysis

Intermittent hemodialysis has been the mainstay of KRT in AKI for 7 decades. Patients typically undergo dialysis treatments for 3 to 5 hours on a daily or thrice-weekly, alternate-day schedule depending on catabolic demands, electrolyte disturbances, and volume status. Just as with the timing of initiation of dialysis in AKI, the most appropriate dosing strategy for IHD in patients with AKI has been the subject of considerable investigation. The dose of IHD may be adjusted by altering the intensity of each individual dialysis session, usually quantified as the product of urea clearance and dialysis duration normalized to volume of distribution of urea (Kt/V_{urea}), or by changing the frequency of the dialysis sessions. In an observational study, Paganini and colleagues⁶²⁰ demonstrated a survival benefit in patients with intermediate severity of illness scores when the delivered Kt/V_{urea} was >1.0 per treatment as compared with a delivered $Kt/V_{urea} <1.0$ per treatment. However, there have been no prospective clinical trials evaluating the relationship between the delivered Kt/V_{urea} and outcomes when dialysis is provided on a constant treatment schedule.

Schiffl and colleagues⁶²¹ reported on a prospective trial of 160 patients with AKI assigned in an alternating fashion to alternate-day or daily intermittent hemodialysis. The more frequent treatment schedule was associated with a reduction in mortality at 14 days after the last dialysis session from 46% in the alternate-day dialysis arm to 28% in the daily treatment arm ($P = 0.01$). The duration of renal failure declined from 16 ± 6 days to 9 ± 2 days ($P = 0.001$). This study has been criticized, however, because the delivered dose of therapy per session was low in both treatment arms ($Kt/V_{urea} <0.95$), resulting in a high rate of symptoms in the alternate-day dialysis arm that may have been related to overtly inadequate dialysis.⁶²² The impact of frequency of IHD was also evaluated in the ATN study.⁶¹⁷ In the ATN study, 1124 critically ill patients were randomized to an intensive or less-intensive strategy for the management of KRT. When patients were hemodynamically stable, they received IHD, and when hemodynamically unstable they received CKRT or SLED, regardless of treatment arm. Patients randomized to the less-intensive treatment strategy received IHD on a thrice-weekly (alternate-day except Sunday) schedule while patients randomized to the intensive arm received six-times IHD per week (daily except Sunday). The 60-day all-cause mortality was 53.6% in the intensive treatment arm compared with 51.5% in the less-intensive arm ($P = 0.47$)⁶¹⁷ (Fig. 28.6). The mean delivered Kt/V_{urea} was 1.3 per treatment after the first IHD session. Although the study was not designed to evaluate outcomes by individual modality of KRT, there were no differences in mortality between groups when evaluated on the basis of percentage of time treated using IHD.⁶²³ According to these results, it does not appear that there is further benefit to routinely increasing the frequency of IHD treatments beyond three times per week as long as the delivered Kt/V_{urea} is at least 1.2 per treatment. More frequent treatments may be necessary if the target dose per treatment cannot be achieved (e.g., in hypercatabolic patients, in patients with severe hyperkalemia or metabolic acidosis, and for issues related to volume management). The KDIGO Clinical Practice Guideline for Acute Kidney Injury recommends delivering a Kt/V_{urea} of 3.9 per week when using IHD in AKI, calculating the weekly Kt/V_{urea} as the arithmetic sum of the delivered dose per treatment.³ It should be recognized, however, that this approach for calculating an equivalent weekly Kt/V_{urea} is not consistent with urea kinetic principles and that rigorous data for the appropriate dose of therapy when treatments are delivered more frequently than three times per week are not available.¹⁵

The selection of IHD dialyzer membrane may also impact clinical outcomes. Exposure to cellulosic membranes results in accentuated leukocyte and complement activation and delayed recovery of kidney function in experimental models of AKI as compared with exposure to more biocompatible synthetic membranes.^{624,625} Clinical trials comparing dialysis membranes have yielded conflicting results. Although some studies demonstrated delayed recovery of kidney function with cellulosic membranes,⁶²⁶⁻⁶²⁸ other studies observed no difference between cellulosic and other synthetic membranes thought to be more biocompatible.⁶²⁹⁻⁶³³ When these data have been aggregated in systematic reviews, a benefit of the synthetic membranes is not convincingly demonstrated.^{634,635} While the effect of membrane type

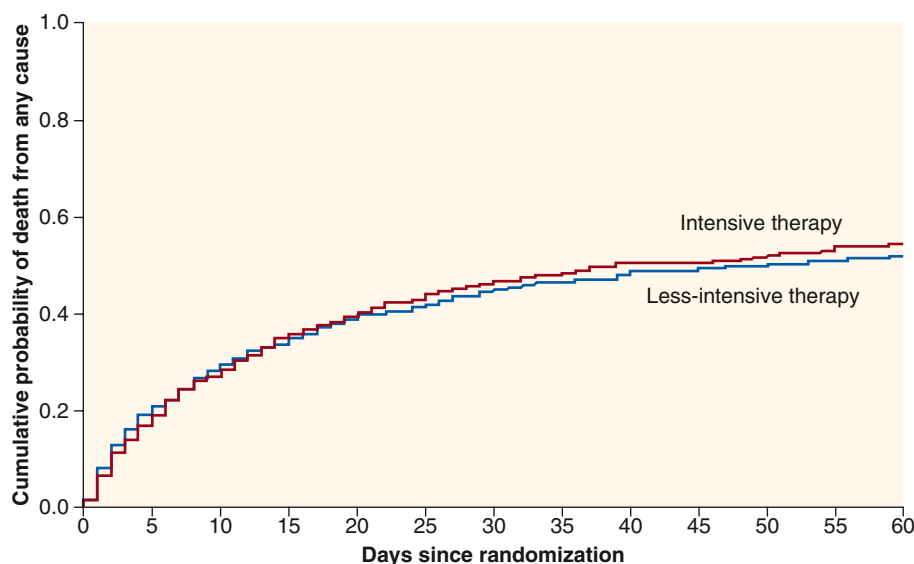


Fig. 28.6 Sixty-day mortality with intensive versus less-intensive kidney replacement therapy in the Acute Renal Failure Trial Network (ATN) study. Kaplan-Meier plot of mortality in 1124 critically ill patients with acute kidney injury randomized to a strategy of more intensive kidney replacement therapy (6x-per week intermittent hemodialysis or continuous venovenous hemodiafiltration at 35 mL/kg per hour) versus less-intensive kidney replacement therapy (3x-per week intermittent hemodialysis or continuous venovenous hemodiafiltration at 20 mL/kg per hour). At 60 days, mortality was 53.6% in the more intensive arm versus 51.5% in the less-intensive arm (OR: 1.09; 95% CI: 0.86–1.40; $P = 0.47$). (From The VA/NIH Acute Renal Failure Trial Network. Intensity of renal support in critically ill patients with acute kidney injury. *N Engl J Med.* 2008; 359:7–20.)

on humoral and cellular activation may still influence recovery of kidney function in AKI, the clinical importance of this issue has diminished as the cost differential between synthetic and cellulosic membranes has narrowed and the use of unsubstituted cellulosic membranes has decreased.

The major complications associated with IHD are related to the need to access the vasculature, the need for anticoagulation to maintain patency of the extracorporeal circuit, and intradialytic hypotension primarily resulting from shifts in solute and volume.^{617,621,636} Many of these issues, particularly the need for vascular access and anticoagulation, are similar for CKRT and SLED.

Vascular access is usually obtained through insertion of a double-lumen catheter into a large-caliber central (internal jugular or subclavian) or femoral vein.⁶³⁷ The major complications associated with vascular access include vascular and organ trauma during insertion, bleeding, catheter malfunction and thrombosis, and infection.⁶³⁷ Although femoral catheters are generally associated with an increased risk of infection compared with catheters in the subclavian or internal jugular veins, an increased risk of infection was observed only when femoral-vein catheters were used in patients with a high BMI in a randomized controlled trial involving patients undergoing acute KRT.⁶³⁸ Prompt transition to tunneled hemodialysis catheters (or placement of tunneled hemodialysis catheters in advance of initiation) has been proposed as a means of decreasing the risk of infection in patients undergoing acute dialysis.^{639,640} However, this strategy has not been rigorously evaluated in prospective clinical trials.

Anticoagulation is used to help maintain patency of the extracorporeal dialysis circuit in IHD, as well as CKRT and SLED.^{641,642} The most commonly used anticoagulant for dialysis is unfractionated heparin, with multiple protocols used to attain sufficient anticoagulation of the dialysis circuit while minimizing systemic effects.^{641,642} Regional heparinization, in which heparin is infused proximal to the dialyzer and protamine is infused into the return line to reverse heparin's effect,⁶⁴³ can be used but has generally been supplanted by low-dose heparin protocols.⁶⁴⁴ Low-molecular-weight heparins (LMWH) may be used as an alternative to

unfractionated heparin, but the benefits of this approach are unclear as LMWH is not associated with enhanced efficacy, drug half-life is variably prolonged with impaired kidney function, and monitoring of the anticoagulant effect is more difficult.⁶⁴¹ In patients with heparin-induced thrombocytopenia (HIT), heparin administration is contraindicated. Alternative anticoagulation strategies include regional citrate^{641,645-647}; the serine protease inhibitor nafamostat⁶⁴⁸; the direct thrombin inhibitors hirudin, leprudin, and argatroban⁶⁴⁹⁻⁶⁵³; and, rarely, the prostenoids epoprostenol and iloprost.^{641,642} In many patients, particularly those with underlying coagulopathy or thrombocytopenia, and in patients with active hemorrhage or recent postoperative status, acute KRT can be provided in the absence of anticoagulation.^{617,654,655}

Intradialytic hypotension is common in patients undergoing acute IHD.^{584,617,623,636,656} Episodes of hypotension may impair solute clearance and the efficiency of dialysis and can further compromise renal perfusion and delay recovery of kidney function.^{584,657-659} Intradialytic hypotension is typically triggered by intercompartmental fluid shifts or excessive fluid removal, leading to decreased intravascular volume, and may be exacerbated by altered vascular responsiveness related to the underlying acute process.^{636,660} Hypotension may be particularly problematic in critically ill patients in whom sepsis, cardiac dysfunction, hypoalbuminemia, malnutrition, or large third-space losses may accompany the development of AKI. Prevention of intradialytic hypotension requires careful assessment of intravascular volume; prescription of realistic ultrafiltration targets; extension of treatment time so as to minimize the ultrafiltration rate; increasing the dialysate sodium concentration; and decreasing the dialysate temperature.^{656,660-662} It is noteworthy that intradialytic hypotension can develop even among patients in whom no ultrafiltration is prescribed; the reason(s) for hypotension are not entirely clear, although many have attributed hemodynamic instability to the rapid exchange of solutes induced by high flux/high-efficiency hemodialysis with extracellular to intracellular shifting of body water.

Although there is a tendency to reduce the extracorporeal blood flow in patients prone to hypotension, there is little evidence that this provides any benefit. Reducing blood flow decreased the volume of the extracorporeal circuit in the past when parallel plate and coil dialyzers were used; however, there is little change in the volume of the extracorporeal circuit in response to changes in blood flow when hollow fiber dialyzers are used. Reducing blood flow may, however, result in reduction of the delivered dose of dialysis.

Continuous kidney replacement therapy

Continuous kidney replacement therapies represent a spectrum of treatment modalities. In their initial description, the continuous therapies were provided using arteriovenous extracorporeal circuits.⁶⁶³⁻⁶⁶⁷ While this approach provided technical simplicity, blood flow was dependent on the gradient between mean arterial and central venous pressure and there was an increased risk of complications from prolonged arterial cannulation.⁶⁶⁸ As a result, the continuous arteriovenous therapies have largely been supplanted by pump-driven, venovenous CKRT.⁶⁶⁹⁻⁶⁷² The modalities of venovenous CKRT vary predominantly based on their mechanism of solute removal. With continuous venovenous hemofiltration (CVVH), solute transport occurs by convection; with continuous venovenous hemodialysis (CVVHD) by diffusion; and with continuous venovenous hemodiafiltration (CVVHDF) by a combination of the two.^{615,672-674} Although, at the same level of urea clearance, convective therapies provide enhanced clearance of higher molecular weight solutes compared with diffusive therapies, no clear clinical benefit has been demonstrated for CVVH or CVVHDF as compared with CVVHD.⁶⁷⁵

The clearance of urea and other small solutes during CKRT is generally proportional to the total effluent flow rate (the sum of ultrafiltrate and dialysate flow rates),^{667,672,673} and dose of therapy is usually expressed as the effluent volume indexed to body weight. This approach to estimating solute clearance is based on the assumption of near-complete solute equilibration between blood and effluent and may overestimate the actual solute clearance.^{676,677} Several single-center randomized controlled trials demonstrated an improvement in survival when doses of CVVH were increased from 20 to 25 mL/kg per hour to doses in excess of 35 to 45 mL/kg per hour.^{678,679} However, other small studies did not find a similar benefit.^{602,680} Two large multicenter, randomized controlled trials also did not find a survival benefit associated with more intensive CKRT.^{617,681} In the previously described ATN study, 1124 patients were randomized to two intensities of KRT.⁶¹⁷ In both treatment arms, patients received IHD when hemodynamically stable and CVVHDF or SLED when hemodynamically unstable. In the less-intensive arm, CVVHDF was provided at an effluent flow rate of 20 mL/kg per hour, and in the more intensive arm at 35 mL/kg per hour. Sixty-day all-cause mortality was 51.5% in the less-intensive arm and 53.6% in the more intensive arm ($P = 0.47$) (Fig. 28.6).⁶¹⁷ In the Randomized Evaluation of Normal versus Augmented Level (RENAL) Replacement Therapy study, 1508 patients were randomized to CVVHDF at either 25 mL/kg per hour or 40 mL/kg per hour.⁶⁸¹ Ninety-day all-cause mortality was 44.7% in both treatment arms ($P = 0.99$) (Fig. 28.7).⁶⁸¹ On the basis of

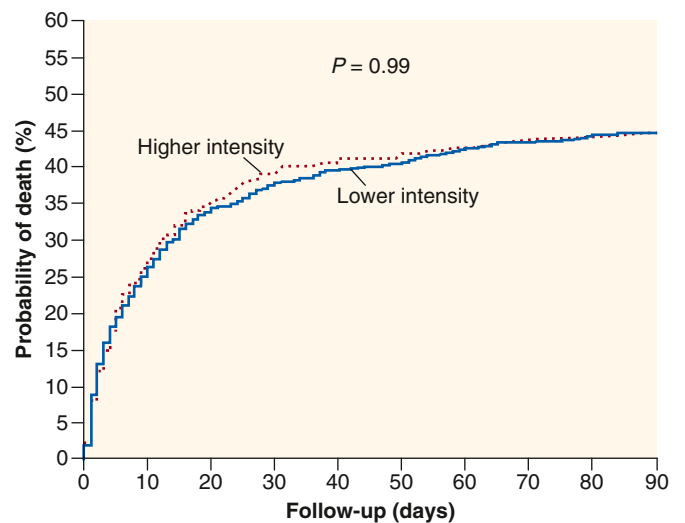


Fig. 28.7 Ninety-day mortality with intensive versus less-intensive continuous venovenous hemodiafiltration (CVVHDF) in the Randomized Evaluation of Normal versus Augmented Level (RENAL) Replacement Therapy Study Kaplan-Meier plot of mortality in 1508 critically ill patients with acute kidney injury randomized to CVVHDF at 35 mL/kg per hour versus 20 mL/kg per hour. At 90 days, mortality was 44.7% in both treatment groups (OR: 1.00; 95% CI: 0.81–1.23; $P = 0.99$). (From The RENAL Replacement Therapy Study Investigators. Intensity of continuous renal-replacement therapy in critically ill patients. *N Engl J Med*. 2009;361:1627–1638.)

these data, the KDIGO Clinical Practice Guideline for Acute Kidney Injury recommends delivering an effluent volume during CKRT of 20 to 25 mL/kg per hour, recognizing that a slightly higher dose may need to be prescribed in order to achieve the target delivered dose in order to compensate for interruptions in treatment.³ Although both the ATN and RENAL studies demonstrated an absence of benefit with more intensive dosing of KRT, neither were designed to establish the minimal adequate dose of CKRT. Observational data from Japan have demonstrated that mortality does not increase until the delivered dose is <13 to 5 mL/kg per hour,⁶⁸² suggesting that the KDIGO target of 20 to 25 mL/kg per hour may be higher than necessary.

Given the improved hemodynamic tolerance of CKRT as compared with IHD, particularly in patients with underlying hemodynamic instability, it has been postulated that CKRT would be associated with improved clinical outcomes. Five randomized controlled trials comparing outcomes with CKRT and IHD have been published. In a multicenter, randomized controlled trial of 166 patients with AKI, Mehta and colleagues⁶⁸³ observed ICU and hospital mortality rates of 59.5% and 65.5%, respectively, in patients randomized to CKRT as compared with 41.5% and 47.6%, respectively, in patients randomized to IHD ($P < 0.02$). As the result of an imbalance in randomization, patients in the CKRT arm had higher severity of illness as measured by APACHE III score and a higher rate of liver failure. Adjusting for the imbalanced randomization in a post hoc analysis, the investigators found no difference in mortality attributable to modality of KRT. In another single-center randomized trial ($n = 80$), Augustine and colleagues⁶⁸⁴ reported more effective fluid removal and greater hemodynamic stability associated with CVVHD as compared with IHD but no difference in survival. Similarly,

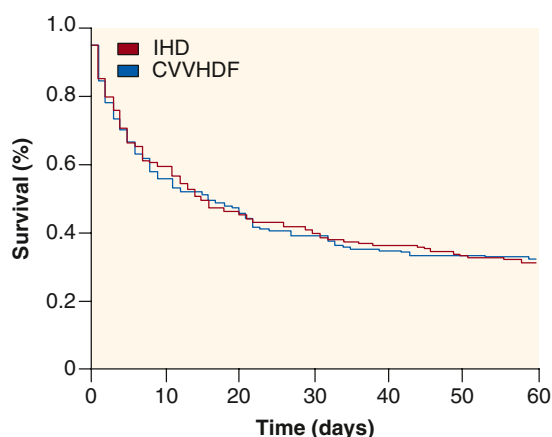


Fig. 28.8 60-Day Survival with Intermittent Hemodialysis (IHD) versus Continuous Venovenous Hemodiafiltration (CVVHDF) in the Hemodiafe Study. Kaplan-Meier plot of survival among 359 critically ill patients with acute kidney injury randomized to intermittent hemodialysis versus continuous venovenous hemodiafiltration. At 60 days survival was 31.5% among patients randomized to IHD versus 32.6% among patients randomized to CVVHDF ($P = 0.98$). (From Vinsonneau C, Camus C, Combes A, et al. Continuous venovenous haemodiafiltration versus intermittent haemodialysis for acute renal failure in patients with multiple-organ dysfunction syndrome: a multicentre randomised trial. *Lancet* 2006;368:379–385.)

in another single-center RCT from Switzerland, Uehlinger and colleagues⁶⁸⁵ observed no difference in survival in 70 patients randomized to continuous venovenous hemodiafiltration (CVVHDF) as compared with 55 patients assigned to IHD. In the Hemodiafe study, a multicenter RCT conducted in 21 ICUs in France, Vinsonneau and colleagues⁶⁵⁶ reported 60-day survival rates of 31.5% in 184 patients randomized to IHD as compared with 32.6% in 175 patients randomized to CVVHDF ($P = 0.98$) (Fig. 28.8). Similarly, Lins and colleagues observed hospital mortality rates of 62.5% in 144 patients randomized to IHD and 58.1% in 172 patients randomized to CKRT ($P = 0.43$).⁶⁸⁶ Multiple meta-analyses have concluded that there is no difference in survival among patients by KRT.^{687–689} Although several studies have suggested that CKRT is associated with improved rates of recovery of kidney function in surviving patients as compared with IHD,^{683,690–694} all of these studies are confounded by higher mortality rates in the CKRT group. When analyzed across studies in which there were no differences in mortality, rates of recovery of kidney function do not appear to be affected by modality of KRT.^{584,687,689,695}

Prolonged intermittent kidney replacement therapy

Prolonged intermittent KRT (PIKRT) represents a treatment modality in which conventional hemodialysis equipment is modified to provide extended-duration hemodialysis using lower blood flow rates and dialysate flow rates.^{696–698} A variety of terms have been developed to describe these therapies, including SLED,^{699,700} extended daily dialysis,⁷⁰¹ and sustained low-efficiency daily dialysis (SLEDD-f).⁷⁰² By extending the duration of the dialysis treatment while providing slower ultrafiltration and

solute clearance, these therapies are generally associated with enhanced hemodynamic tolerability compared with IHD. The degree of metabolic control attained with these treatments is comparable with that observed with CKRT.⁷⁰³ In an observational study performed in three ICUs in New Zealand, Australia, and Italy that changed from using CKRT to PIKRT, there was no difference in observed outcomes following the change in modality of KRT.⁶⁹⁷ Similarly, in a single-center, prospective, randomized controlled trial that included 232 patients, 90-day survival rates were similar in the PIKRT and CKRT groups (PIKRT: 50.4%; CKRT: 44.4%; $P = 0.43$), although overall resource utilization was lower with PIKRT.⁷⁰⁴ In a meta-analysis, there were no differences in mortality or recovery of kidney function comparing PIKRT with CKRT.⁷⁰⁵

Peritoneal dialysis

The use of peritoneal dialysis (PD) in the management of AKI has diminished as the use of continuous and hybrid therapies have increased.^{706–708} PD has the advantage of requiring minimal technology, facilitating its use in remote or resource-constrained areas.^{709,710} As a result, it is still used in the treatment of AKI in regions where access to IHD or CKRT is not possible. Access for acute PD can be obtained either by percutaneous placement of an uncuffed temporary peritoneal catheter or through surgical placement of a tunneled cuffed catheter. PD has the advantage of avoiding the need for vascular access or anticoagulation. Solute clearance and control of metabolic parameters may be inferior to that achieved with other modalities of KRT.⁷¹¹ Although systemic hypotension is less of an issue than with other modalities of KRT, ultrafiltration cannot be as tightly controlled. Other limitations include the relative contraindication in patients with acute abdominal processes or recent abdominal surgery, the risk of visceral organ injury during catheter placement, the risk of PD-associated peritonitis, and an increased tendency toward hyperglycemia due to the high glucose concentrations in peritoneal dialysate, which in other acute settings has been associated with adverse outcomes.⁷¹⁰ Several trials have compared outcomes using PD with other modalities of KRT in AKI.^{711–714} In a study of 70 patients with infection-associated AKI in Vietnam, 58 of whom had severe falciparum malaria, peritoneal dialysis was associated with less adequate metabolic control and higher mortality than continuous hemofiltration.⁷¹¹ In contrast, in a study of 120 patients in Brazil who were randomized to high-volume PD or daily hemodialysis, indices of metabolic control, recovery of kidney function, and survival were similar with both modalities.⁷¹² In a meta-analysis that included 8 observational studies and four clinical trials, Chinoh and colleagues⁷¹⁵ observed similar survival rates with PD as compared with extracorporeal KRT in patients with AKI.

SUMMARY OF ACUTE KIDNEY INJURY MANAGEMENT

AKI remains a common and serious disease with protean etiologies and variable clinical courses. Management of AKI begins with prevention in those clinical circumstances

in which evidence-based preventive interventions are available. While pharmacologic therapy is available for select etiologies of AKI, the treatment of established ATI is largely supportive with pharmacologic and kidney replacement therapies designed to ameliorate the adverse metabolic and clinical complications of this condition. Additional research is needed to identify treatments that decrease the risk of developing AKI, and in patients with established AKI, to reduce the severity of AKI and/or facilitate recovery. See Chapter 64 for further discussion on critical care nephrology.

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QUESTIONS

- Which of the following is true regarding the role of early goal-directed therapy (EGDT) for the management of sepsis?
 - EGDT is associated with a reduction in dialysis-requiring AKI but not mortality.
 - EGDT is associated with a reduction in dialysis-requiring AKI and mortality.
 - Recent clinical trials demonstrated that EGDT is not associated with a reduction in dialysis-requiring AKI.
 - Early antibiotic therapy and volume resuscitation are not associated with improved patient outcomes.

Answer: c

Rationale: Three randomized controlled trials, ProCESS, ARISE, and ProMISE, did not demonstrate a reduction in the need for KRT with EGDT.

- A 63-year-old woman with diabetes mellitus complicated by retinopathy and nephropathy is admitted with a non-ST-elevation myocardial infarction. Her serum creatinine is 1.68 mg/dL, and her urine albumin excretion is 762 mcg per mg creatinine. Which ONE of the following interventions is associated with the strongest evidence for reducing her risk of developing contrast-associated acute kidney injury after her planned coronary angiography?
 - Oral N-acetylcysteine
 - Periprocedural hemofiltration
 - Periprocedural infusion of isotonic sodium chloride
 - Use of iodixanol

Answer: c

Rationale: While the level of evidence documenting that periprocedural infusion of isotonic sodium chloride prevents development of contrast-associated AKI is low, trials in high-risk patients suggest that volume expansion with isotonic crystalloid is beneficial. None of the other interventions (oral N-acetylcysteine, periprocedural hemofiltration or use of iodixanol as compared to low-osmolal contrast agents) is associated with a reduction in the risk of contrast-associated AKI.

- Which ONE of the following strategies for management of kidney replacement therapy in patients with acute kidney injury and multisystem organ failure in the setting of sepsis is most strongly supported by evidence from clinical trials?

Use of continuous renal replacement therapy rather than intermittent hemodialysis

Use of continuous hemofiltration rather than continuous hemodialysis

Use of lower doses (20–25 mL/kg per hour) rather than higher doses (>35 mL/kg per hour) of continuous renal replacement therapy

Use of daily intermittent hemodialysis with a delivered kt/v_{urea} of 1.2 to 1.4 per treatment rather than alternate-day hemodialysis with the same delivered kt/v_{urea} per treatment

Answer: c

Rationale: Per the results of the ATN and RENAL trials, the KDIGO Clinical Practice Guidelines for Acute Kidney Injury recommend the use of lower doses (20–25 mL/kg per hour) rather than higher doses (>35 mL/kg per hour) of continuous renal replacement therapy in prescribing kidney replacement therapy in critically ill patients with acute kidney injury.

- With regard to the optimal time to initiate kidney replacement therapy (KRT) in the setting of acute kidney injury, which of the following is true?
 - Early initiation of KRT is associated with lower mortality as compared with delayed initiation of KRT.
 - Early initiation of renal replacement therapy is associated with improved recovery of kidney function as compared with delayed initiation of KRT.
 - In the absence of specific clinical indications (e.g., hyperkalemia and volume overload), initiation of kidney replacement therapy should be delayed until the BUN is >140 mg/dL.
 - In the absence of specific clinical indications (e.g., hyperkalemia and volume overload), a strategy of delayed initiation of KRT is associated with similar patient survival and reduced health care utilization than early initiation of KRT.

Answer: d

Rationale: Per the results of the STARRT-AKI, AKIKI, and IDEAL-ICU trials, a strategy of delayed initiation of KRT, is associated with similar patient survival, lower rates of dialysis-dependence and more KRT-free days as compared with accelerated initiation of KRT. In the AKIKI-2 trial, however, prolonged delay in initiation of KRT, until the BUN was >140 mmol/L, was associated with higher risks of death.